



INTRODUCTORY ANATOMY

**Department of Anatomy
School of Medical Sciences**

Access course 2026

Graduate Entry Medicine Programme

REFERENCE MATERIALS

Kenneth, S. S. (2020). *Anatomy and physiology: The unity of form and function.* McGraw Hill.

Scanlon, V. C. and Sanders, T. (2018). *Essentials of anatomy and physiology.* FA Davis.

Tortora, G. J. and Derrickson, B. H. (2018). *Principles of anatomy and physiology.* John Wiley & Sons.

Agur, A. M. and Dalley, A. F. (2022). *Moore's essential clinical anatomy.* Lippincott Williams & Wilkins.

Learning outcomes

- Define anatomy and describe its subdisciplines.
- Discuss why understanding human anatomy is essential to the science of medicine
- Define the anatomical position.
- Use anatomical terminology to describe body directions, regions, and planes
- Locate the major body cavities and their subdivisions.
- Describe the medical imaging techniques that are used to visualize structures inside the body.

HISTORY OF ANATOMY

- **275 BC:** Alexandria
- **Herophilus**
- **600-1100** Knowledge of Greek anatomy transferred to Western Europe
- Islamic scholars translate Greek anatomical treatises into Arabic
- **1100s-1500s**, Galen translates from Arabic into Latin, later from the Greek originals

- **1235**, First European medical school founded at Salerno, Italy; human bodies are publicly dissected
- **1490**, Anatomical theatre was opened in Padua, Italy
- **1491**, First illustrated printed medical book published in Venice, Johannes de Ketham, *Fasciculus medicinae*
- **1600-1900**, Anatomy plays an important role in medical education and research

An Overview of Anatomy

■ Anatomy

- the study of body structures and the relationships among these structures
- derived from the Greek word *anatome*, (*ana* -"up," and *tome*-"a cutting") that mean "a cutting up." = **dissection**

What is physiology?

Branches of Anatomy

- **Gross anatomy (Macroscopic anatomy)** - *study of structures visible to the naked eye*. Includes regional, systemic and clinical anatomy
- **Microscopic anatomy (histology)** - *study of structures too small to be seen by the naked eye*. Subdivisions include **cytology** (study of cells) and **histology** (study of body tissues).
- **Developmental anatomy**- study of structural changes in an individual throughout the life span. **Embryology** focuses on developmental changes occurring before birth.

Other types of anatomical studies

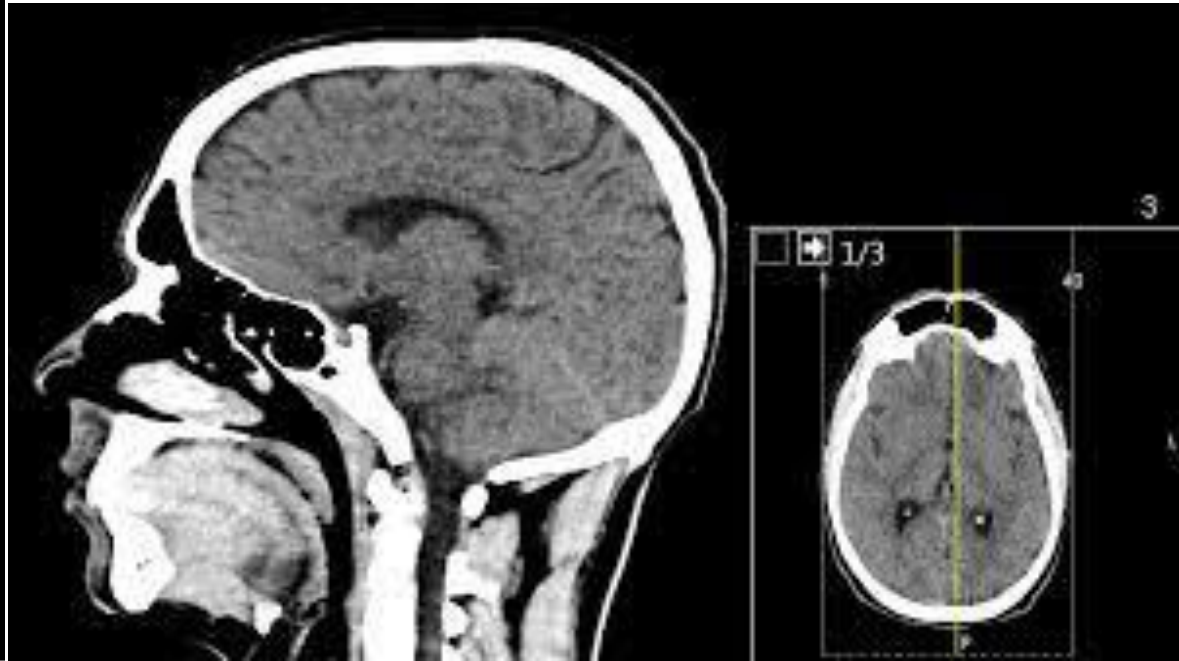
- **Surface (topographical) anatomy** – *study of the surface markings of the body to understand internal anatomy through visualization and palpation*



- **Radiographic anatomy**-*the study of internal body structures by means of imaging techniques such as X-rays, MRI etc*



MRI SCAN



CT SCAN



5.6" Groove Director



4" Splinter Tweezer Forceps Curved



4" 1x2 Teeth Tissue Forceps



4.5 " Splinter Tweezer Forceps



5" Strait Tip Teasing Needle



5" Curved Tip Teasing Needle



5" Scalpel Knife Handle



5.1" Scalpel Knife Handle



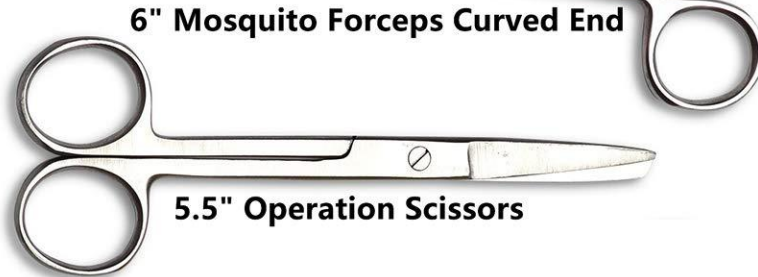
Dissecting Spatula



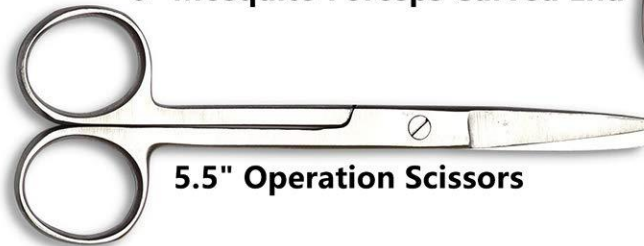
**6 " Mall Probe Seeker With Curved End (Steel)
Yellow Powder Coated Handle**



4.5" Iris Scissors



6" Mosquito Forceps Curved End



5.5" Operation Scissors



11 Blades For (Handle #3) X11



20 Blades For (Handle #4) X11



X2 Suture & Threads



Magnifying glass



6 Inches Stainless Steel Measuring Ruler



OVERVIEW OF ANATOMY CONTD.

ANATOMICAL TERMINOLOGY

- Anatomy has its own vocabulary/terms i.e anatomical nomenclature

Examples

- **Thorax**-chest
- **Axilla**-armpit
- **Carpal**-wrist
- **Brachial**-arm
- **Gluteal**-buttocks
- **Crural**-leg (front side)
- **Tarsal**-ankle

Eponyms-e.g Adam's apple, Eustachian tube, Sertoli cells, Achilles Tendon

OVERVIEW OF ANATOMY CONTD.

ANATOMICAL TERMINOLOGY

- **Prefixes**
- **Epi-** upon or over e.g epicardium i.e over heart
- **En- Endo-** within e.g endocardium i.e within the heart
- **Gastro-**stomach e.g gastritis i.e inflammation of the stomach
- **Colic-** Large eg. colon
- **Infra-** below e.g infraorbital i.e below the orbit
- **Myo-** muscle eg. Myometrium, Myosin,
- **Retro-** behind eg. Retrocaecal,retroperitoneal

ANATOMICAL TERMINOLOGY

- Prefixes
- A/an: no,not,without eg. Aphagia, Anaemia, A~~ph~~asia, Anesthesia/A~~pa~~thy
- Ante: before eg. Antepartum, Anterior, Antecubital, Antever~~s~~ion
- Anti: against eg. Antibiotic, Antidepressant, Antidiuretic
- Dys: difficult eg. Dyspareunia, Dysopia, Dysmenorrhea, Dysphagia
- Endo: within eg. Endoscope, Encephalitis, Endotracheal
- Epi: above eg. Epiglottis, Epidermis, Epistaxis
- Ex/Exo: out eg. Excretion, Exopathy, Exfoliate
- Hemi: half eg. Hemiplegia, Hemicolectomy, Hemis~~ph~~ere

ANATOMICAL TERMINOLOGY

- Hyper: above eg. Hyperactive, Hyperglycemia, Hyperesthesia
- Hypo; below eg. Hypoactive, Hypodermic, Hypokalemia
- Inter: between eg. Intercostal, Interfemoral, Intermission, Interval
- Intra: within eg. Intraabdominal, Intracranial, Intracostal, Intramammary
- Peri: around, surrounding eg. Peritoneum, Peritonsillar, Peristalsis
- Poly: many eg. Polycythemia, Polyarthritis
- Post: after eg. Postnatal, Postoperative, Postmortem

ANATOMICAL TERMINOLOGY

- Pro : before eg. Procedure, Procreation, Prophylaxis
- Retro: behind eg. Retroaction, Retroversion, Retropharynx
- Trans: across eg. Transabdominal, Transference, Transfusion

ANATOMICAL TERMINOLOGY

- **Suffixes**

algia- pain e.g neuralgia(nerve pain), cephalgia

-ectomy- cutting out e.g hepatectomy , removal of the liver,
hysterectomy, gastrectomy, appendectomy

-itis-inflammation

-ology- study of

-pathy- suffering from

-scopy- to view

-otomy/-stomy-surgical opening

-dynia- pain eg Gastrodynia, Mastodynia

ANATOMICAL TERMINOLOGY

- Suffixes
- emia: blood condition eg. Leukaemia, anaemia
- Oma: tumour (benign/malignant). Eg. Gastroma, Lymphoma, Melanoma, Sarcoma
- Pathy: disease of eg. Hepatopathy, Neuropathy, Cardiopathy
- Plasty: surgical repair eg. bronchoplasty, rhinoplasty
- rrhagia/rrhage: severe bleeding eg. Hemorrhage, Menorrhagia

ANATOMICAL TERMINOLOGY

- Scope: visual exam instrument eg. Ophthalmoscope, Gastroscope, Proctoscope, Bronchoscope
- Spasm: Contraction eg. Pylorospasm, Bronchiospasm, Vasospasm
- Stenosis: tightening structure eg. Pylorostenosis, Aortostenosis, Pulmonarystenosis, Valvularstenosis
- Tomy: to cut into Gastrotomy, Laparotomy
- Um: structure: Epigastrium, Hypogastrium

Combining forms

- Arthro: joint eg. Arthritis, Arthroscopy, Arthroplasty, Arthrocentesis
- Carcin/o: Cancer: Carcinoma, Carcinogen, Carcinostatic, Carcinolysis
- Cyst/o: fluid filled eg. Cystic, Cystitis
- Cyto: cell eg. Cytology, Cytogenic, Cytonecrosis
- Erythro: red eg. Erythrocytosis, Erythrophobia, Erythroopsia, Erythroderma
- Thrombo: clot eg. Thrombosis, Thrombocytopenia
- Viscera: internal organ eg. Visceral, Visceromegaly

Plural forms

- **Words ending with**

“a” The plural becomes “e” eg. Vertebra- Vertebrae, Bursa – Bursae

“ix” The plural becomes becomes “ices” eg. Appendix – Appendices

“is” The plural becomes becomes “es” eg. Diagnosis – Diagnoses

“nx” The plural becomes becomes “g” eg. Phalanx – Phalanges

“um” The plural becomes becomes “a” eg. Diverticulum – Diverticula

“us” The plural becomes becomes “i” eg. Alveolus –alveoli, uterus- uteri, malleolus -malleoli

OVERVIEW OF ANATOMY CONTD.

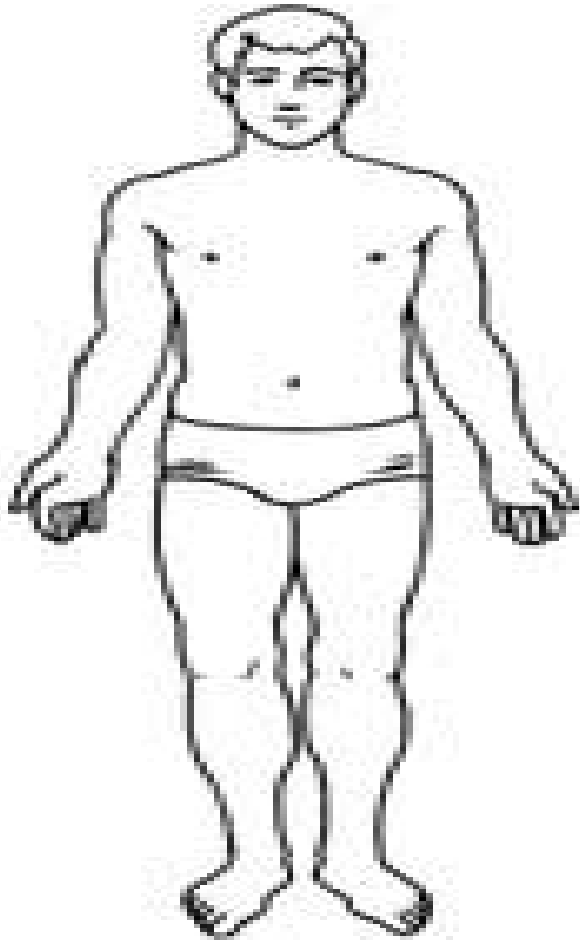
ANATOMICAL POSITION

A body position as if the person were standing upright with:

- **Head, eye and toe directed forward**
- **Arms adjacent to sides with palms facing forward**
- **Lower limbs close together with feet parallel and toes directed forward**

OVERVIEW OF ANATOMY CONTD.

ANATOMICAL POSITION



OVERVIEW OF ANATOMY CONTD.

ANATOMICAL PLANES

Median Plane

Vertical plane longitudinal through the body/body part dividing it into left and right halves

Sagittal Plane

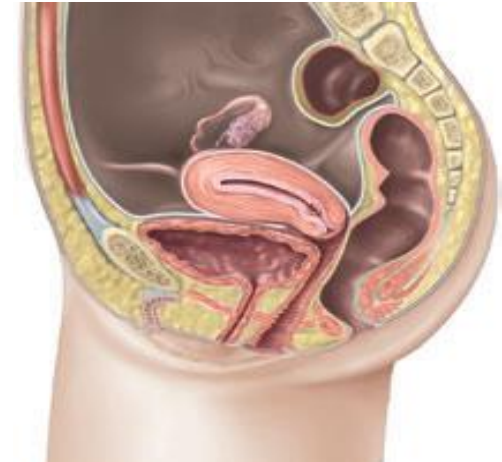
Vertical plane parallel to median plane

Coronal (Frontal) Plane

Vertical plane (90° to median plane) through the body/body part dividing it into front and back.

Transverse (Horizontal) Plane

Horizontal plane through the body/body part dividing it into upper and lower segments.



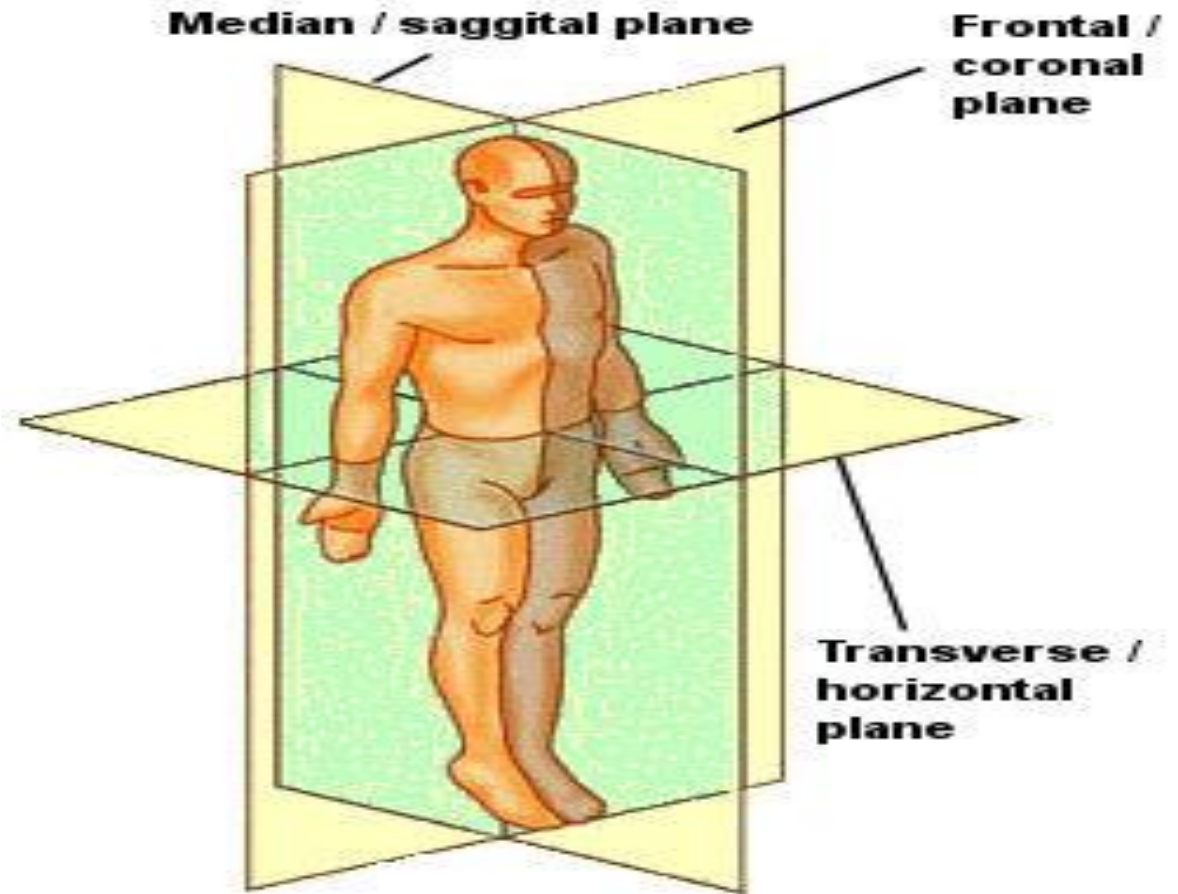
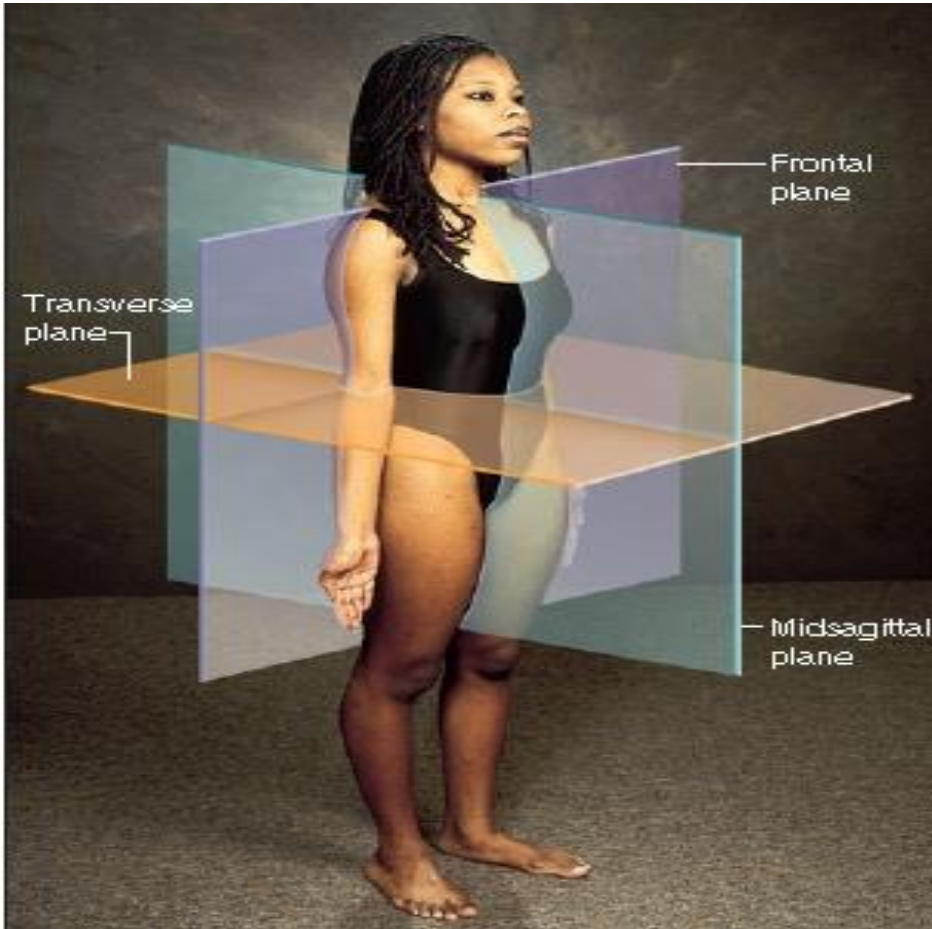
Sagittal Section



Coronal Section

OVERVIEW OF ANATOMY CONTD.

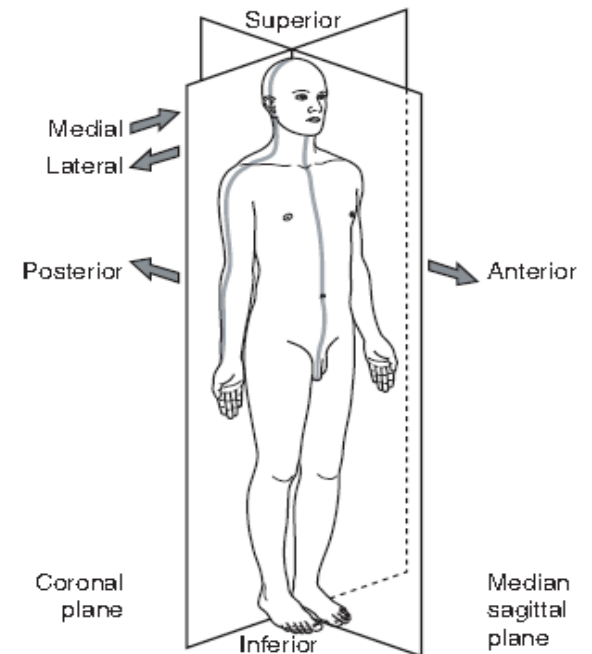
ANATOMICAL PLANES



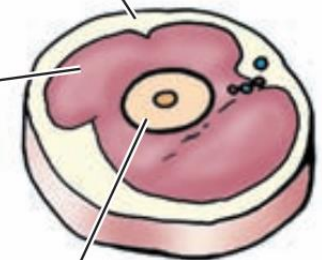
OVERVIEW OF ANATOMY CONTD.

TERMS OF RELATIONSHIP, COMPARISON AND LATERALITY

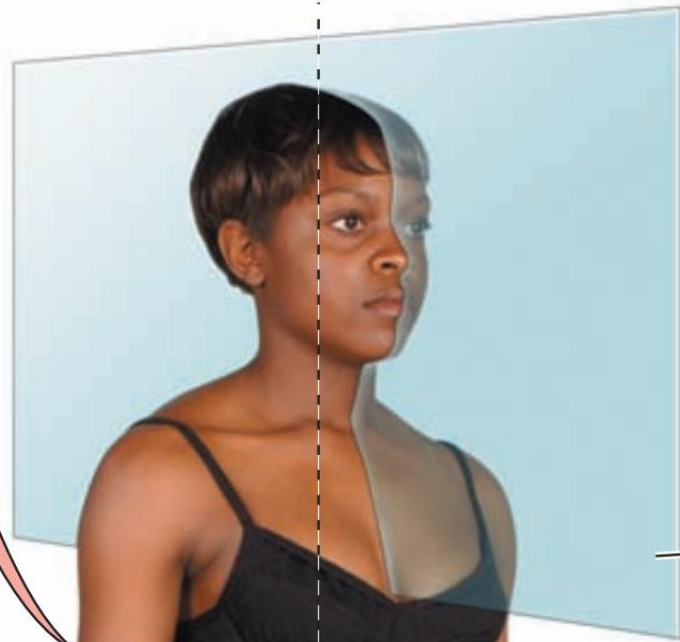
- **Superficial, Intermediate and Deep**
- **Proximal and Distal**
- **Medial and Lateral**
- **Posterior (Dorsal) and Anterior (Ventral)**
- **Inferior (Caudal) and Superior (Cranial)**
- **Bilateral and Unilateral**
- **Ipsilateral and Contralateral**



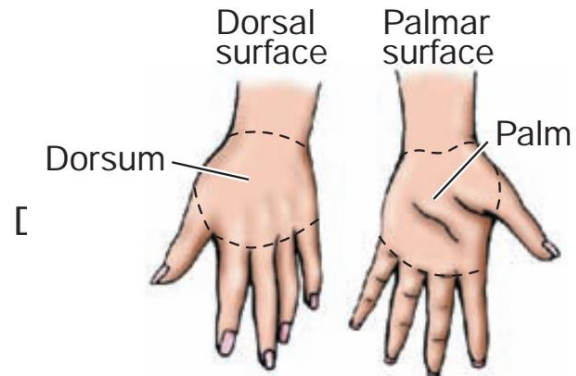
- * **Superficial**
 - Nearer to surface
 - The muscles of the arm are superficial to its bone (humerus).
- * **Intermediate**
 - Between a superficial and a deep structure
 - The biceps muscle is intermediate between the skin and the humerus.
- * **Deep**
 - Farther from surface
 - The humerus is deep to the arm muscles.



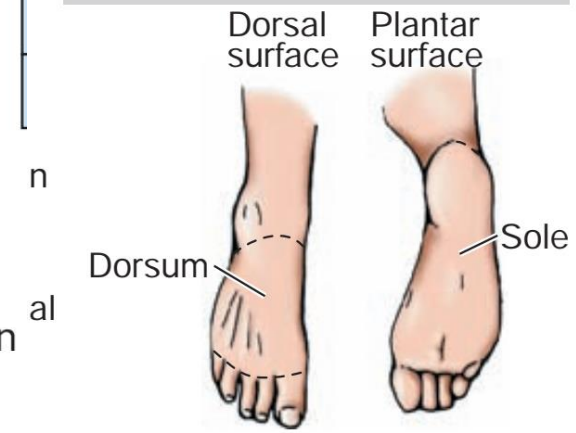
- Superior (cranial)**
 - Nearer to head
 - The heart is superior to the stomach.



- Palmar vs. Dorsal**
 - Anterior hand (palm)
 - Posterior hand (dorsum)



- Plantar vs. Dorsal**
 - Inferior foot surface (sole)
 - Superior foot surface (dorsum)



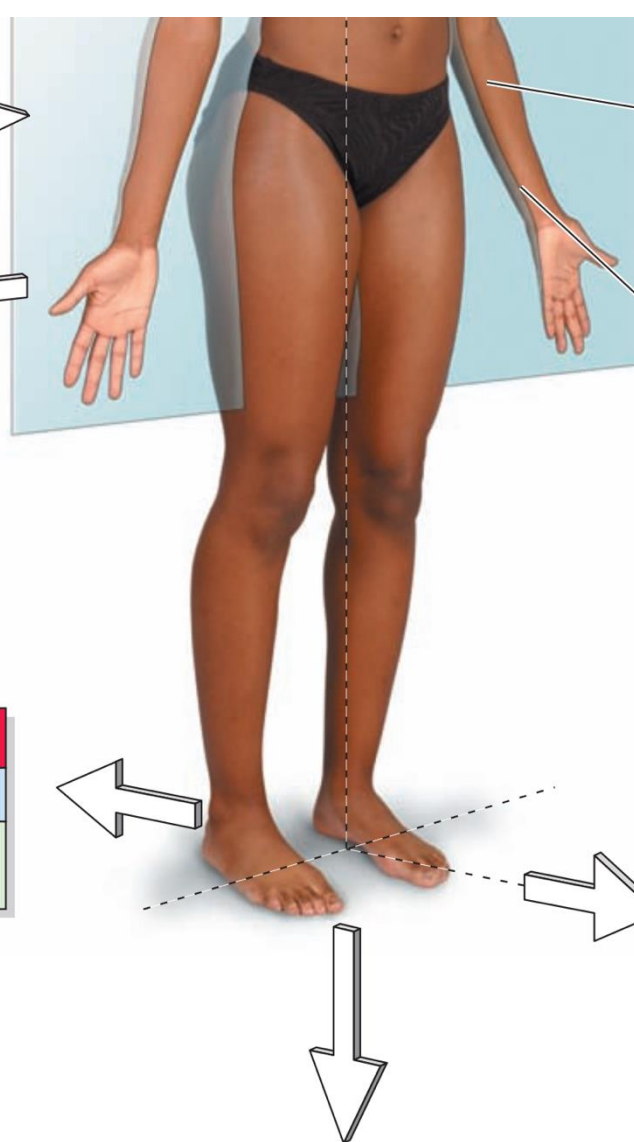
Medial
Nearer to median plane
The 5th digit (little finger) is on the medial side of the hand.

Lateral
Farther from median plane
The 1st digit (thumb) is on the lateral side of the hand.

Posterior (dorsal)
Nearer to back
The heel is posterior to the toes.

* Terms independent of anatomical position

Key
■ Terms applied to entire body
■ Terms specific for hands and feet



* Proximal
Nearer to trunk or point of origin (e.g., of a limb)
The elbow is proximal to the wrist, and the proximal part of an artery is its beginning.

* Distal
Farther from trunk or point of origin (e.g., of a limb)
The wrist is distal to the elbow, and the distal part of the upper limb is the hand.

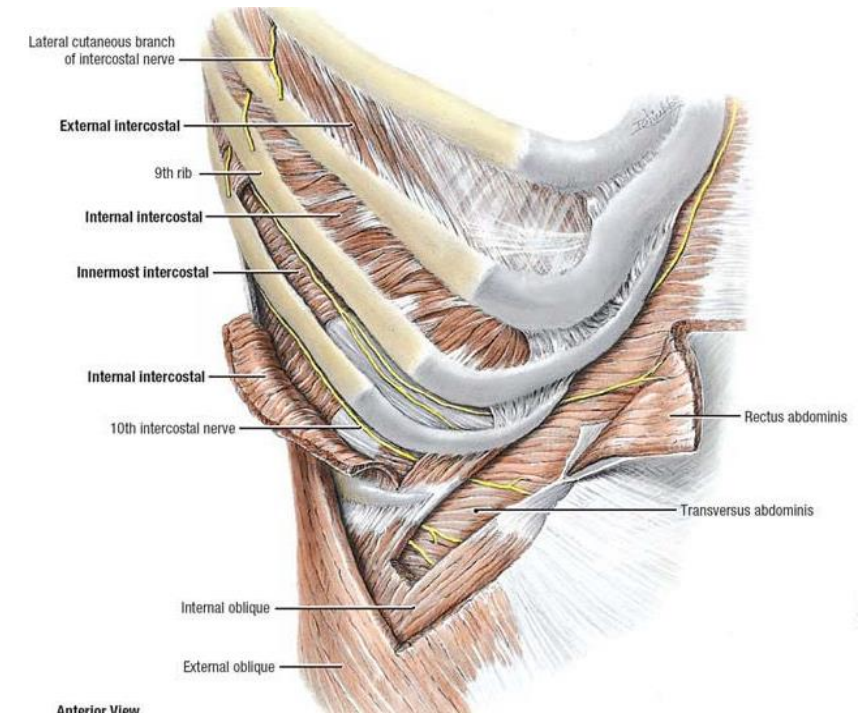
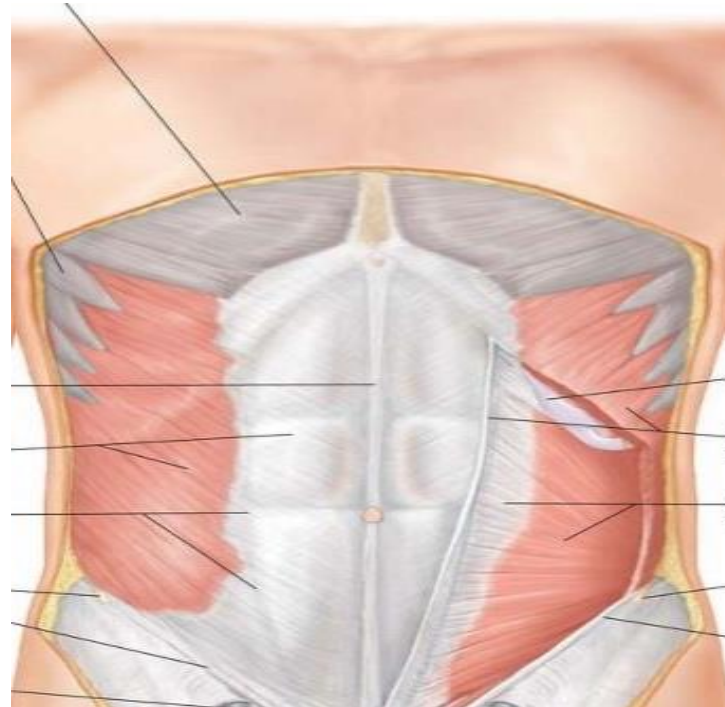
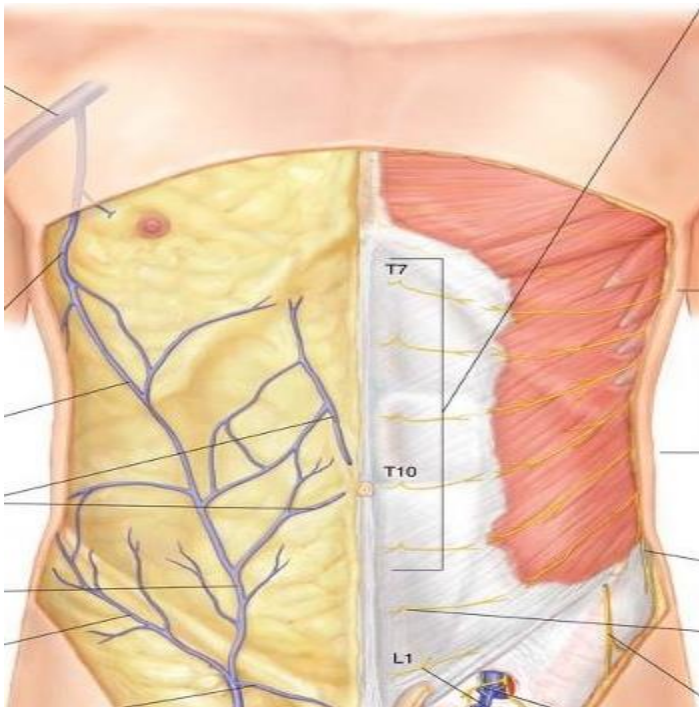
Anterior (ventral)
Nearer to front
The toes are anterior to the ankle.

Inferior (caudal)
Nearer to feet
The stomach is inferior to the heart.

OVERVIEW OF ANATOMY CONTD.

TERMS OF DIRECTION

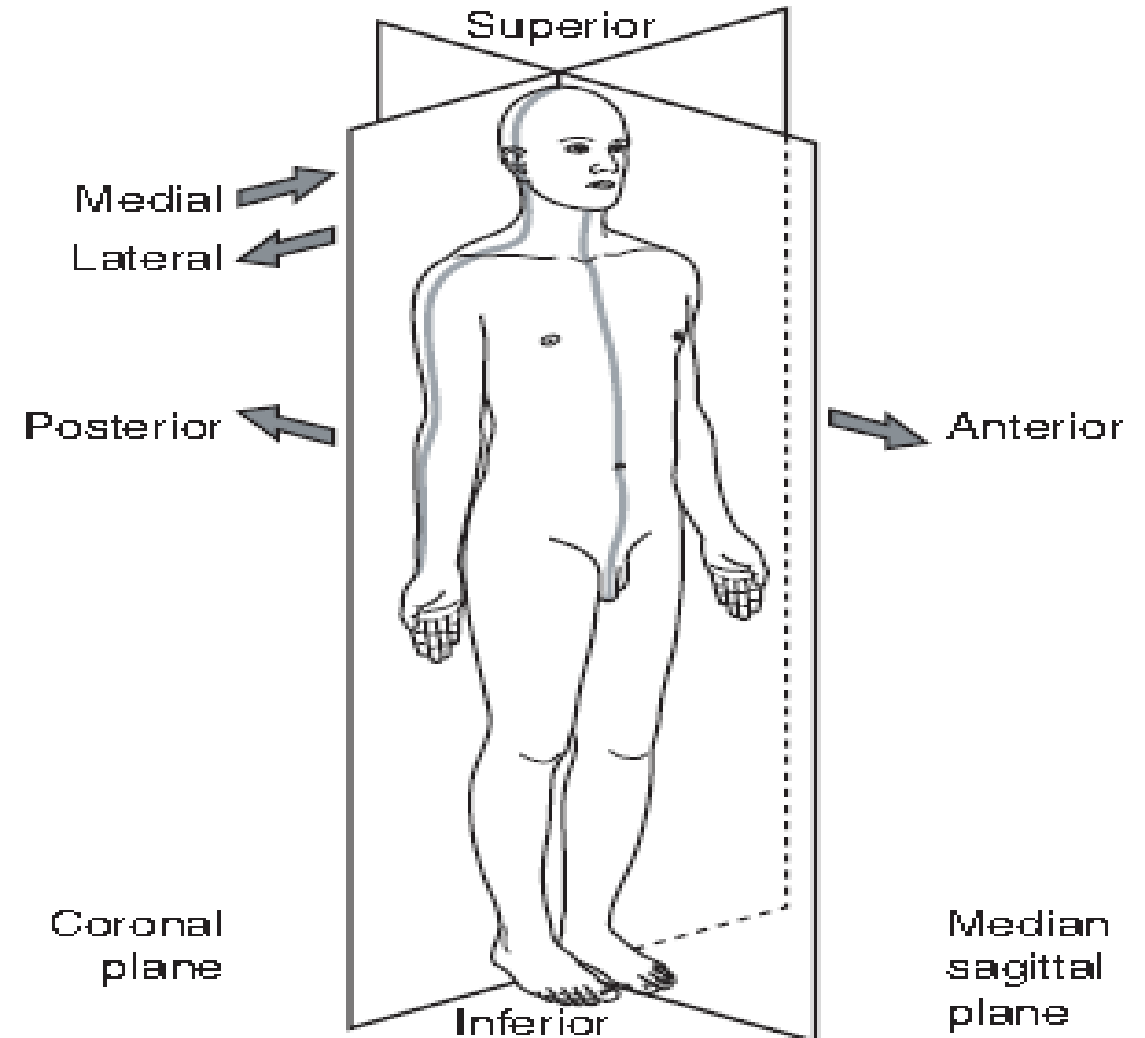
Superficial, Intermediate & Deep Structures



OVERVIEW OF ANATOMY CONTD.

TERMS OF DIRECTION

Superior & Inferior

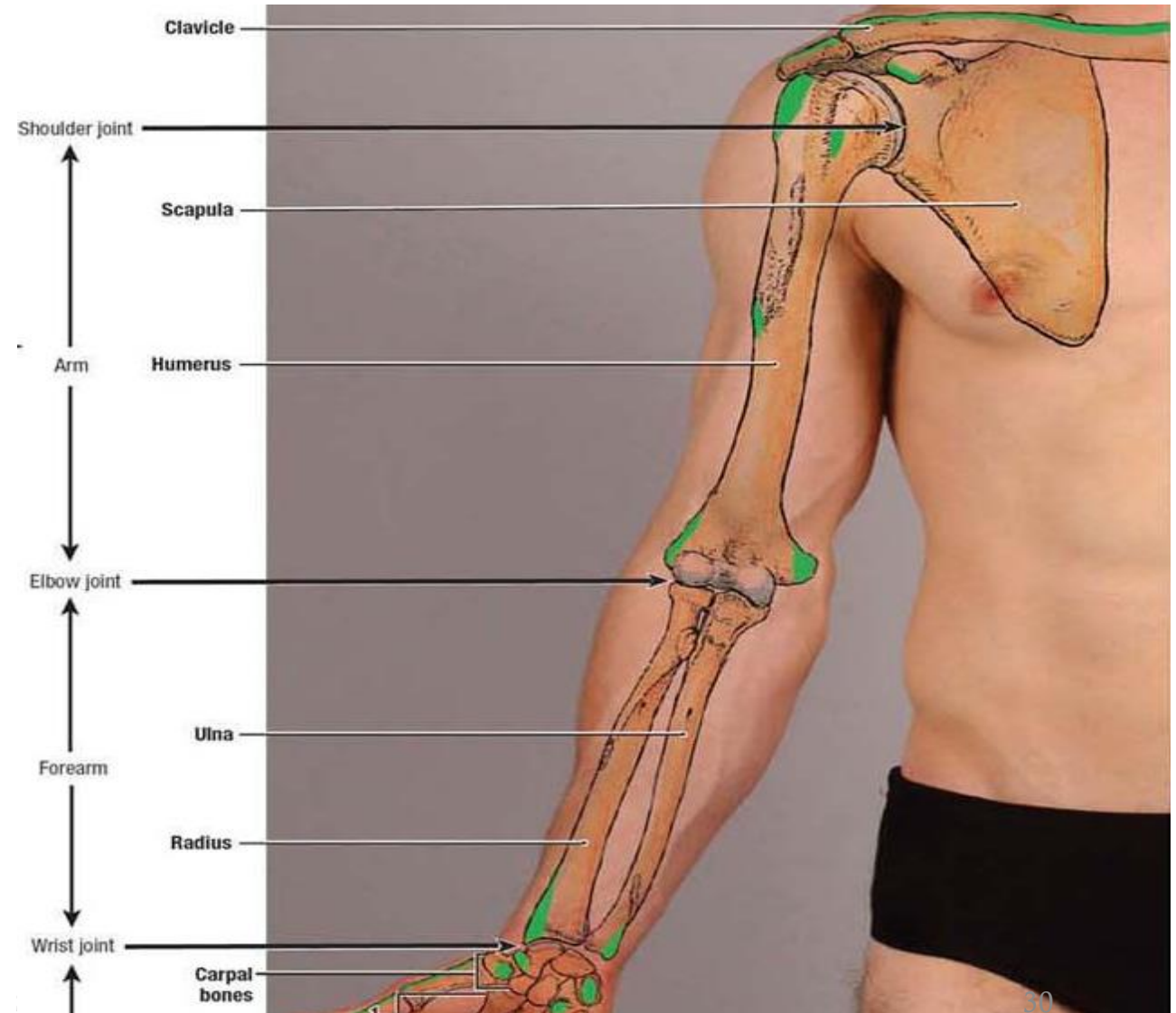


OVERVIEW OF ANATOMY CONTD.

TERMS OF DIRECTION

Proximal & Distal

Medial & Lateral



OVERVIEW OF ANATOMY CONTD.

TERMS OF DIRECTION

Anterior



5/2/2026

Posterior

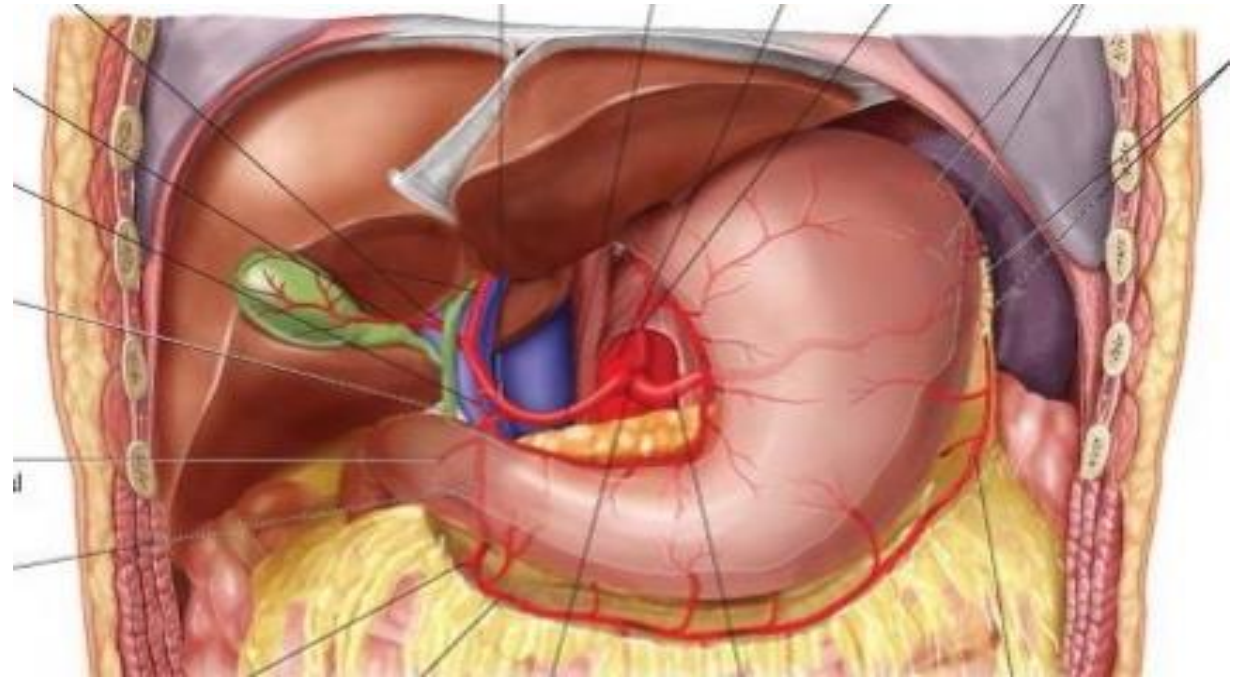
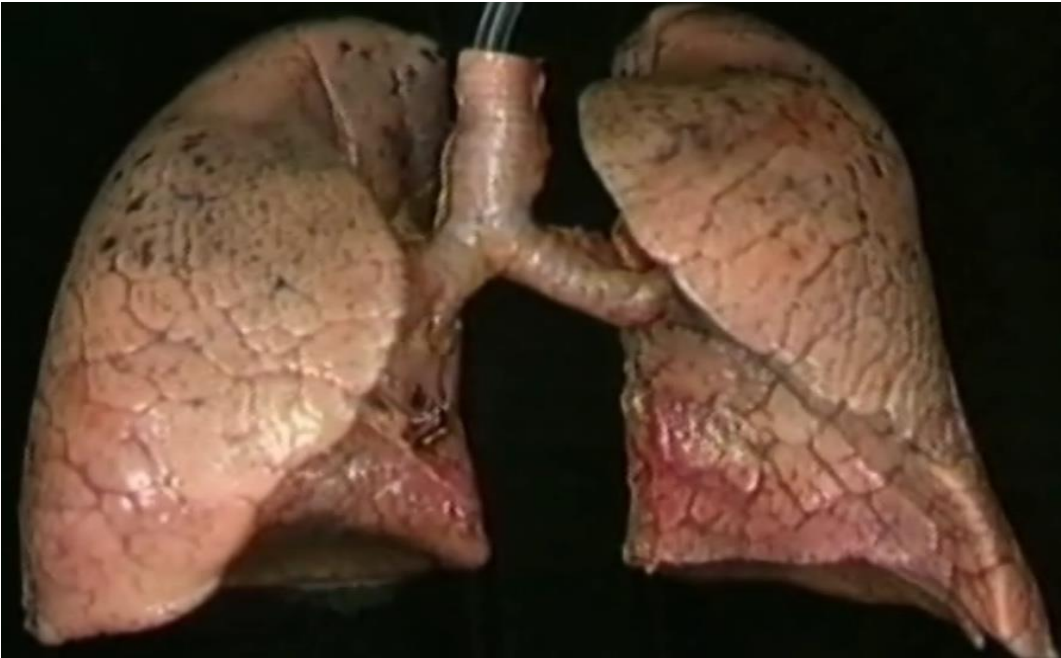


SMS 186

OVERVIEW OF ANATOMY CONTD.

TERMS OF LATERALITY

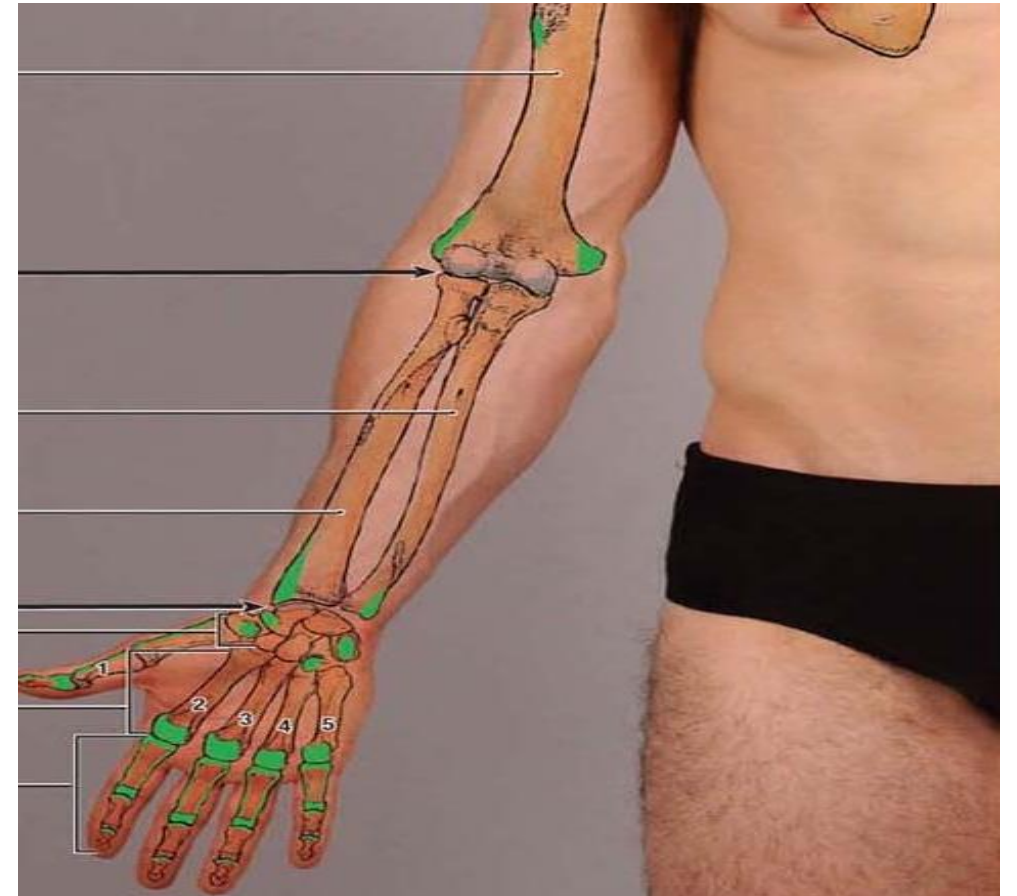
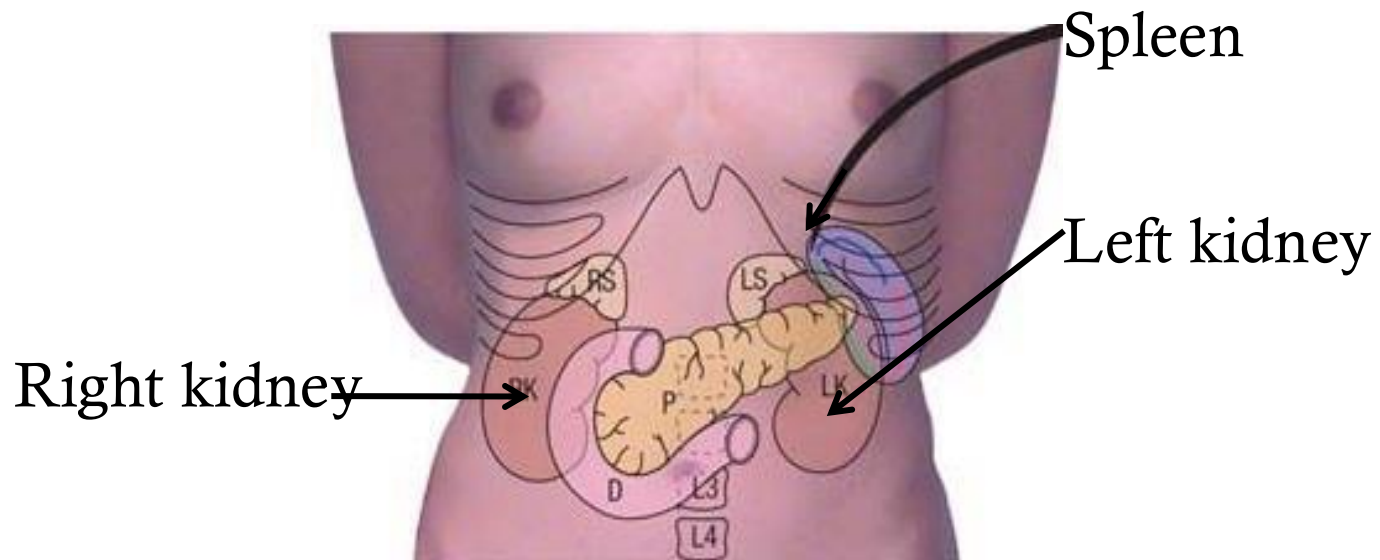
Bilateral & Unilateral



OVERVIEW OF ANATOMY CONTD.

TERMS OF LATERALITY

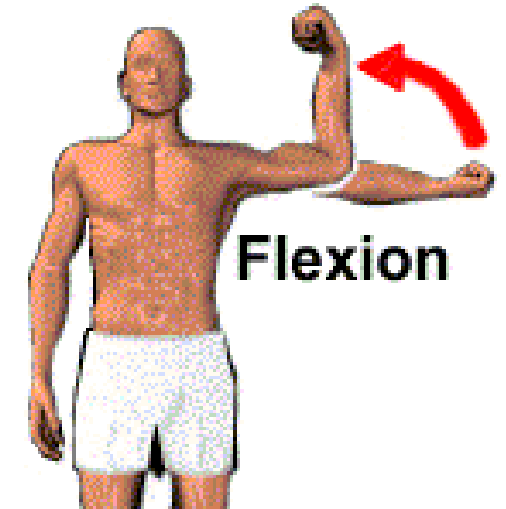
Ipsilateral & contralateral



Movements

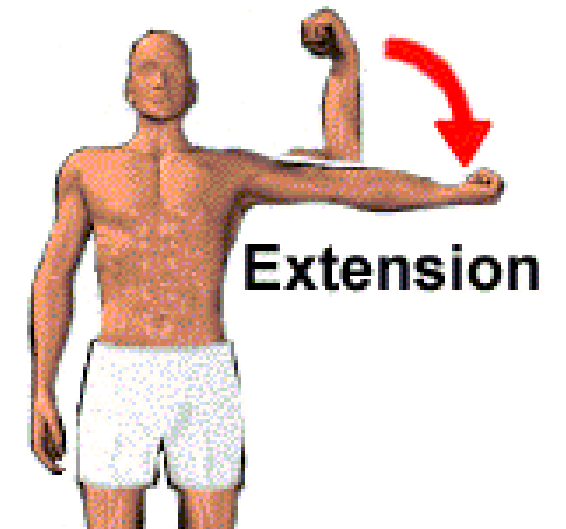
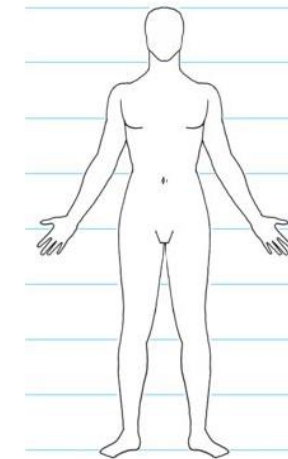
Flexion

- Bending a joint or decreasing the angle between two bones
 - In the Fetal Position we are flexing our joints



Extension

- Straightening a joint or increasing the angle between two bones
 - In the Anatomical Position we are extending our joints

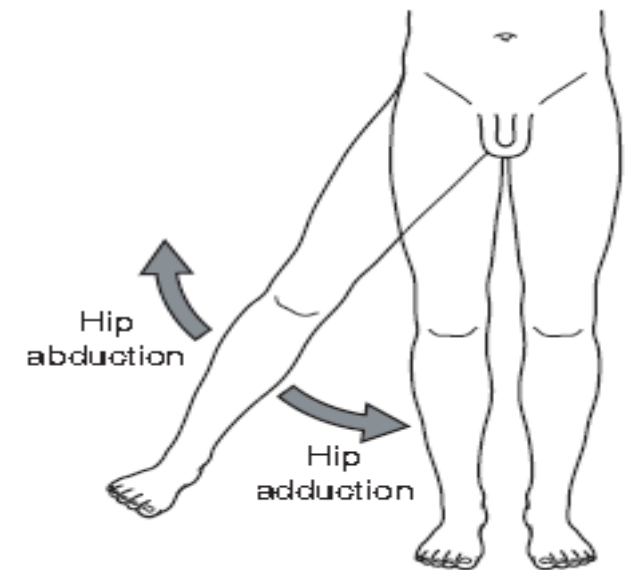
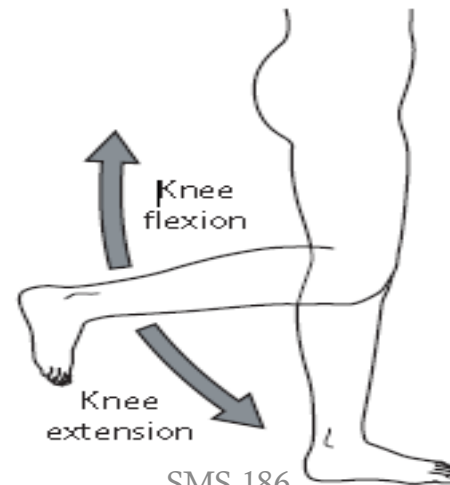
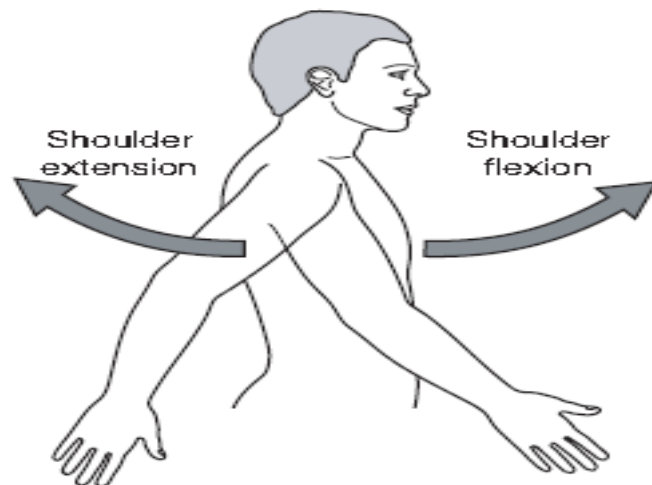
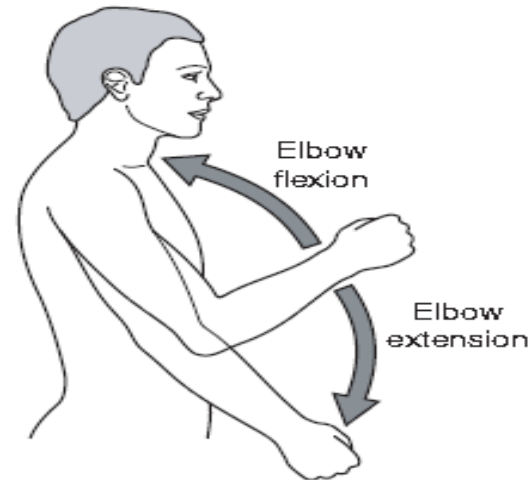
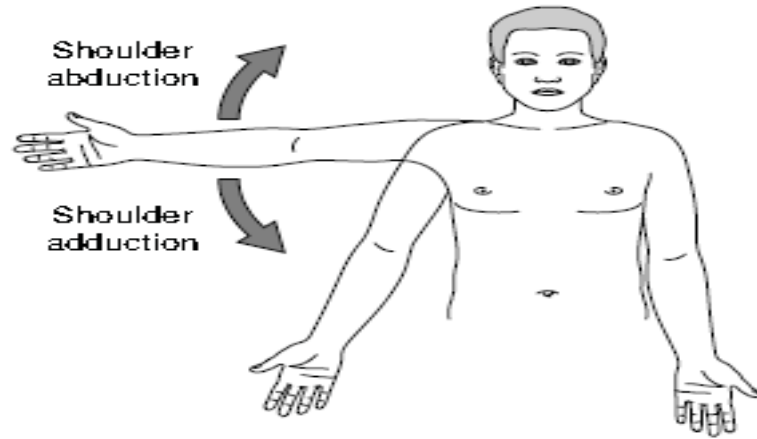


Hyperextension

- Excessive extension of the parts at a joint beyond anatomical position.

OVERVIEW OF ANATOMY CONTD.

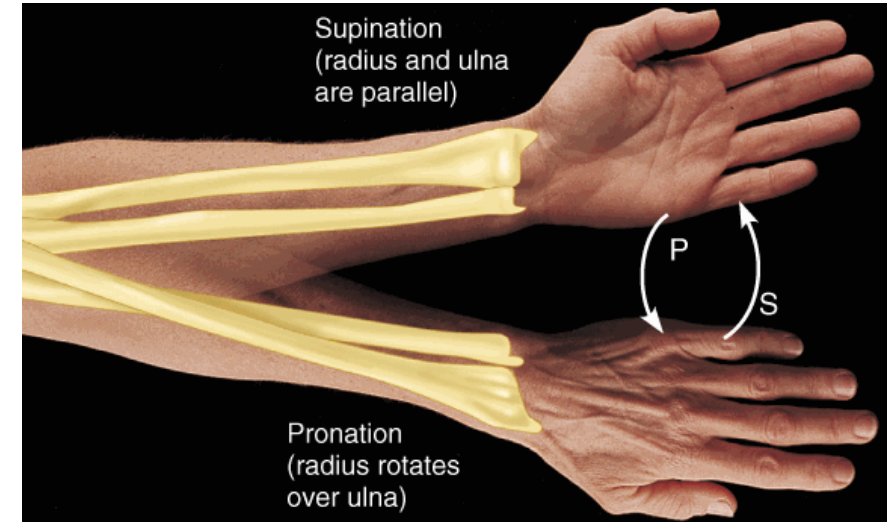
Flexion & Extension; Abduction & Adduction



Movements

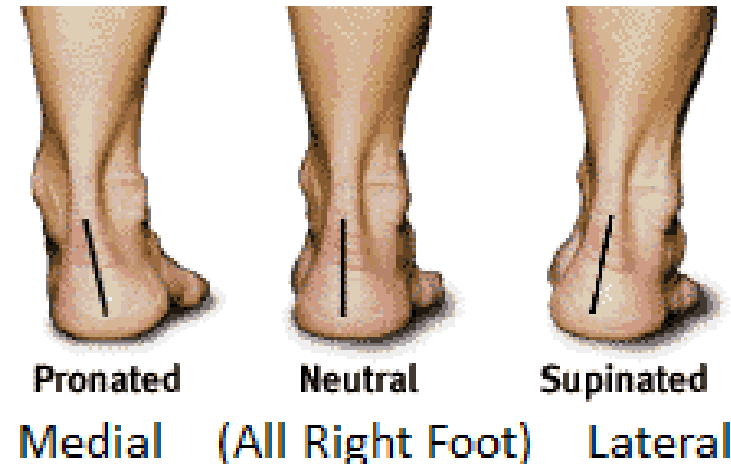
Pronation

- Turning the arm or foot downward
- (palm or sole of the foot - down)
 - Prone



Supination

- Turning the arm or foot upward
- (palm or sole of the foot - up)
 - Supine



Movements of the Foot

Inversion

- Turning the sole of the foot inward

Eversion

- Turning the sole of the foot outward

Dorsiflexion

- Ankle movement bringing the foot towards the shin

Plantarflexion

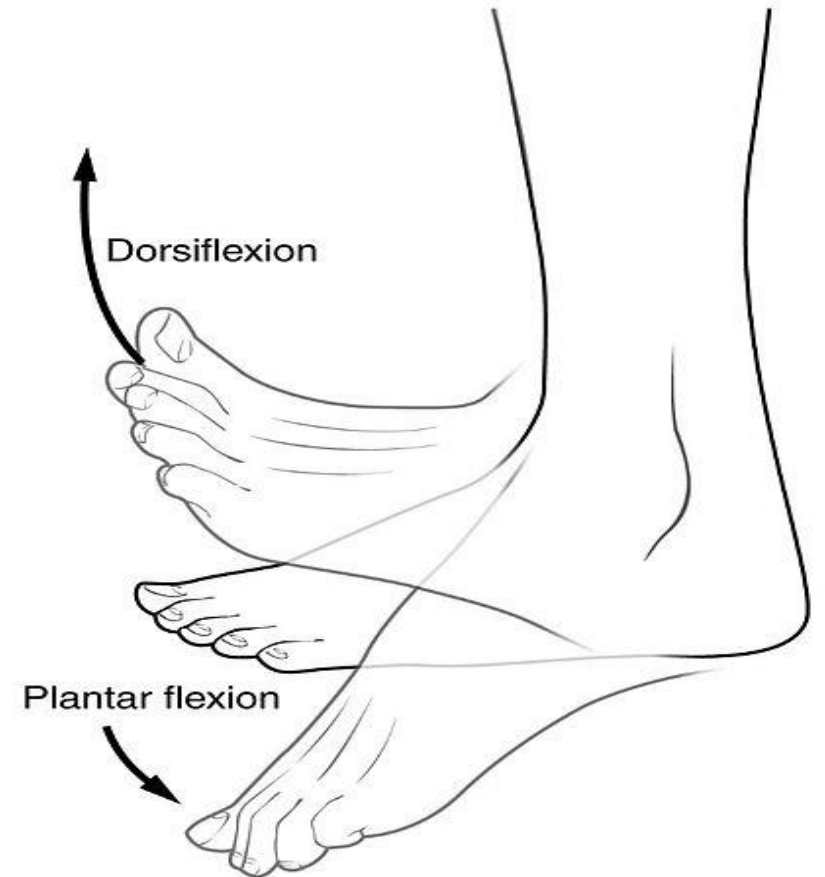
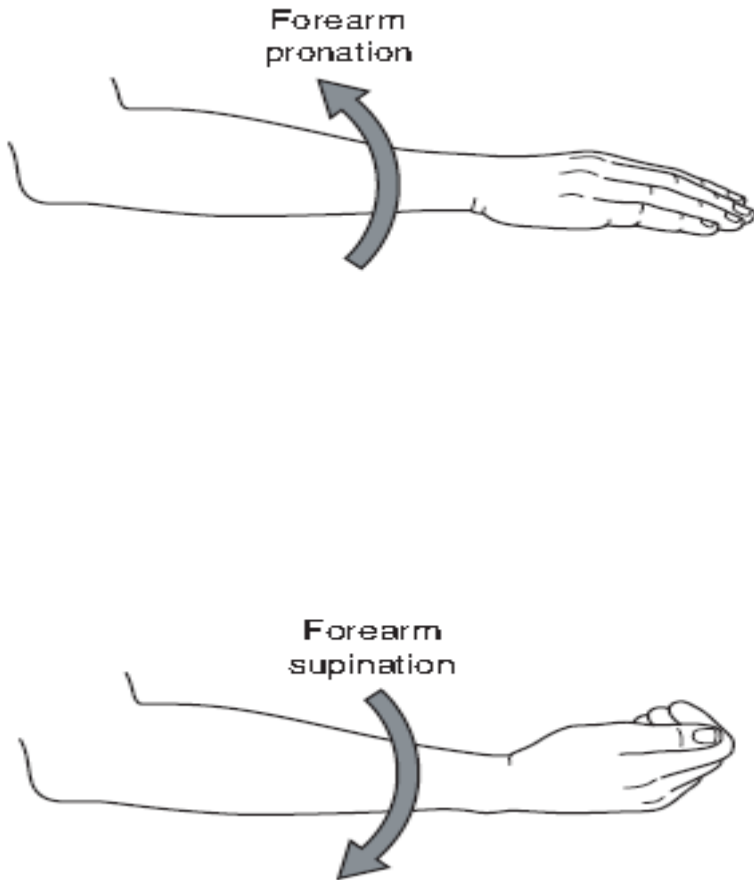
- Ankle movement pointing the foot downward



OVERVIEW OF ANATOMY CONTD.

Pronation & Supination

Dorsiflexion & plantarflexion



Movements

Retraction

- Moving a part backward

Protraction

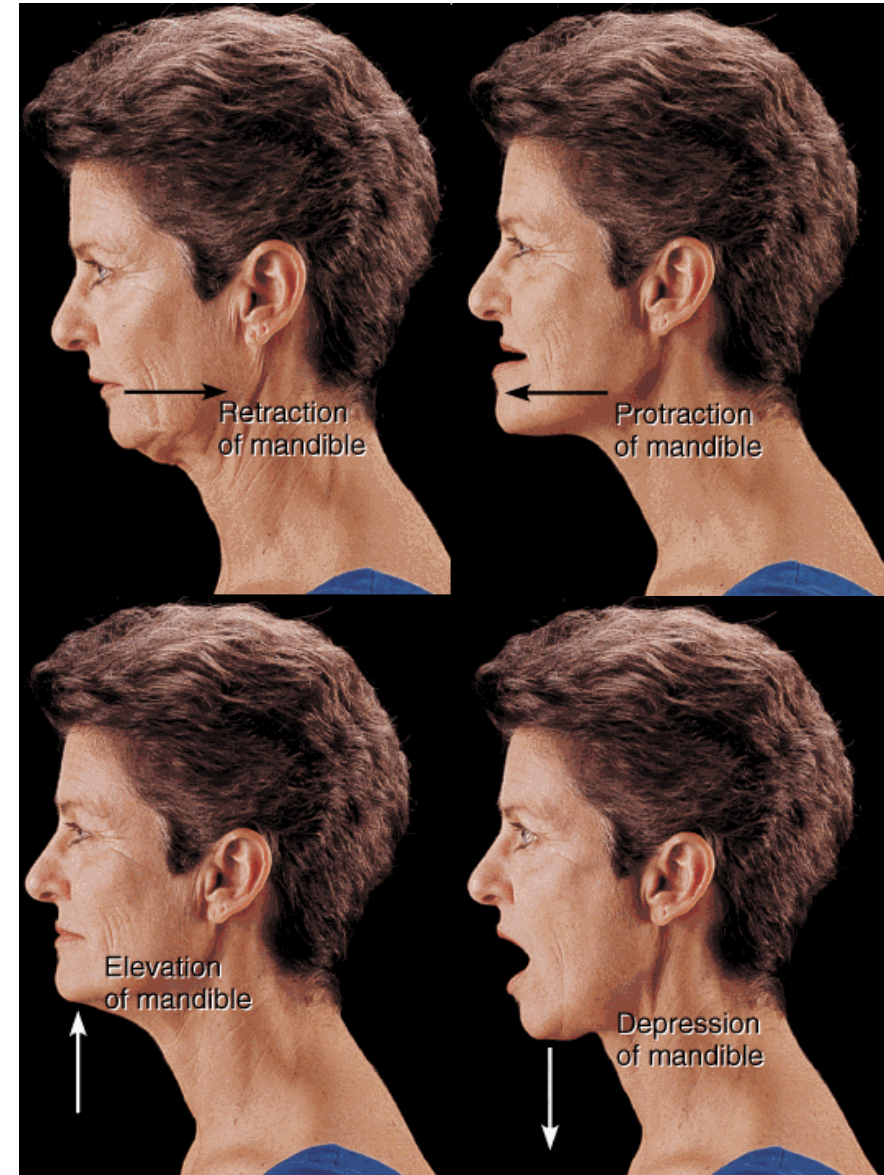
- Moving a part forward

Elevation

- Raising a part

Depression

- Lowering a part



Movements

Rotation

- Turning on a single axis

Circumduction

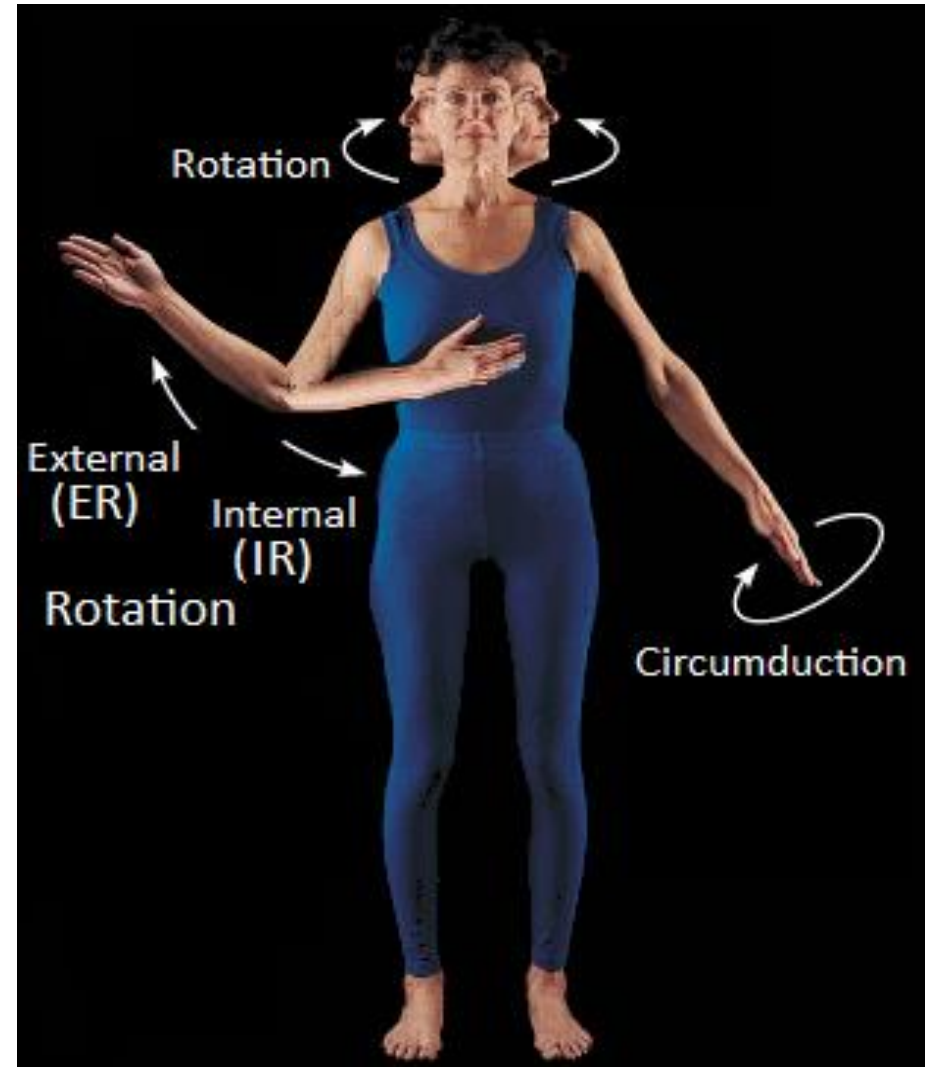
- Tri-planar, circular motion at the hip or shoulder

Internal rotation

- Rotation of the hip or shoulder toward the midline

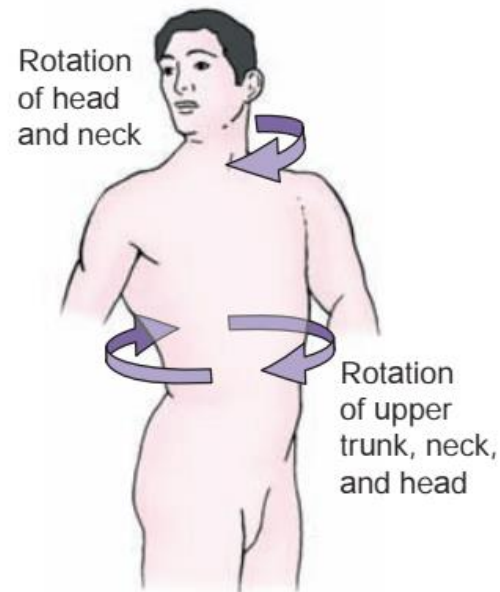
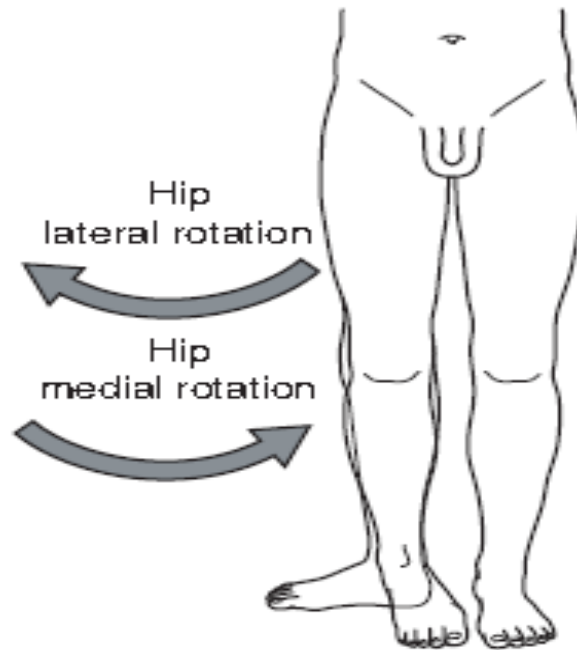
External rotation

- Rotation of the hip or shoulder away from the midline

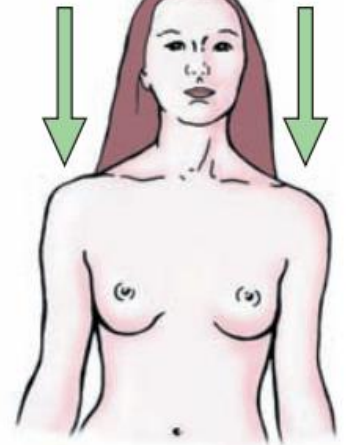


OVERVIEW OF ANATOMY CONTD.

Circumduction & Rotation; Elevation & Depression



Elevation



Depression

Movements of the Wrist & Thumb

Radial Deviation

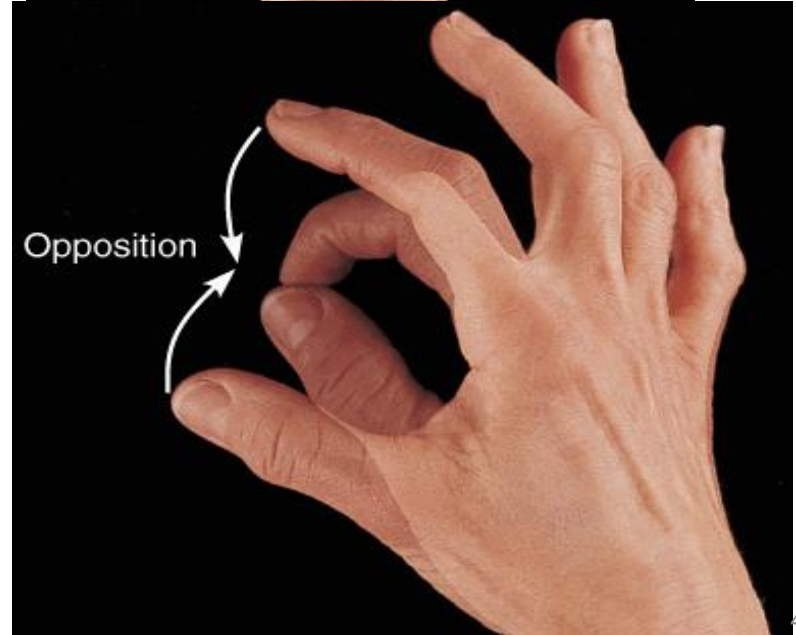
- Movement of the wrist towards the radius or lateral side.

Ulnar Deviation

- Movement of the wrist towards the ulna or medial side.

Opposition

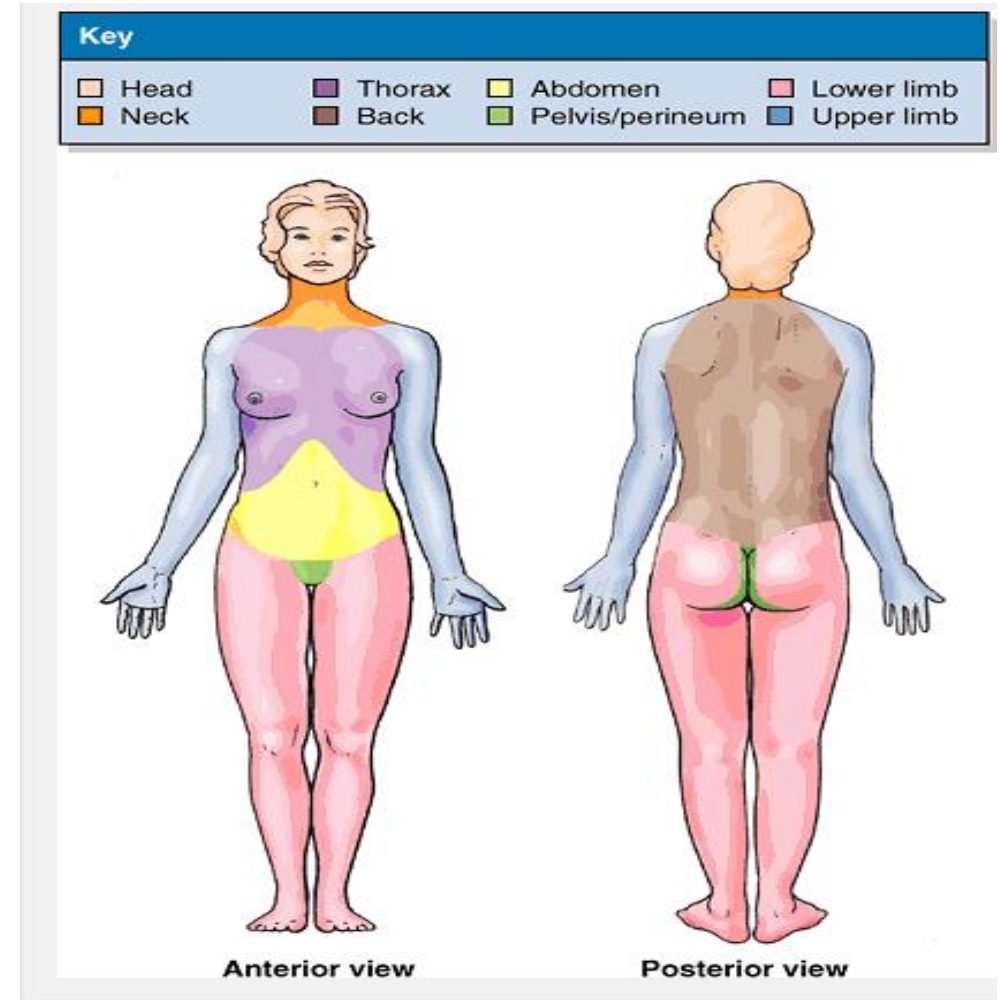
- Movement of the thumb across the palm of the hand.



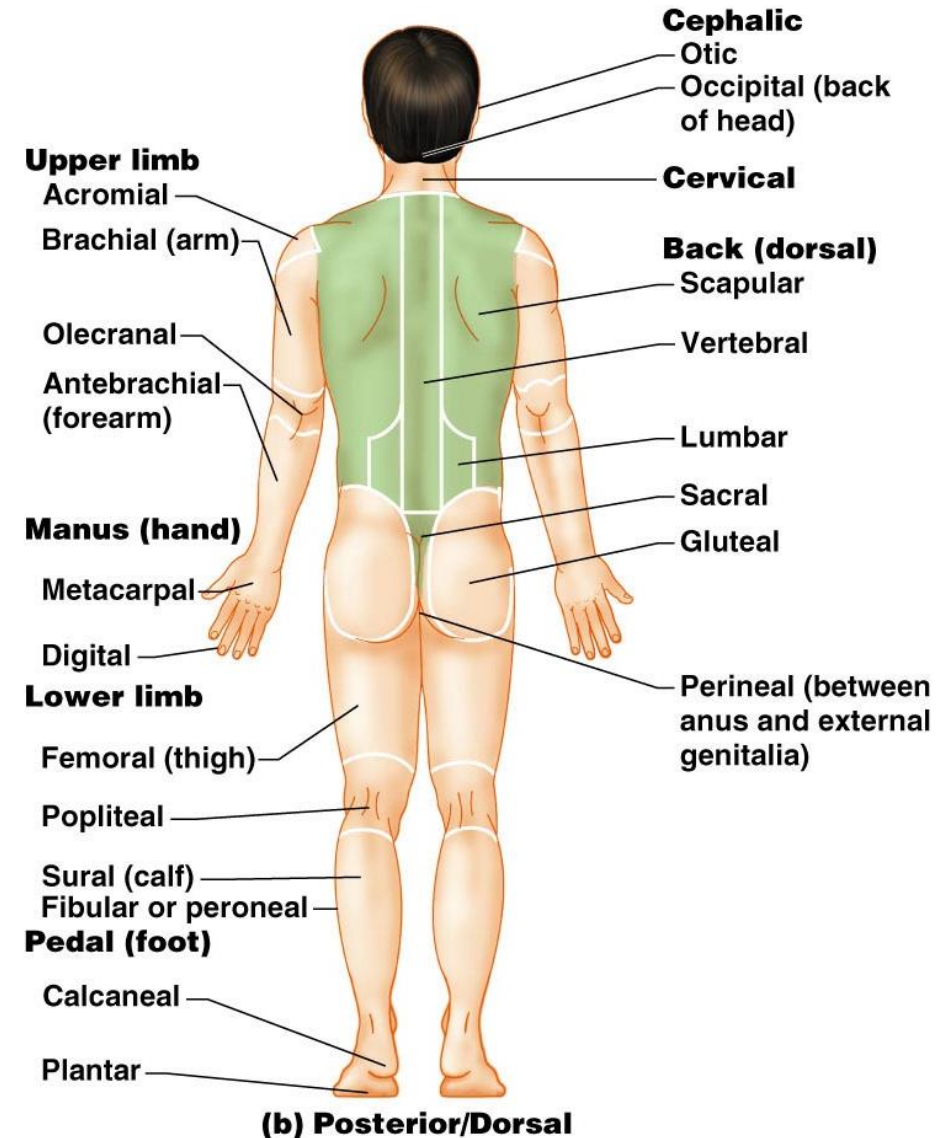
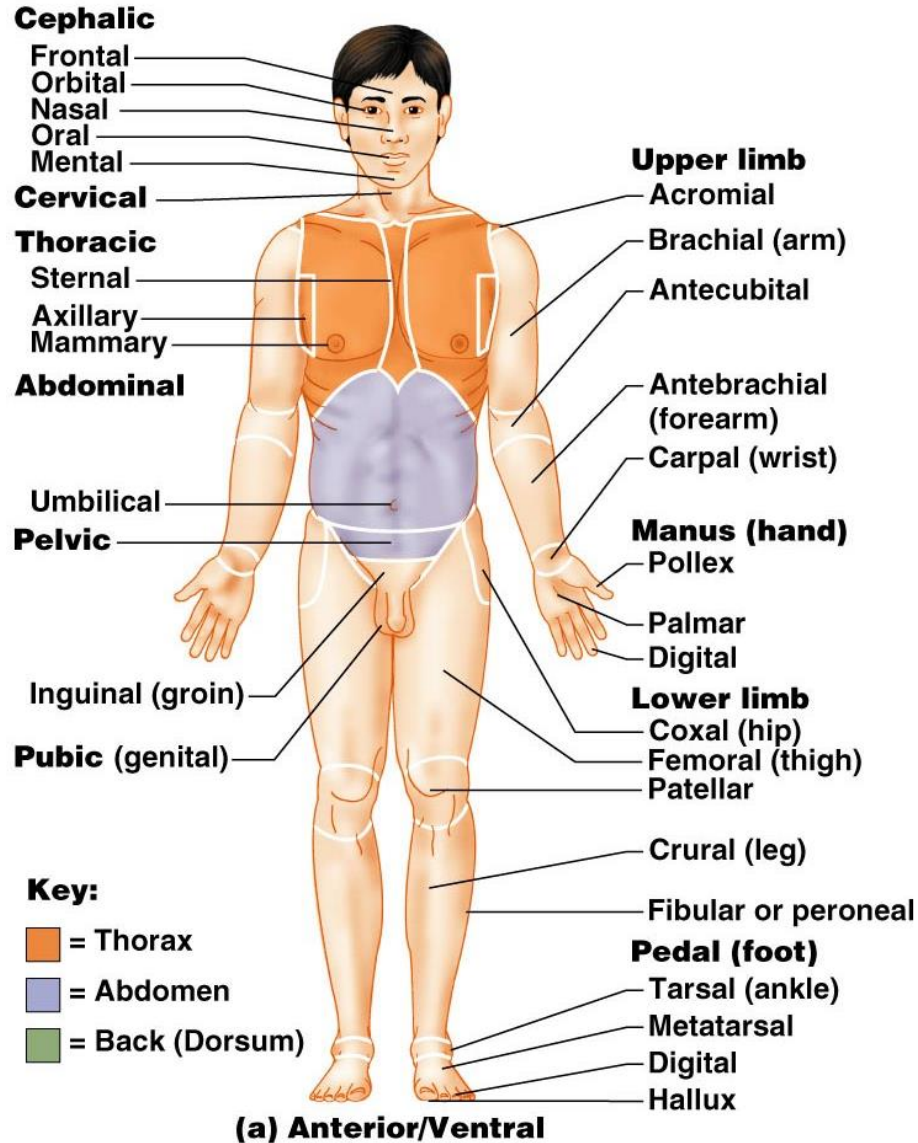
BODY REGIONS

- Body Regions

- ✓ Head
- ✓ Neck
- ✓ Thorax
- ✓ Abdomen
- ✓ Pelvis/perineum
- ✓ Lower limb
- ✓ Upper limb
- ✓ Back



Regional Terms



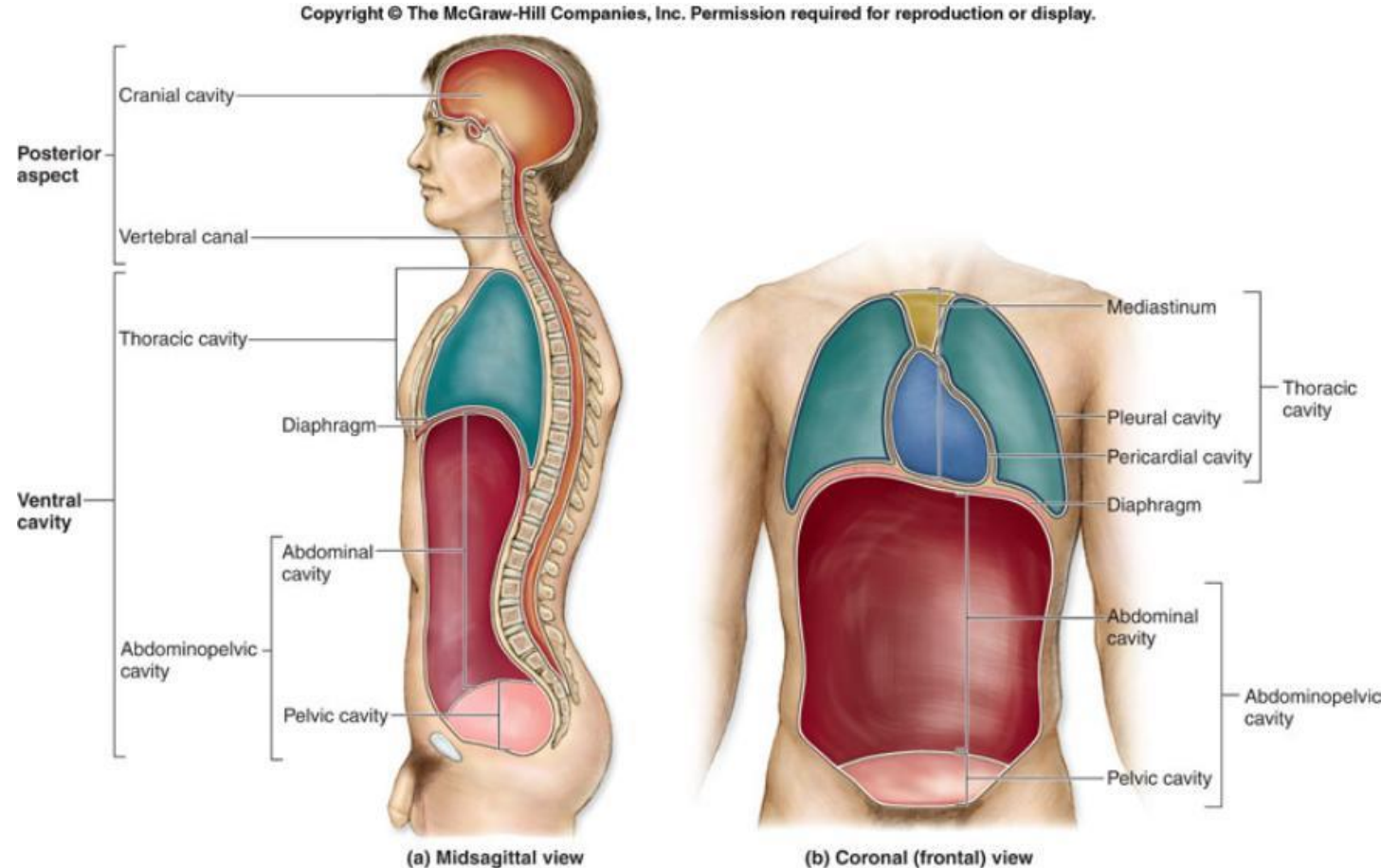
BODY CAVITIES

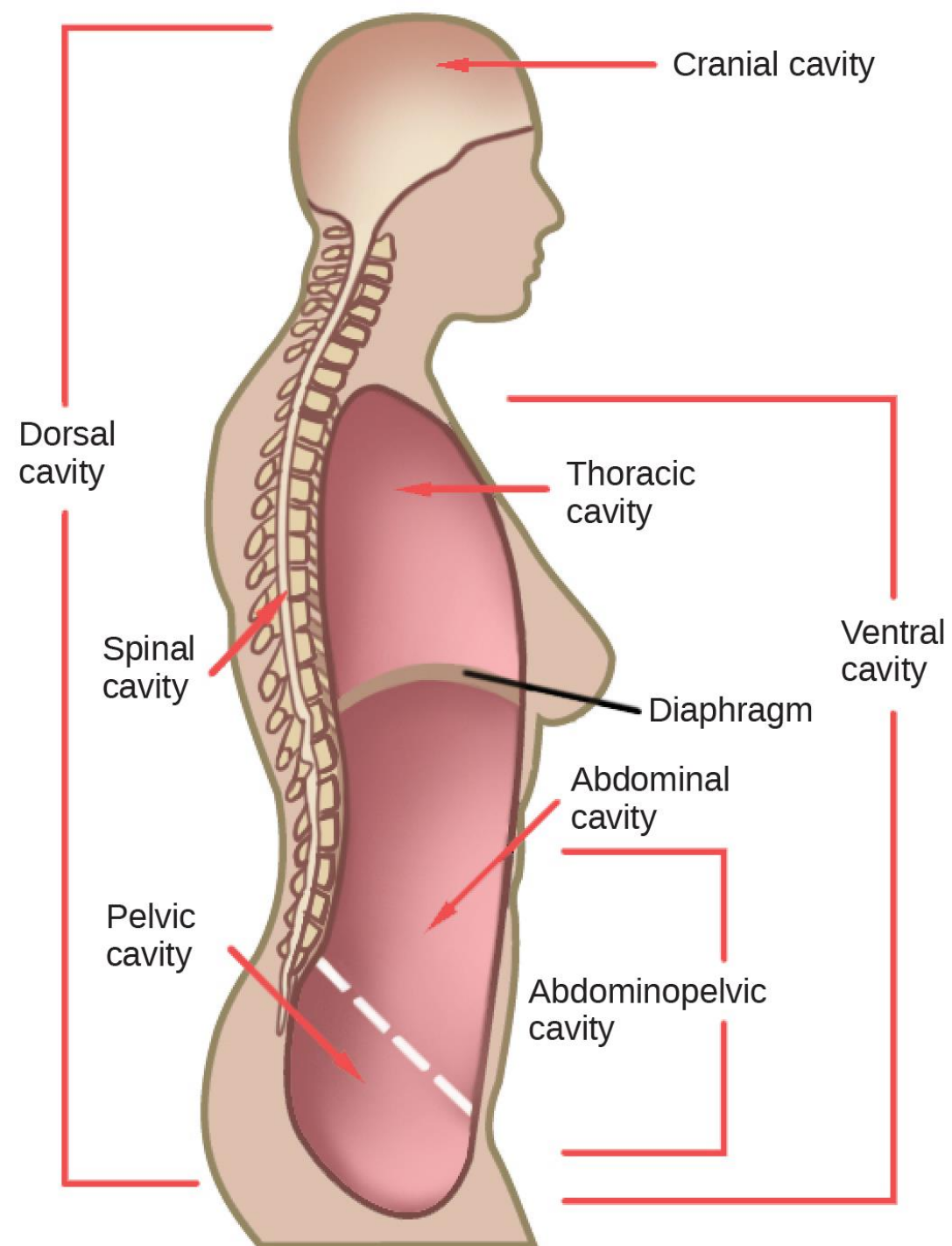
Dorsal body cavity

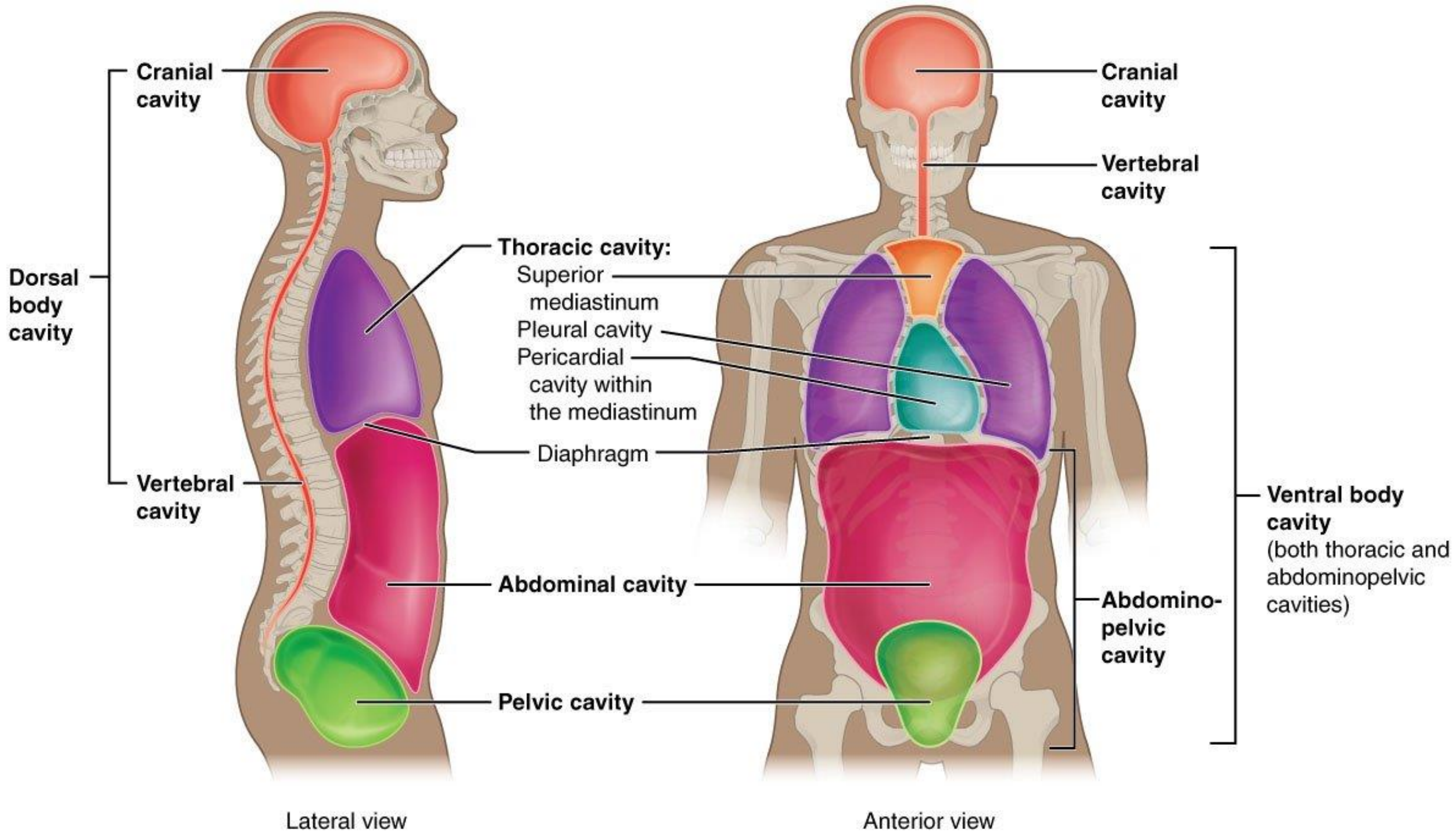
- cranial cavity
- Spinal/vertebral cavity

Ventral body cavity

- Pericardial cavity
- Pleural cavity
- Abdominopelvic cavity (abdominal + pelvic cavities)







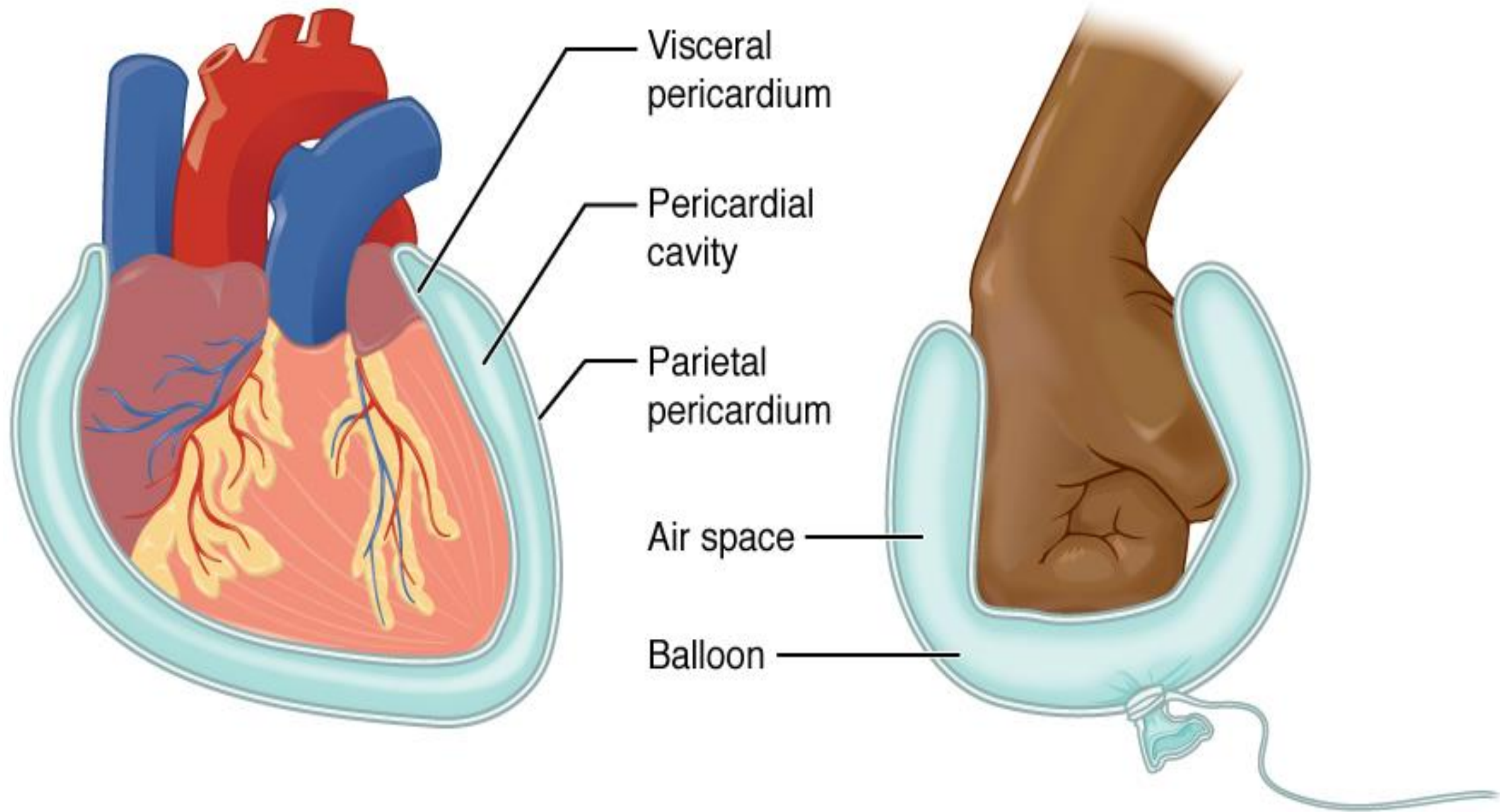
BODY CAVITIES

Dorsal body cavity-lined three membrane layers called meninges

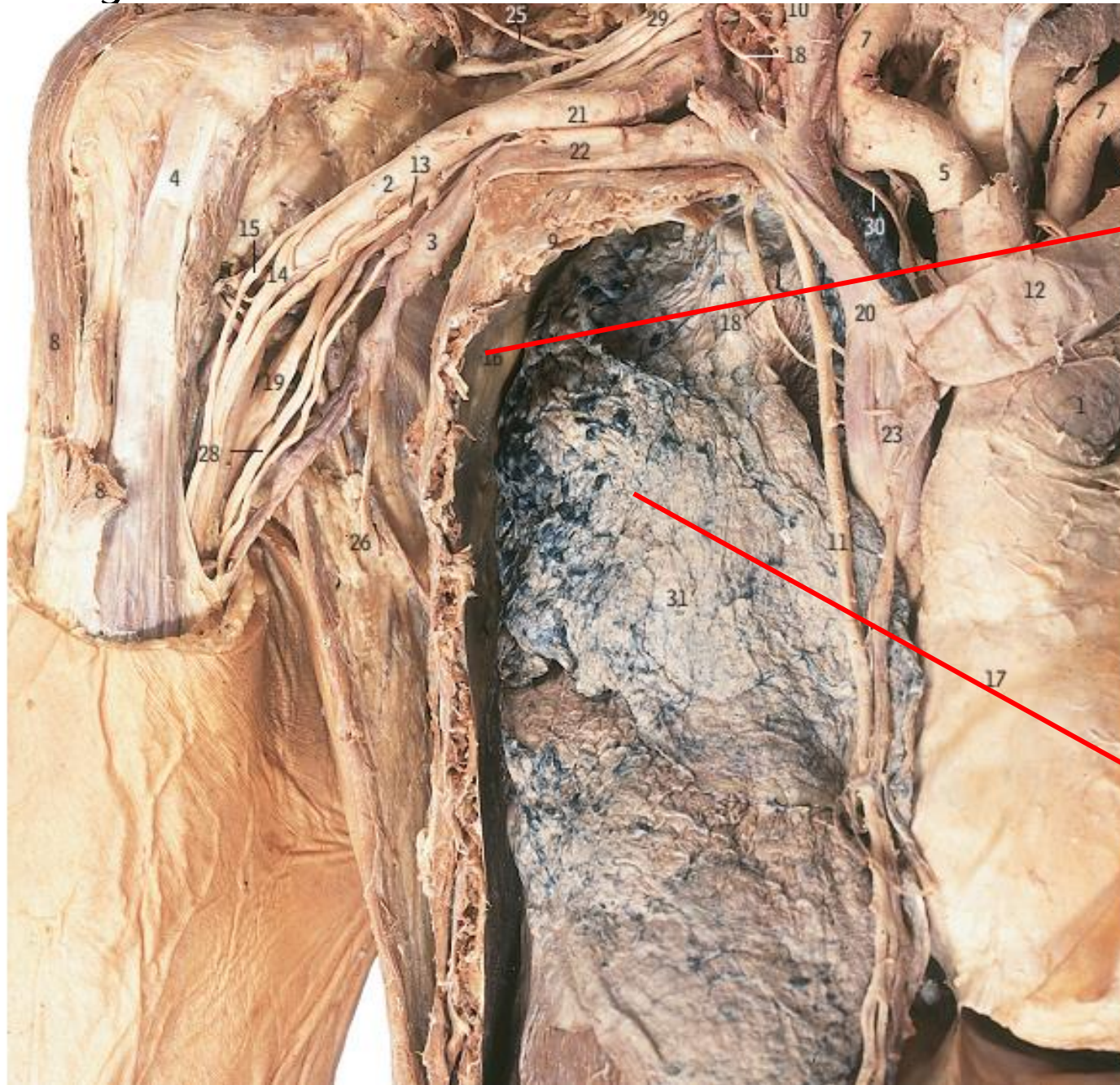
- **cranial cavity-brain**
- **Spinal/vertebral cavity-spinal cord**

Ventral body cavity-lined by serous membrane

- **Pericardial cavity-lined by pericardium; contains the heart**
- **Pleural cavity-lined by pleurae; contains the lungs**
- **Abdominopelvic cavity (abdominal + pelvic cavities)-lined by peritoneum; contains digestive organs, spleen, kidneys; rectum, bladder and reproductive organs**



Layers of Pleura



Parietal pleura

Visceral pleura

OVERVIEW OF BODY SYSTEMS

- **Integumentary system: Skin, nails, hairs and cutaneous glands**
- **Protection, containment for body structures, water retention, vitamin D synthesis, thermoregulation, cutaneous sensation**

OVERVIEW OF BODY SYSTEMS

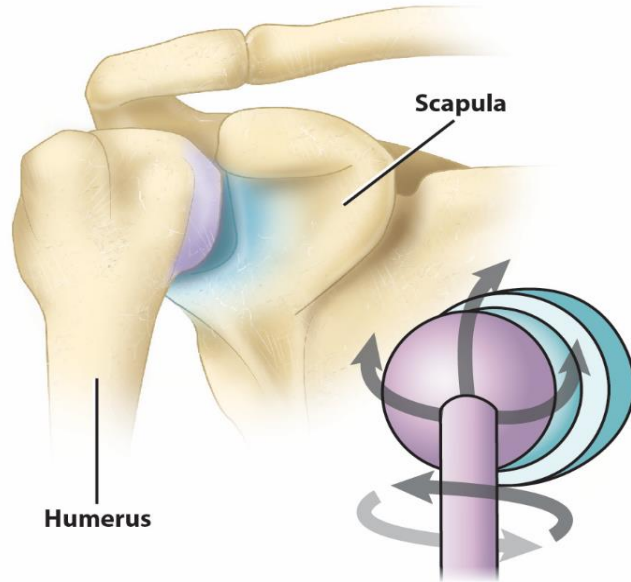
- **Skeletal system: Bones and cartilages**
- **Support, movement, protection of viscera, storage of minerals, haematopoiesis, electrolyte and pH balance**

OVERVIEW OF BODY SYSTEMS

- **Articular System:** joints and their associated ligaments
- **Connecting bony parts, providing sites at which movements occur**

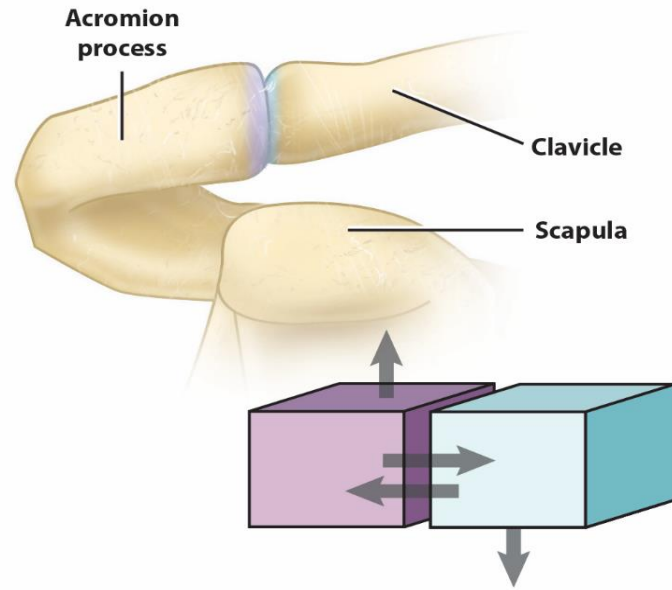
Ball and Socket

Glenohumeral joint



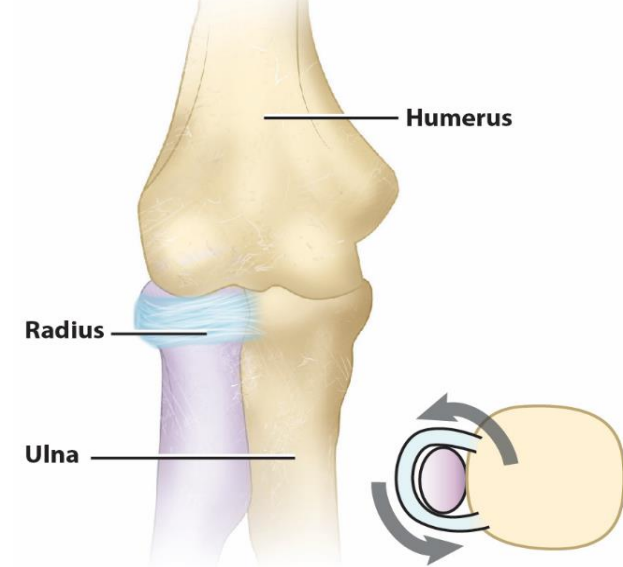
Plane

Acromioclavicular joint



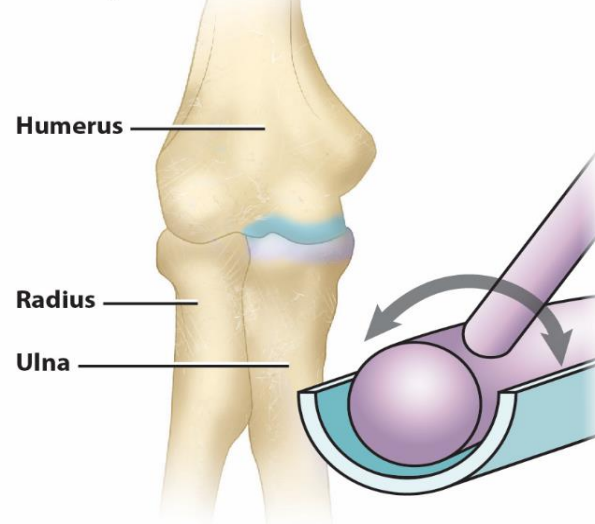
Pivot

Elbow joint



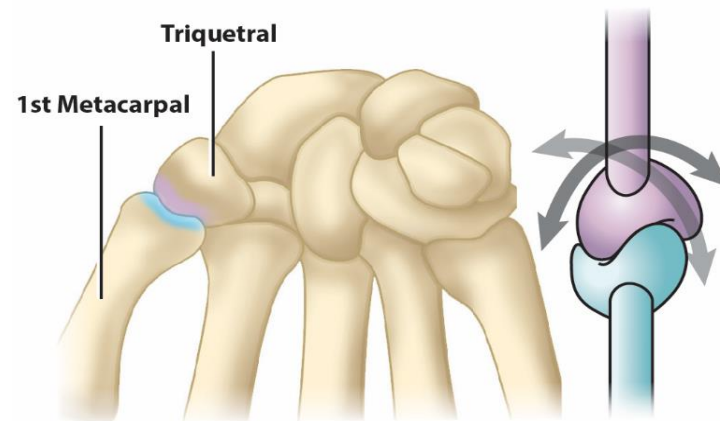
Hinge

Elbow joint



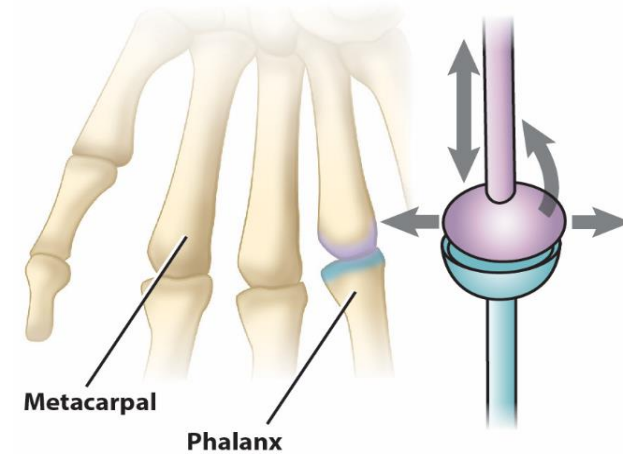
Saddle

Carpometacarpal joint



Condylar

Metacarpophalangeal joint

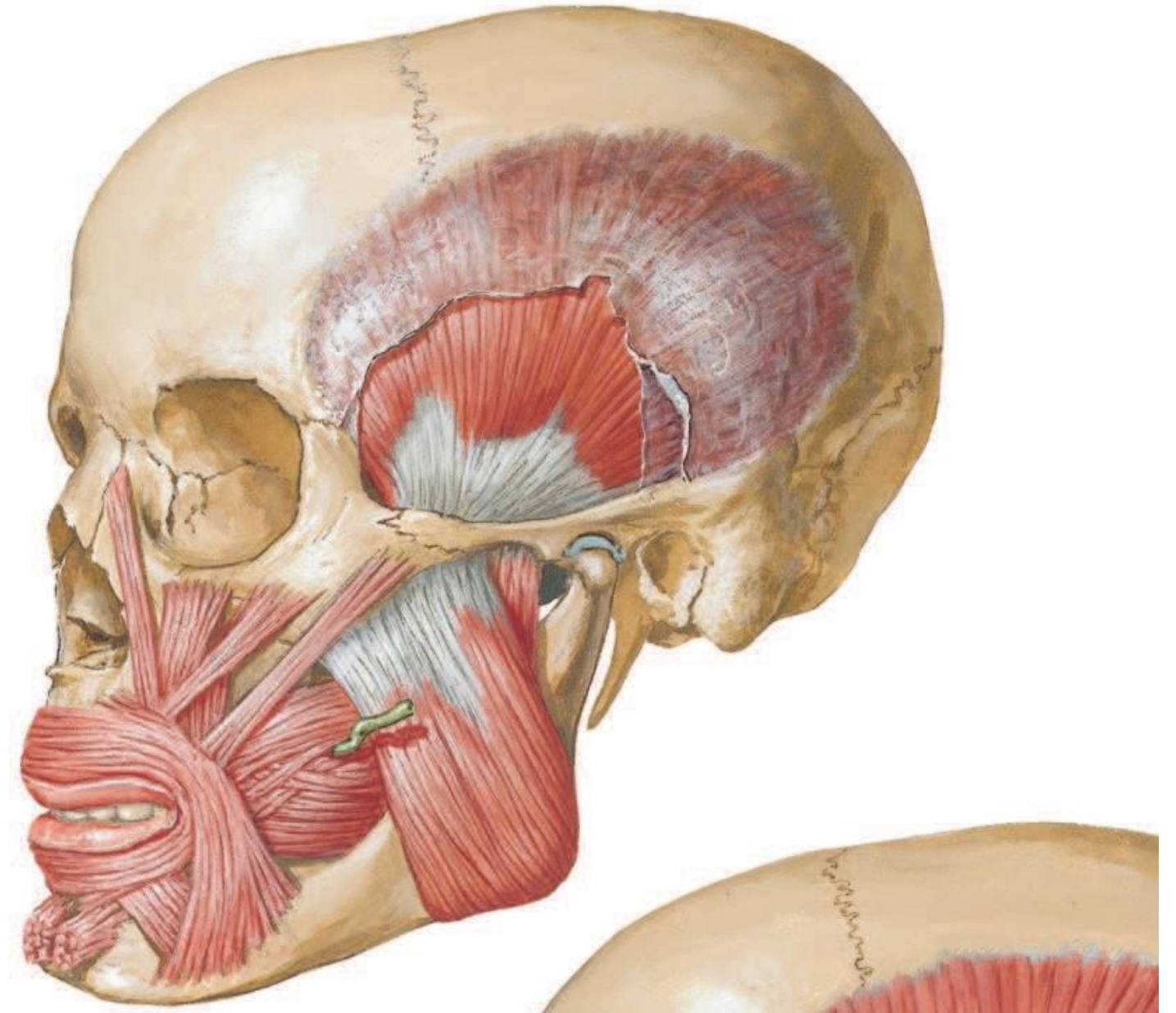


Types of Muscles

- **Agonist/Prime movers:** responsible for producing a particular movement
- **Antagonist:** Muscles that oppose or reverse a movement
- **Synergist:** aid the action of agonists either by assisting with the same movement or by reducing undesirable or unnecessary movement
- **Fixators:** are specialized synergists. They immobilize the origin of a prime mover so that all the tension is exerted at the insertion. Aids in the maintenance of posture.

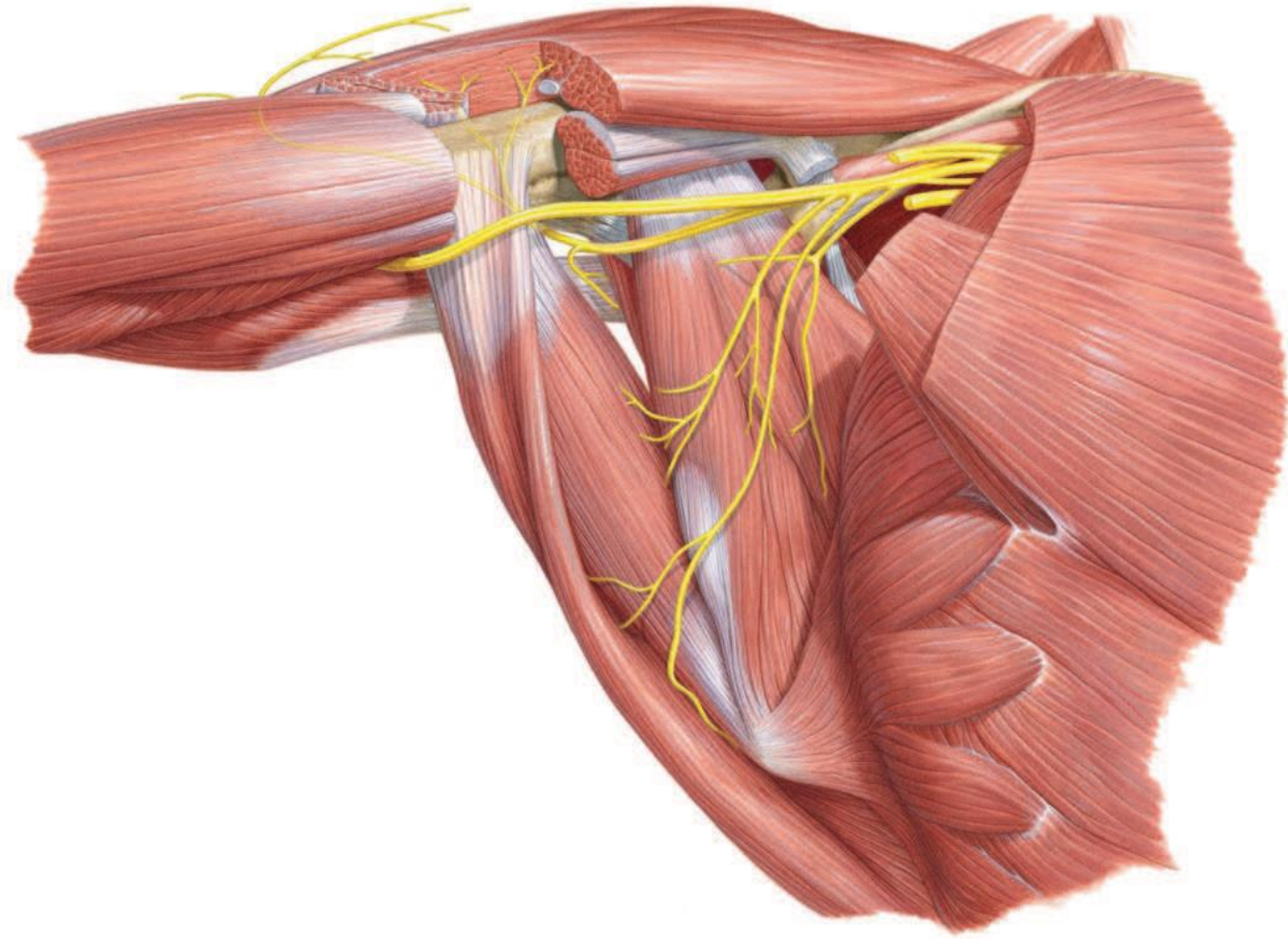
Muscle location

- Some muscles are named for the bone with which they are associated.
- For example, the temporalis muscle overlies the temporal bone



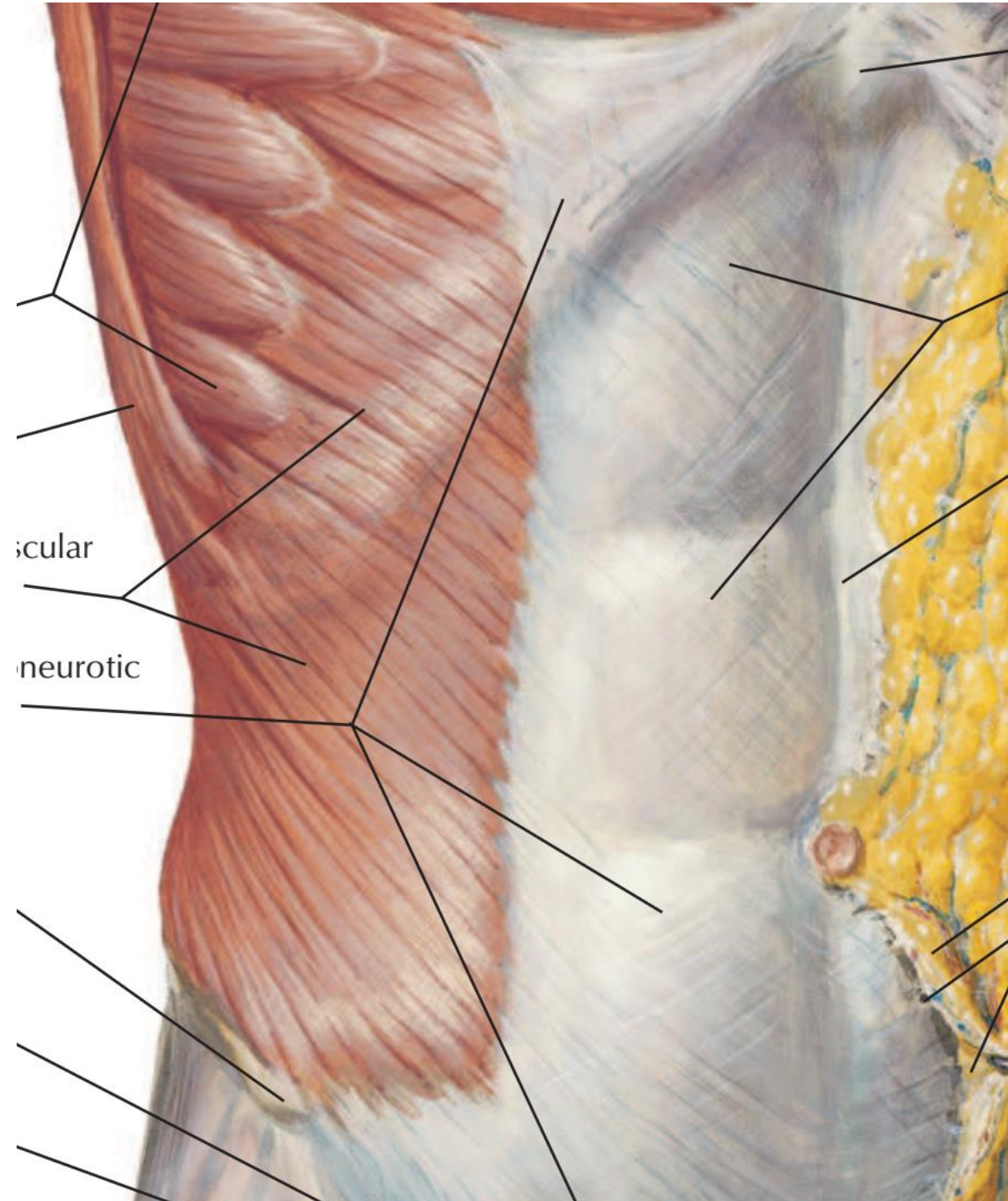
Number of origins

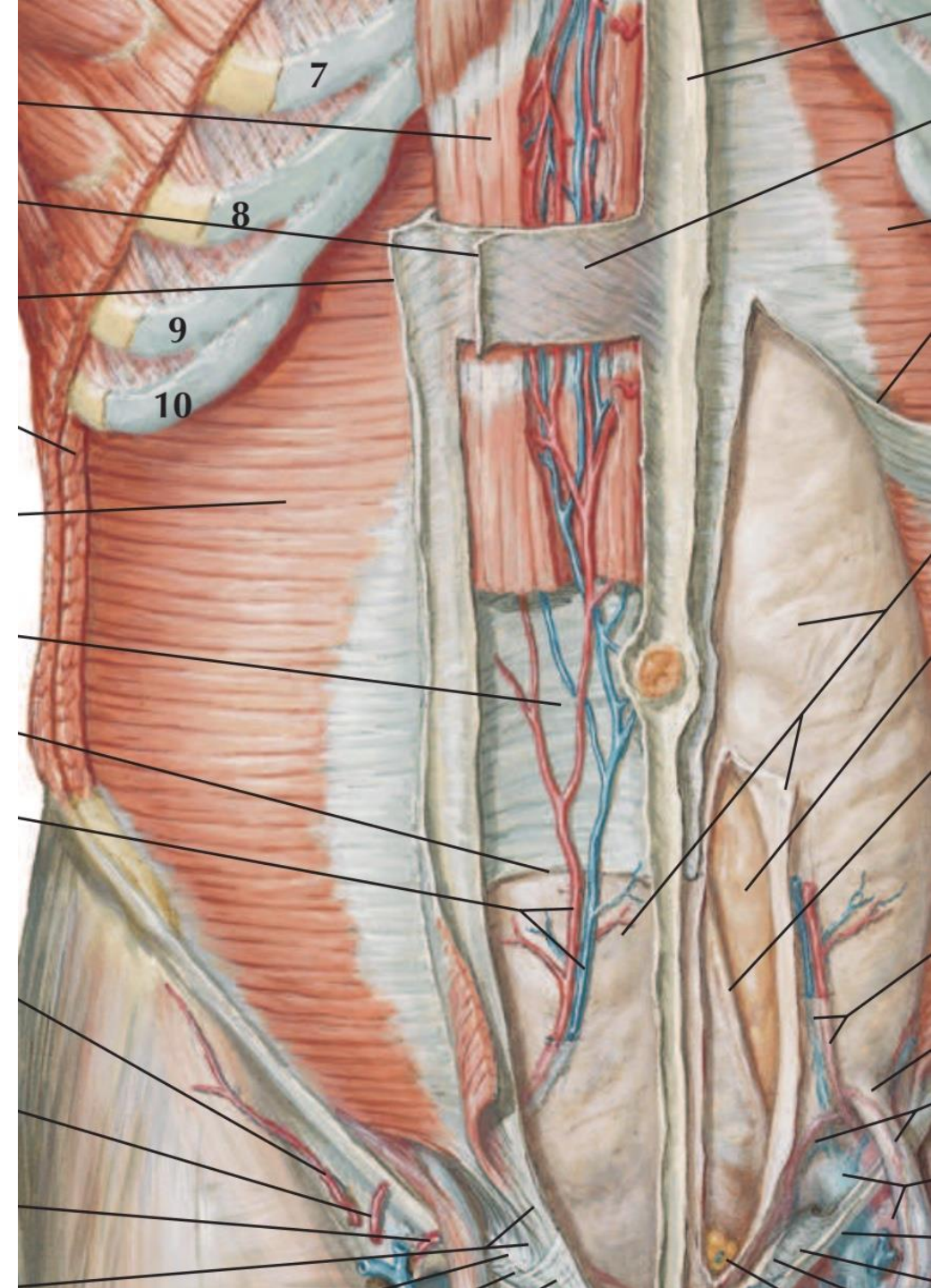
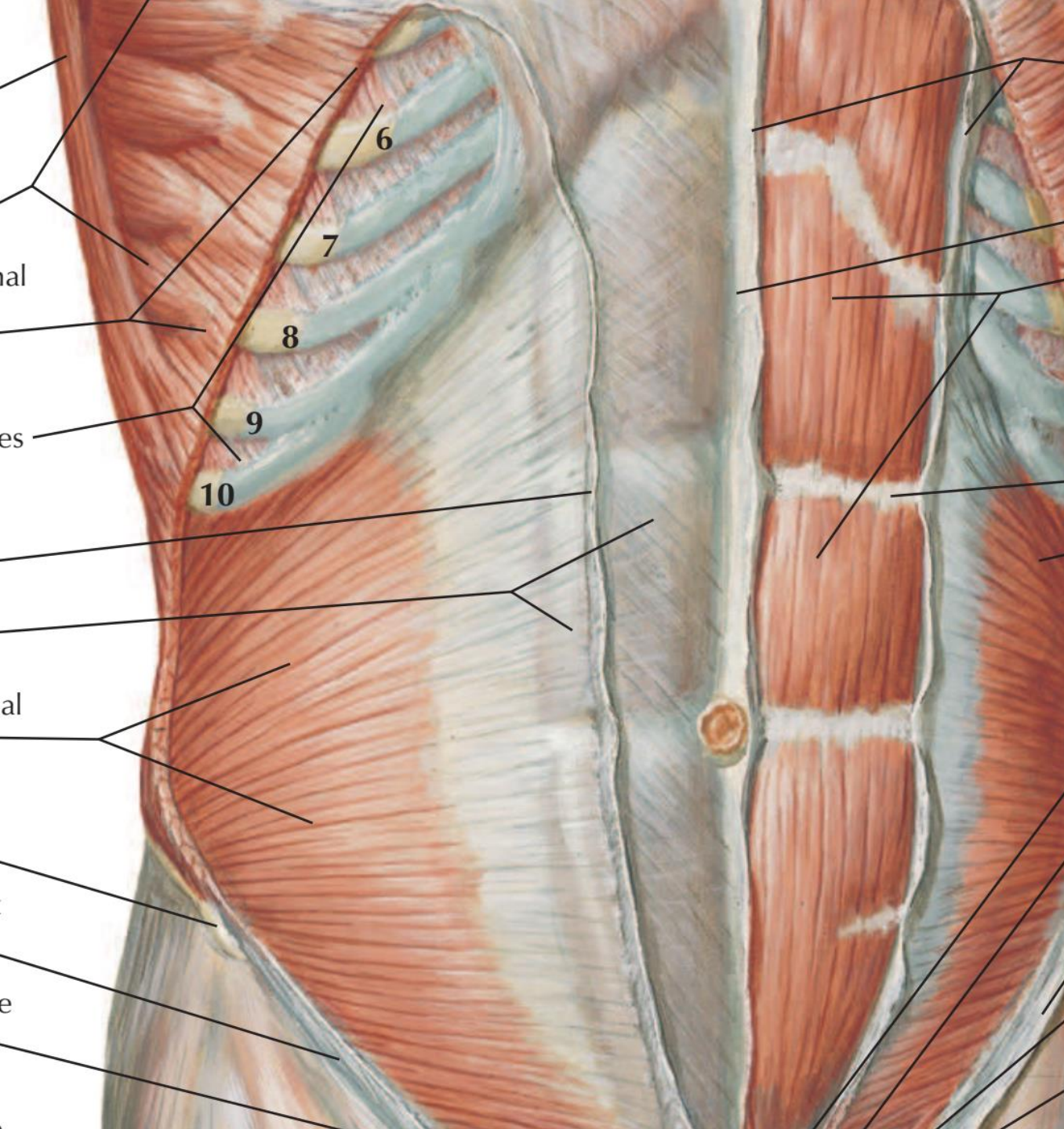
- When the term *biceps*, *triceps*, or *quadriceps* forms part of a muscle name, you can generally assume that the muscle has two, three, or four origins (respectively).
- For example, the biceps brachii has two origins.



Naming of Skeletal Muscles

- **Direction of muscle fibres:**
- Named for the direction in which their muscle fibers run with reference to some imaginary line (esp. midline)
eg: Rectus: straight
 - Transversus – transverse
 - Oblique – obliquely inclined
 - Rectus abdominis – straight of the abdomen
 - Transversus abdominis – transverse muscle of the abdomen





Muscle structure determined by fascicle arrangement. The most common patterns of fascicle arrangement are circular, convergent, parallel, and pennate



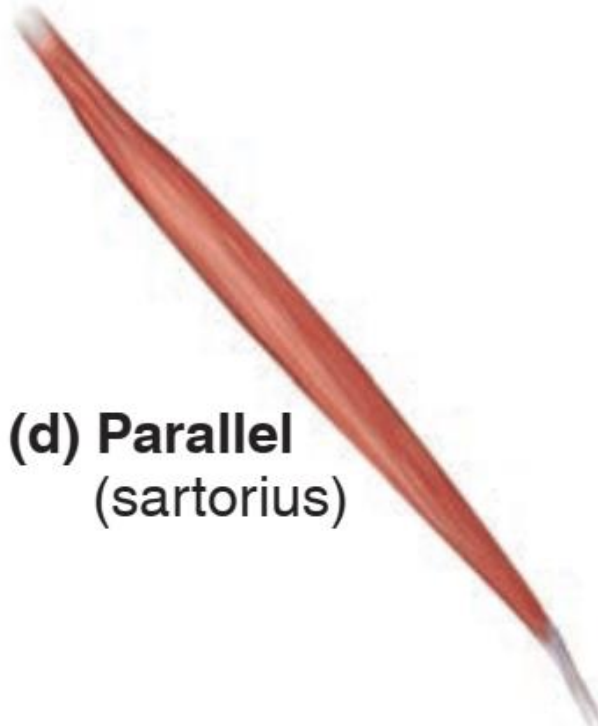
(a) Circular
(orbicularis oris)



(b) Convergent
(pectoralis major)



(c) Fusiform
(biceps brachii)



(d) Parallel
(sartorius)



(e) Multipennate
(deltoid)



(f) Bipennate
(rectus femoris)



(g) Unipennate
(extensor digitorum longus)

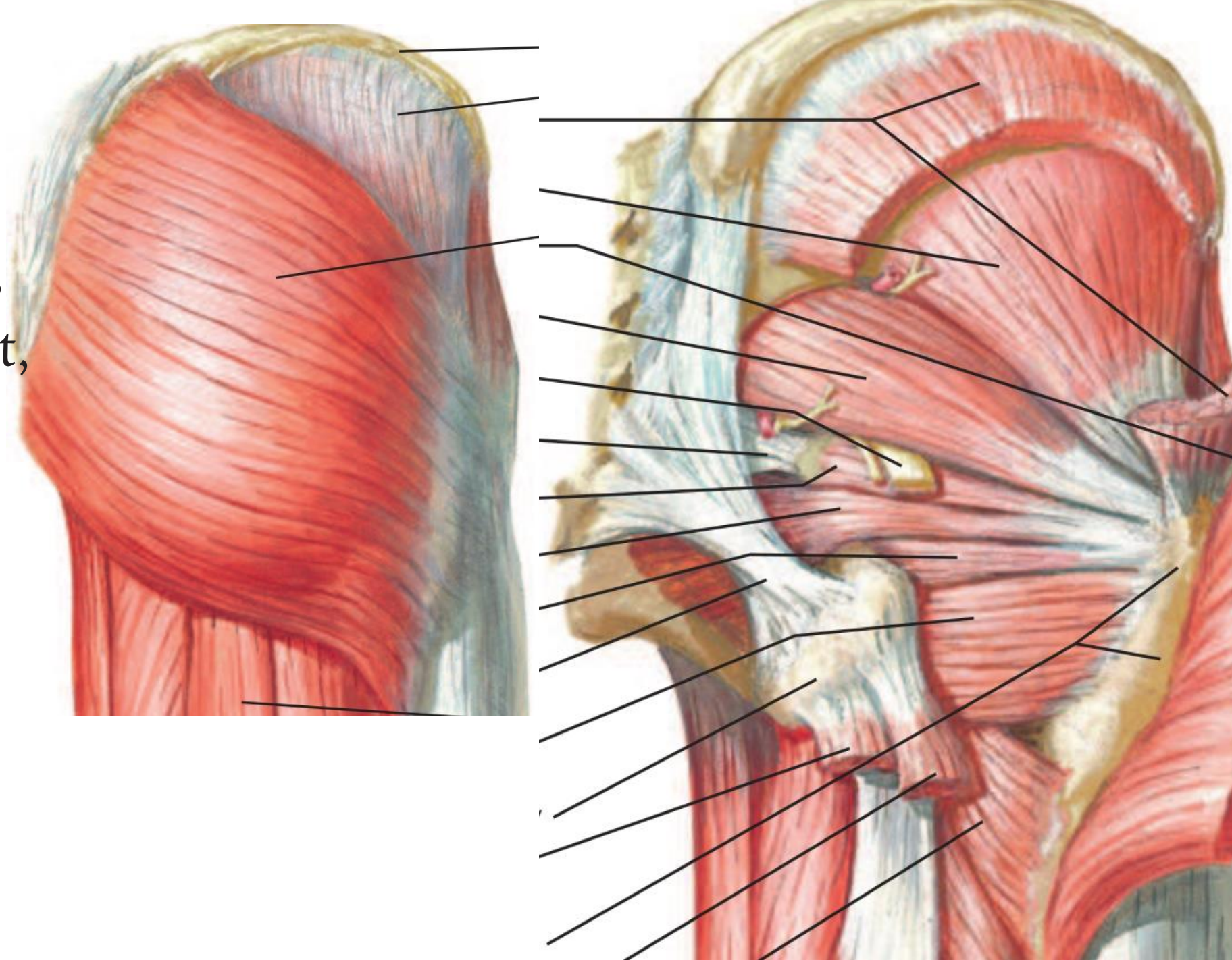
Muscle size

Terms such as

- *maximus* - largest,
- *minimus* - smallest,
- *longus* - long, and
- *brevis* - short

Eg.

gluteus maximus
gluteus minimus.



OVERVIEW OF BODY SYSTEMS

- **Muscular system: muscles**
- **Movement, stability, control of body openings, heat production**

OVERVIEW OF BODY SYSTEMS

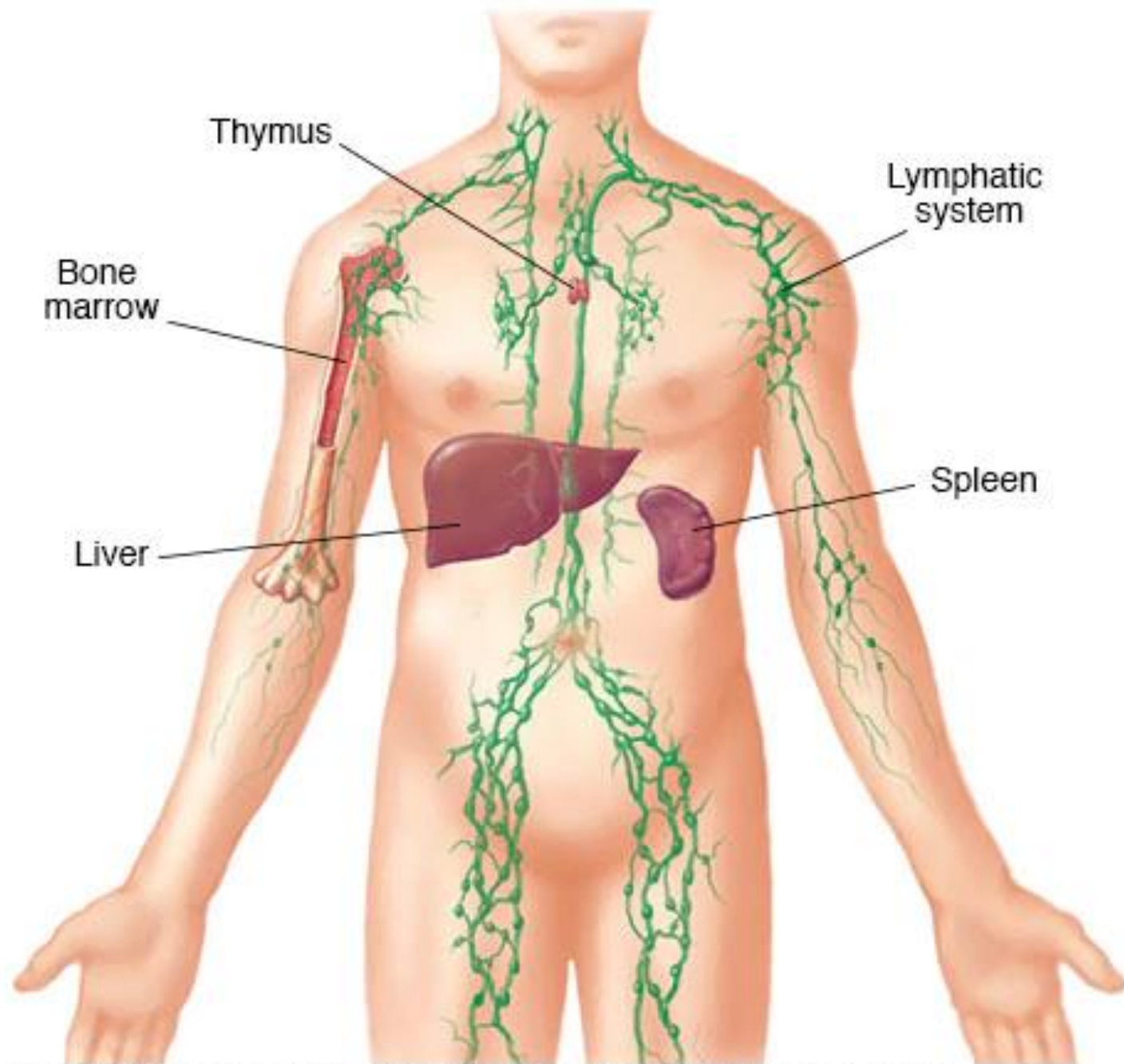
- **Nervous system:** Brain, spinal cord, nerves and ganglia
- **Control and coordination of functions of organ systems, enabling the body's responses to activities within its environment**

OVERVIEW OF BODY SYSTEMS

- **Cardiovascular system:** Heart and blood vessels
- **Distribution of nutrients, oxygen, hormones and removal of waste**

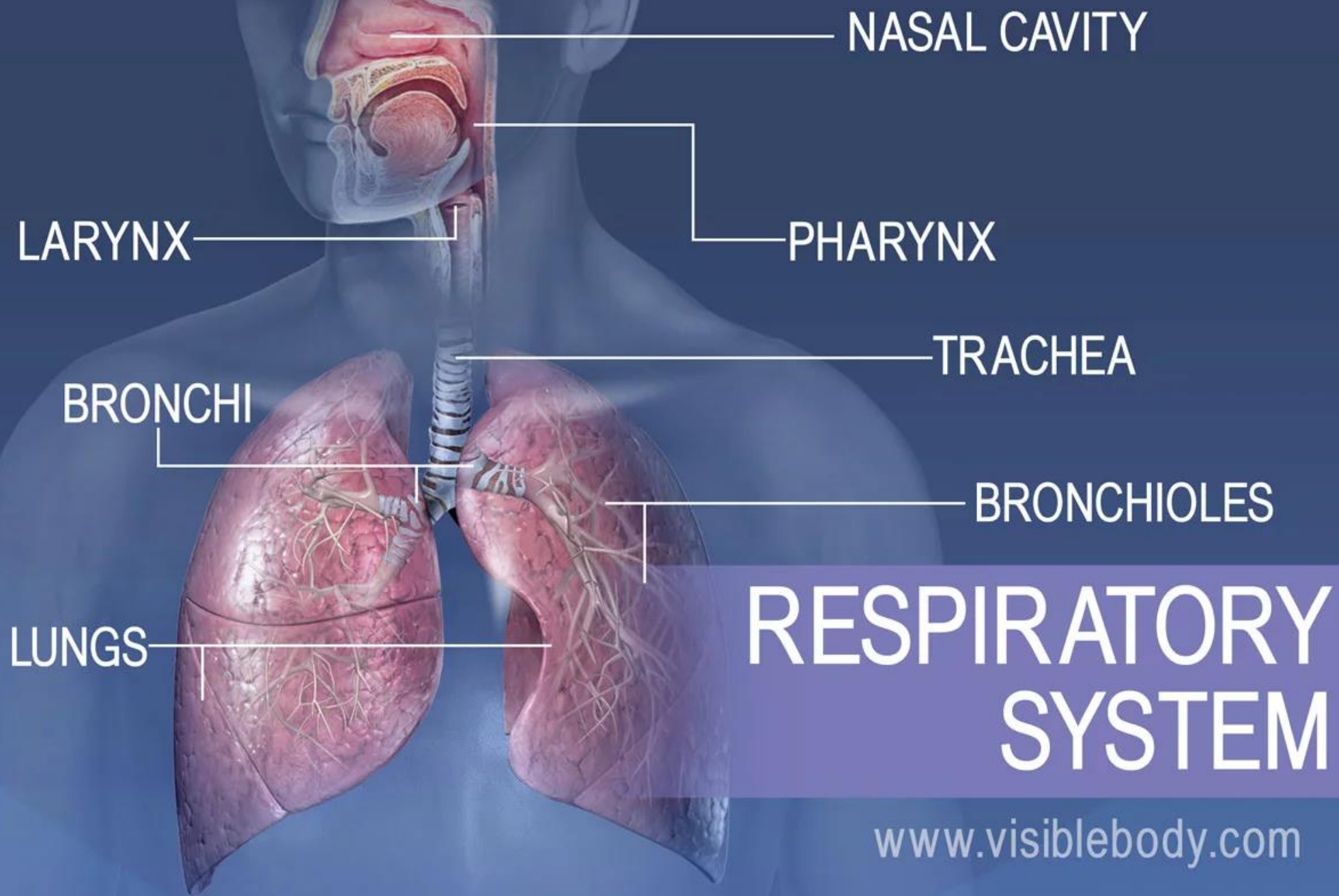
OVERVIEW OF BODY SYSTEMS

- **Lymphatic system:**
- **lymph nodes, lymphatic vessels, thymus, spleen, tonsils**
- **defense, recovery of excess tissue fluid (lymph), detection of pathogens, production of immune cells, transport of fat**



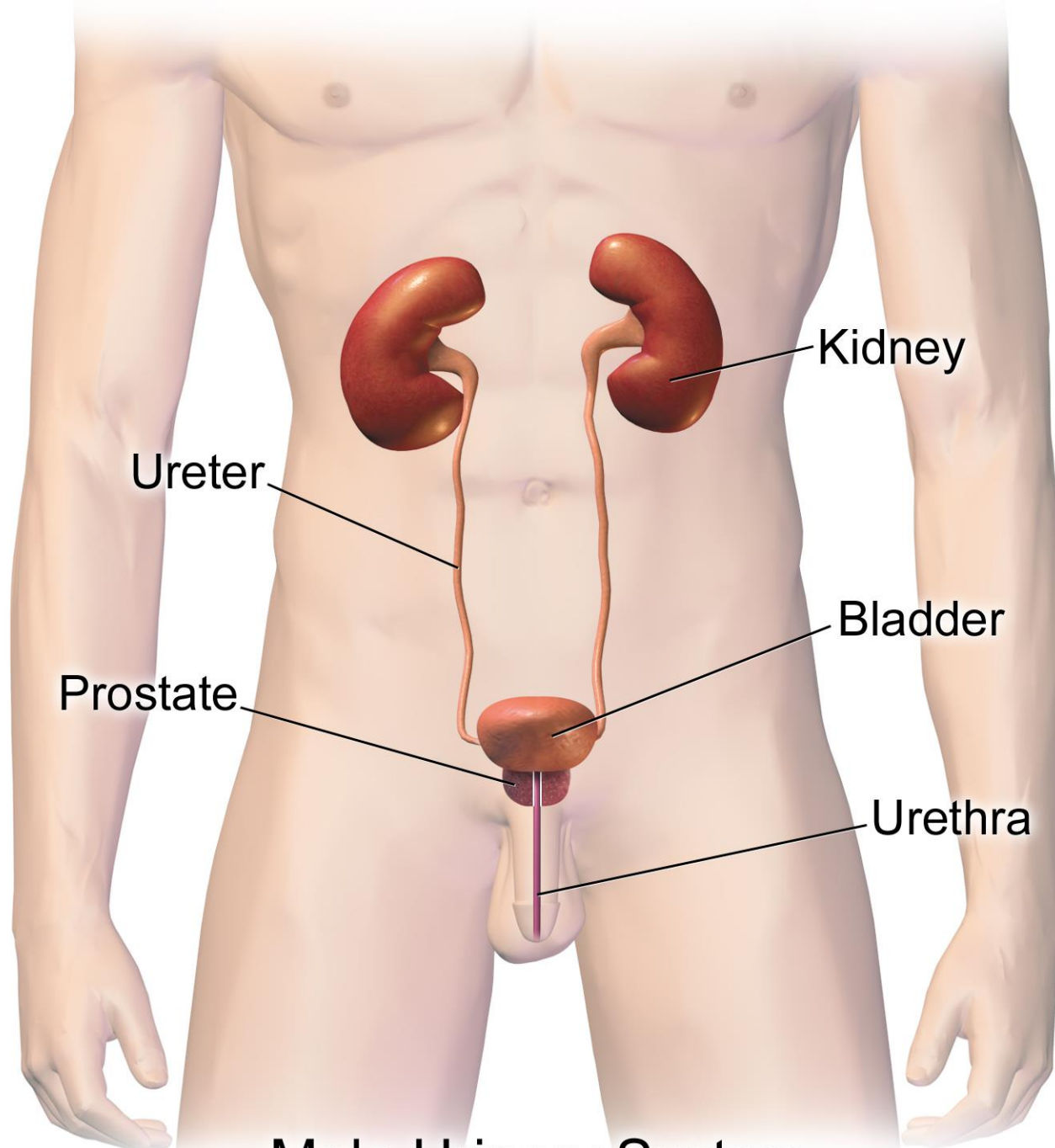
OVERVIEW OF BODY SYSTEMS

- **Respiratory system:** Nose, pharynx, larynx, trachea, lungs
- **Supply of oxygen, removal of carbon dioxide, speech**



OVERVIEW OF BODY SYSTEMS

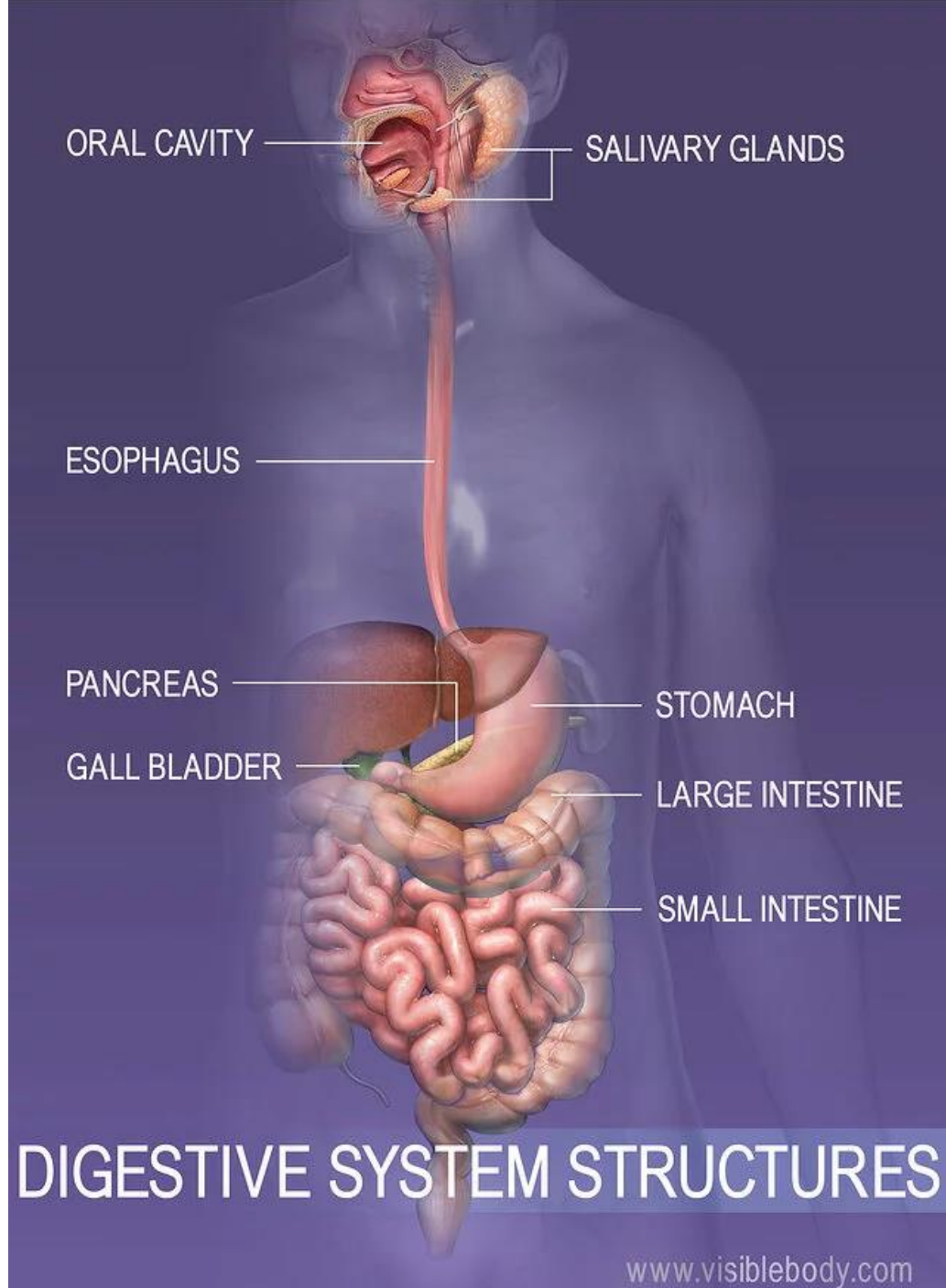
- **Urinary system:** kidneys, ureters, urinary bladder, urethra
- **Elimination of waste, regulation of blood volume & pressure, control of fluid, electrolyte and acid-base balance**



Male Urinary System

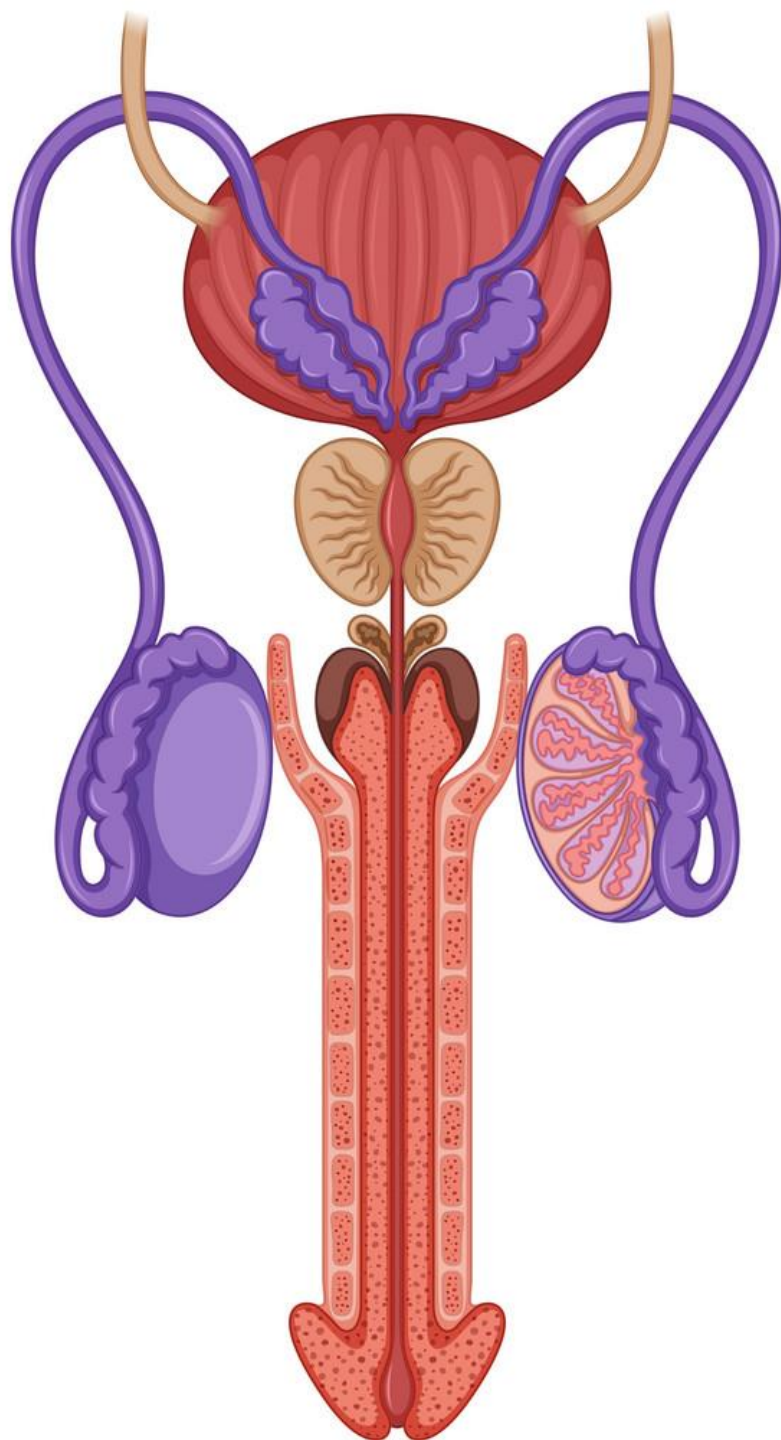
OVERVIEW OF BODY SYSTEMS

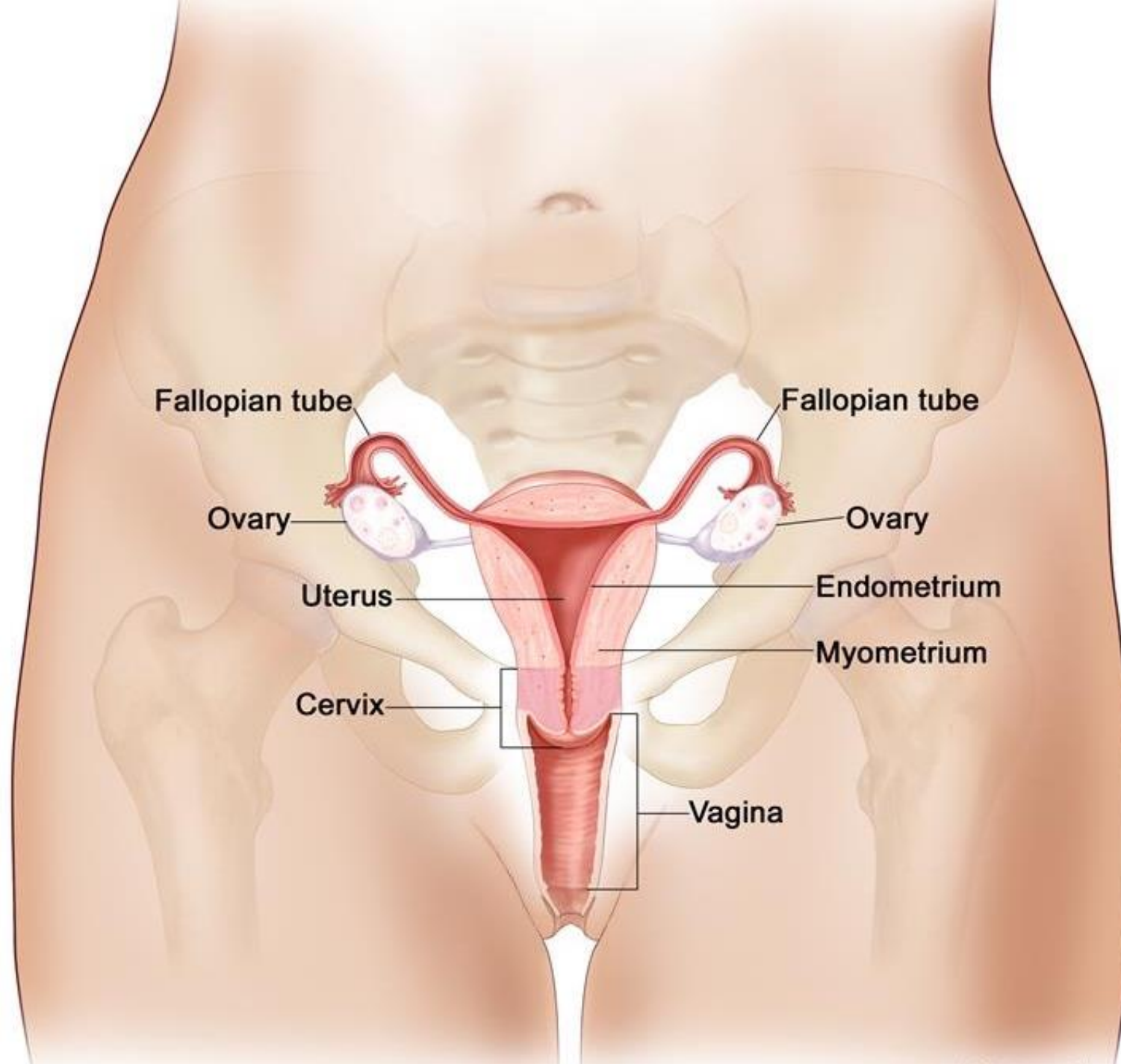
- **Digestive system:** Teeth, tongue, salivary glands, GIT, liver, gall bladder, pancreas
- **Ingestion, mastication, deglutition, digestion, absorption of food & egestion of undigested food**



OVERVIEW OF BODY SYSTEMS

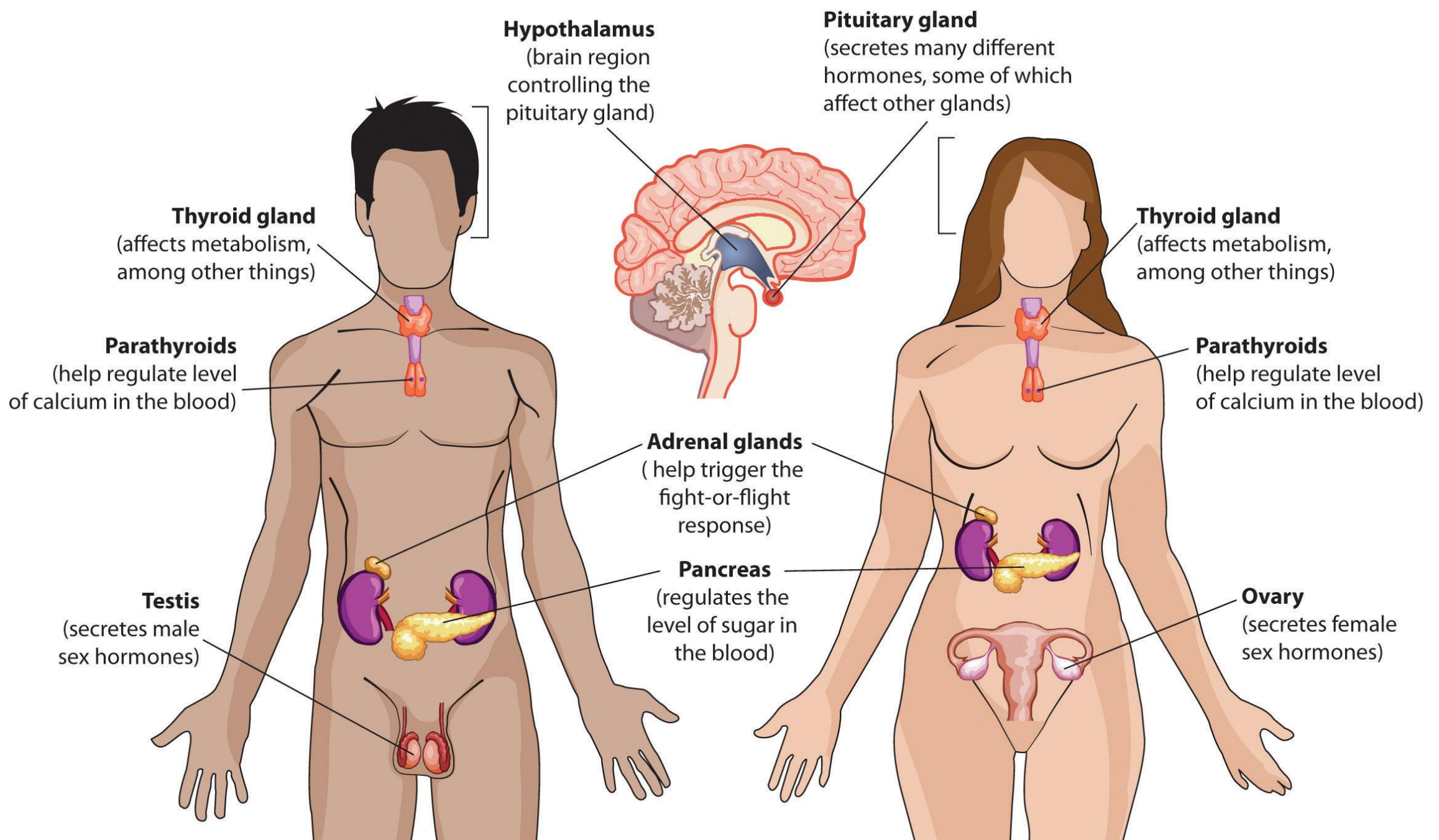
- **Reproductive system:** Gonads, ducts for transporting gonadal products, external genitalia
- **Production of sex gametes and steroid hormones**





OVERVIEW OF BODY SYSTEMS

- **Endocrine system:** Pituitary gland, pineal gland, thyroid, thymus parathyroid, pancreas, adrenal gland, testis and ovaries.
- **Regulation of processes such as growth, metabolism & reproduction**



INTERACTIONS OF INTEGUMENTARY SYSTEM

Skeletal System

- ← Role of skin in vitamin D synthesis promotes calcium absorption needed for bone growth and maintenance
- Supports skin at scalp and other places where bone lies close to surface

Muscular System

- ← Vitamin D synthesis promotes absorption of calcium needed for muscle contraction; skin dissipates heat generated by muscles
- Active muscles generate heat and warm the skin; contractions of skeletal muscles pull on skin and produce facial expressions

Nervous System

- ← Sensory impulses from skin transmitted to nervous system
- Regulates diameter of cutaneous blood vessels; stimulates perspiration and contraction of piloerector muscles

INTERACTIONS OF INTEGUMENTARY SYSTEM

Endocrine System

- ← Vitamin D₃ acts as a hormone
- Sex hormones cause changes in integumentary features at puberty; some hormone imbalances have pathological effects on skin

Circulatory System

- ← Dermal vasoconstriction diverts blood to other organs; skin prevents loss of fluid from cardiovascular system; dermal mast cells cause vasodilation and increased blood flow
- Delivers O₂, nutrients, and hormones to skin and carries away wastes; hemoglobin colors skin

Lymphatic/Immune Systems

- ← Dendritic cells detect foreign substances
- Lymphatic system controls fluid balance and prevents edema; immune cells protect skin from infection and promote tissue repair

INTERACTIONS OF INTEGUMENTARY SYSTEM

Respiratory System

- ← Nasal hairs filter particles that might otherwise be inhaled
- Provides O_2 and removes CO_2

Urinary System

- ← Skin complements urinary system by excreting salts and some nitrogenous wastes in sweat
- Disposes of wastes and maintains electrolyte and pH balance

INTERACTIONS OF INTEGUMENTARY SYSTEM

Digestive System

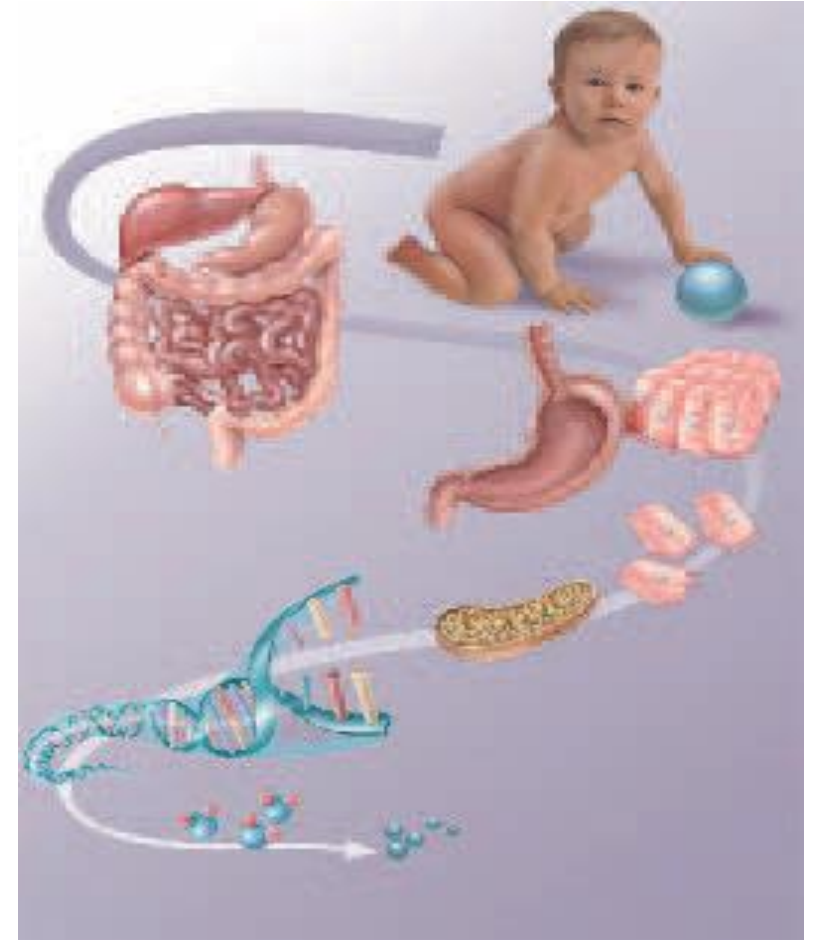
- ← Vitamin D synthesis promotes intestinal absorption of calcium
- Provides nutrients needed for integumentary development and function; nutritional disorders often reflected in appearance of skin, hair, and nails

Reproductive System

- ← Cutaneous receptors respond to erotic stimuli; mammary glands produce milk to nourish infants; apocrine glands produce scents with subtle sexual functions
- Gonadal sex hormones promote growth, maturation, and maintenance of skin

GENERAL STRUCTURE OF THE BODY

- Human body consists of innumerable **cells**
- Groups of cells differentiate and form **tissues**
- Different tissues in various combinations form **organs**
- Several organs make up a **system**
- Organism is made of **systems**



GENERAL STRUCTURE OF THE BODY

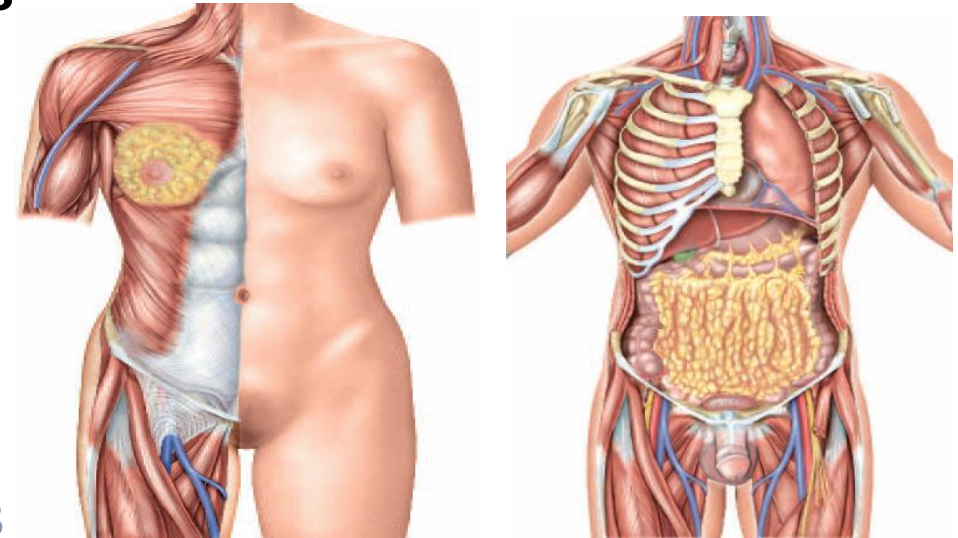
- **Four basic types of tissues essentially make up the body:** epithelium, connective tissue, muscle and nerve tissue
- **Generally, the human is organized as layers enclosing body cavities:**

Layers

- Skin
- Subcutaneous tissue/superficial fascia
- Deep fascia

Deep fascia covers deeper structures including muscles and skeleton

Body cavities contain internal organs (viscera)



ANATOMICAL VARIATION

- Human diversity is expressed in anatomical variation
- Variation in external human form may reflect variation in internal form

Eg. the shape and position of the stomach



- Generally in anatomy, patterns in most common structure and body organization is what is presented

Anatomical Variation

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

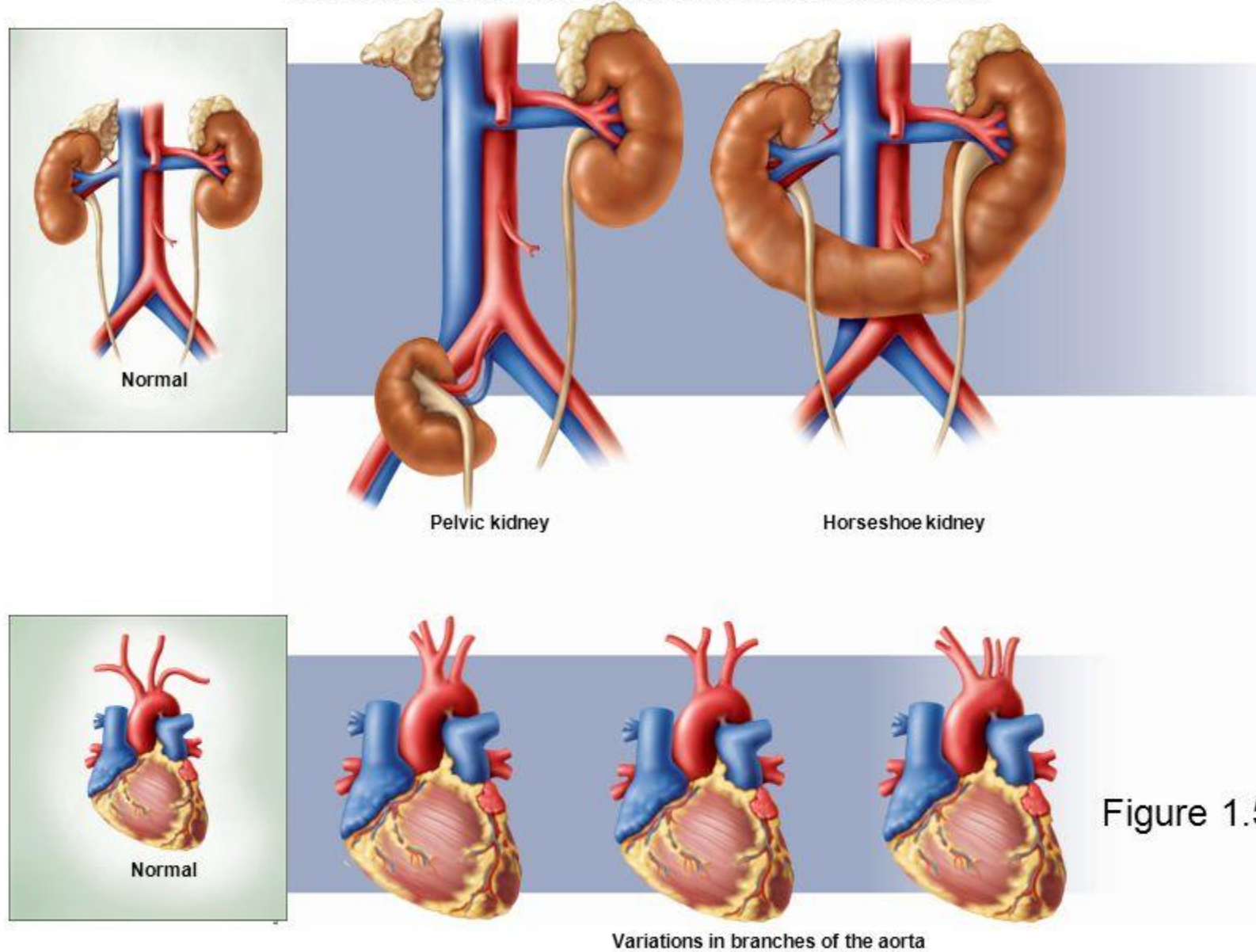


Figure 1.5

1-28

ABDOMINAL REGIONS

- **Four abdominal quadrants & nine abdominal regions are clinically important for describing the position of a mass, localization of a patient's pain or the location of abdominal viscera**
- **The quadrants are formed by median & transumbilical planes**
- **The 9 regions are formed by 2 midclavicular planes, subcostal plane/transpyloric & transtubercular planes**

ABDOMINAL QUADRANTS

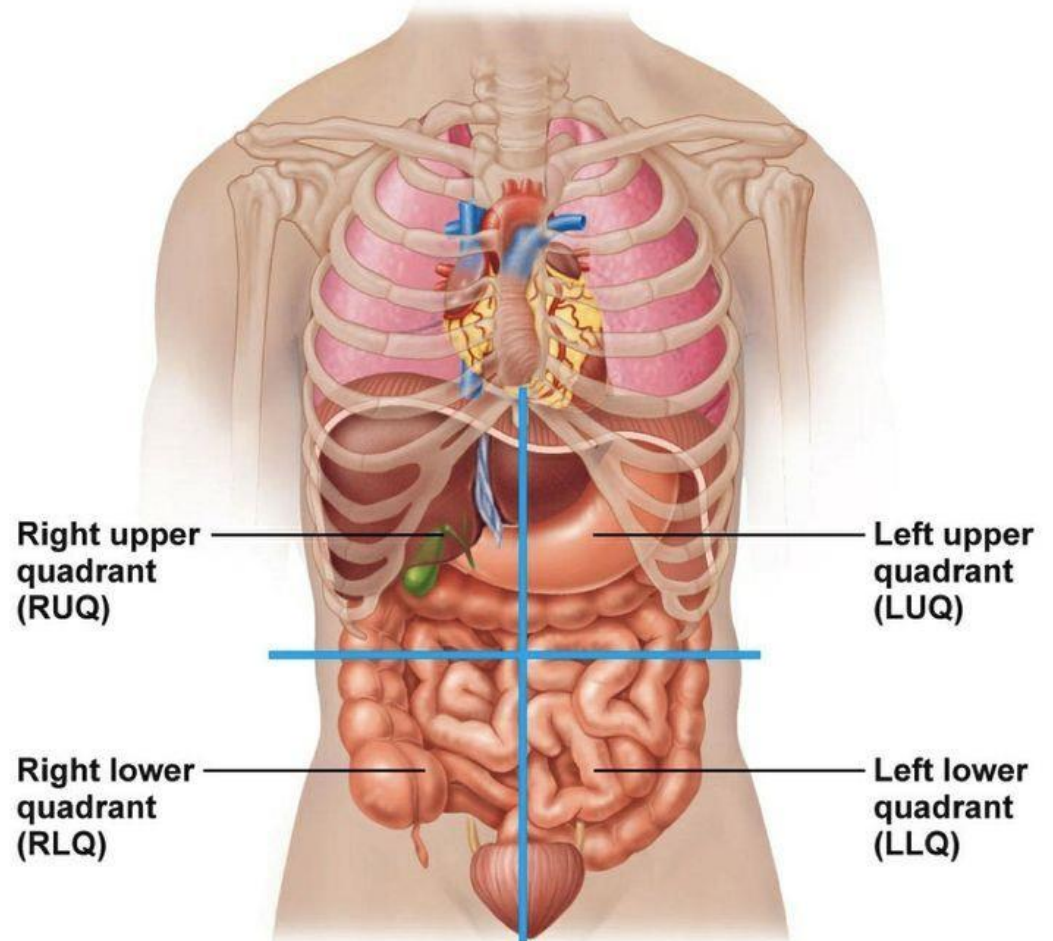
- 4 abdominal quadrants

- ✓ right upper quadrant

- ✓ left upper quadrant

- ✓ right lower quadrant

- ✓ left lower quadrant



(c) The four abdominopelvic quadrants

Regions of Abdominal Area

Right
hypochondriac
region

Epi-
gastric
region

Left
hypochondriac
region

Right
lumbar
region

Umbilical
region

Left
lumbar
region

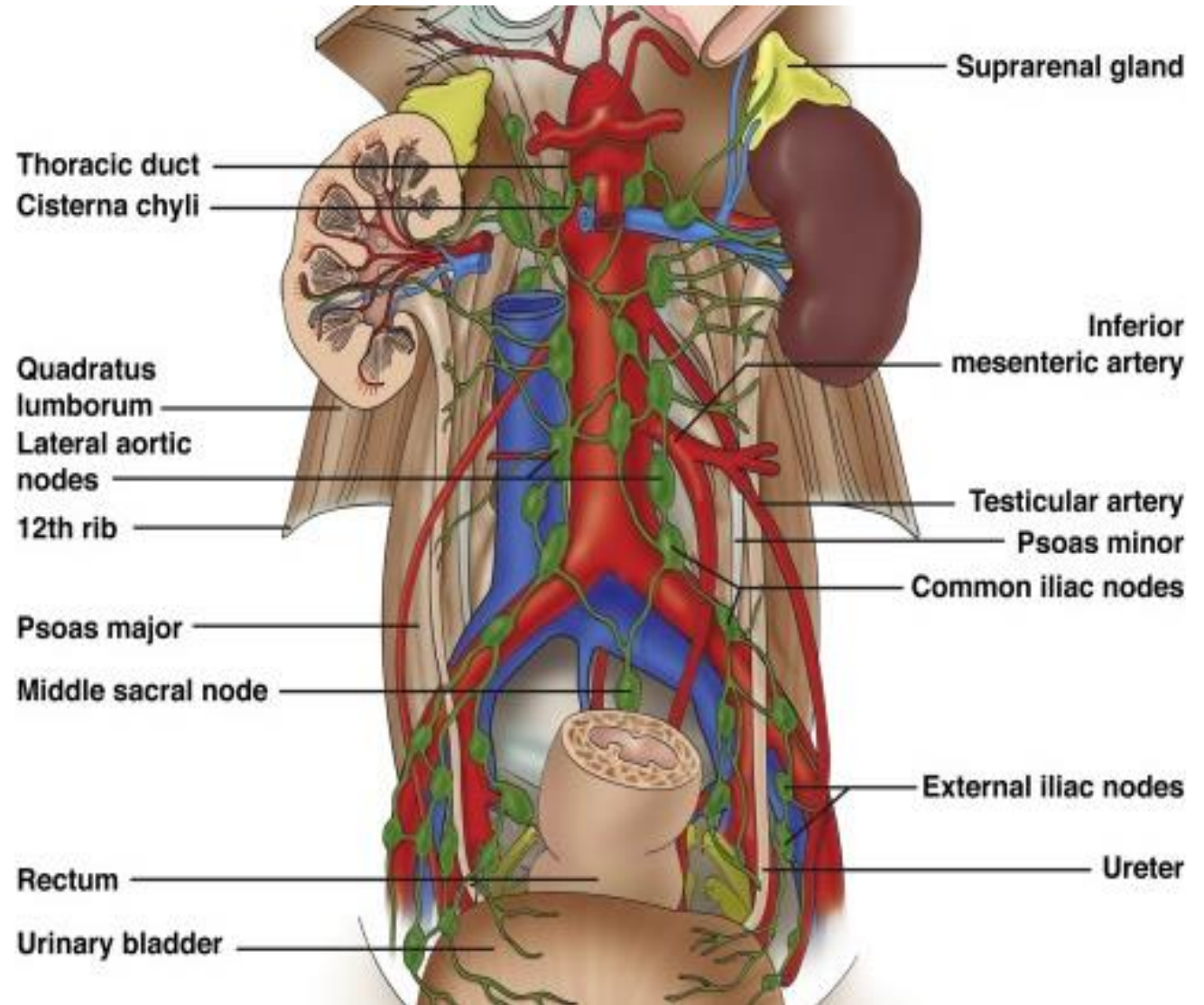
Right
iliac
region

Hypo-
gastric
region

Left
iliac
region

COLOUR CODING FOR ANATOMIC DIAGRAMS

- RED IS USUALLY USED TO COLOUR ARTERIES
- BLUE FOR VEINS
- YELLOW FOR NERVES
- GREEN FOR LYMPHATICS
- PINKISH BROWN FOR MUSCLES
- CREAMY YELLOW FOR BONE
- SKY BLUE FOR CARTILAGE



MEDICAL IMAGING TECHNIQUES-BRIEF OVERVIEW

- The use of different imaging modalities and processes to obtain information about anatomic structures for diagnostic and treatment purposes is a common clinical practice.
- Radiological Anatomy** studies the structure and function of the body using medical imaging techniques.

Medical Imaging (field commonly called radiology)

- techniques and procedures used to create images of the human body. They allow visualization of internal structures to diagnose abnormal anatomy and deviations from normal physiology.
- **Methods of obtaining images to help screen for, diagnose, or monitor medical conditions:**
 - X-rays
 - CT (computed tomography) scan
 - MRI (magnetic resonance imaging)
 - Ultrasound
 - Nuclear medicine scans- **positron emission tomography (PET), Single photon emission computed tomography (SPECT) scanning, Bone scan, Sentinel lymph node mapping/biopsy**

Clinical Anatomy—Medical Imaging Techniques

Plain X-ray (radiograph)— An x-ray is an electromagnetic wave of very short wavelength. They are classed as a carcinogen

- Best for visualizing bones and abnormal dense structures; some tumors and tuberculosis nodules in the lungs
- High density tissue like bone will absorb the x rays more and therefore will appear **white** on an x-ray. Low density tissue like organs such as the lungs will not absorb many x-rays and hence will appear **dark** on the x-ray.



MEDICAL IMAGING TECHNIQUES-BRIEF OVERVIEW

Conventional radiography

- Contrast agents such as barium and iodine salts have been used with this technique to visualize luminal or vascular organs and spaces such the digestive tract that otherwise are not visible on plain x-ray films



Plain x-ray films



Plain x-ray films with contrast agent

Clinical Anatomy—Medical Imaging Techniques

Specialized X-ray techniques include digital mammography that uses low dose radiation to detect breast cancer, and panoramic X-rays are used to detect dental disease.

Bone densitometry (DEXA scan)-use X rays to measure bone strength by measuring the density of minerals, especially calcium in the bone using X-rays or CT scan. eg in osteoporosis (“porous bone”).

Fluoroscopy - uses X-rays in real-time to create moving images of the body

Which of the following will appear black on an x-ray image?

- A. Rib
- B. Heart
- C. Hip prostheses
- D. Trachea

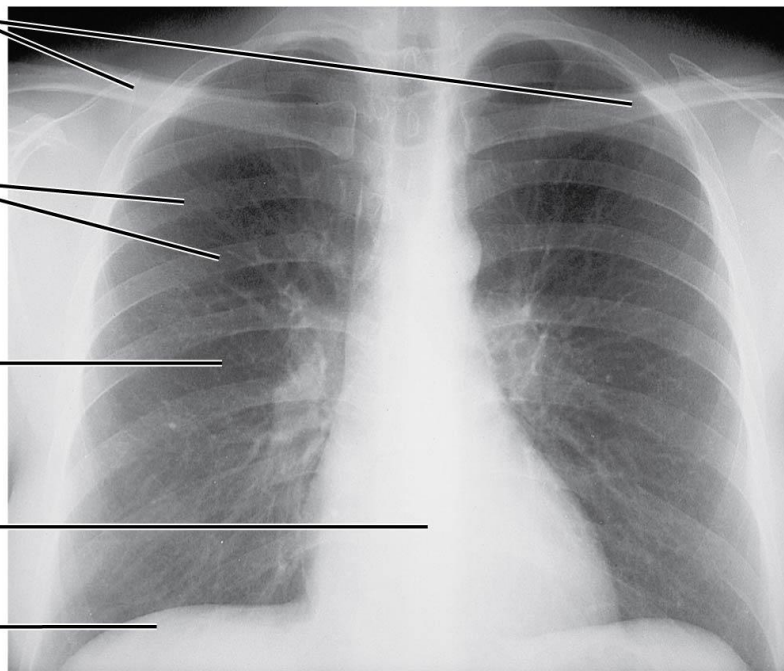
Clavicles
(collarbones)

Ribs

Air in lungs
(black)

Heart

Diaphragm



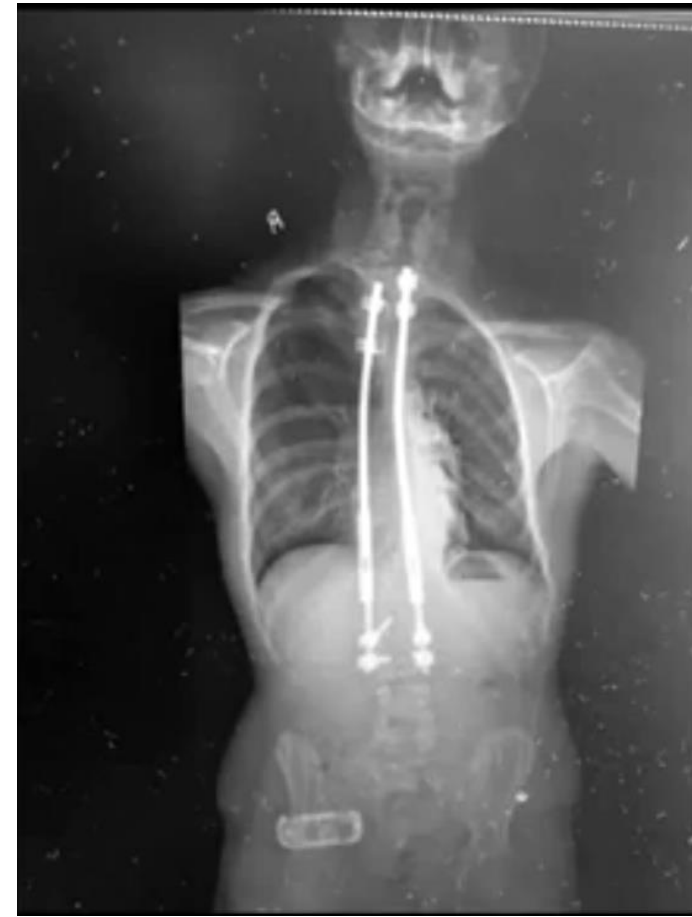
(a) Radiograph of the chest

© 2014 Pearson Education, Inc.



(b) Mammogram (cancerous tumor at arrow)

X-ray



Posteroanterior



Anteroposterior



Lateral.

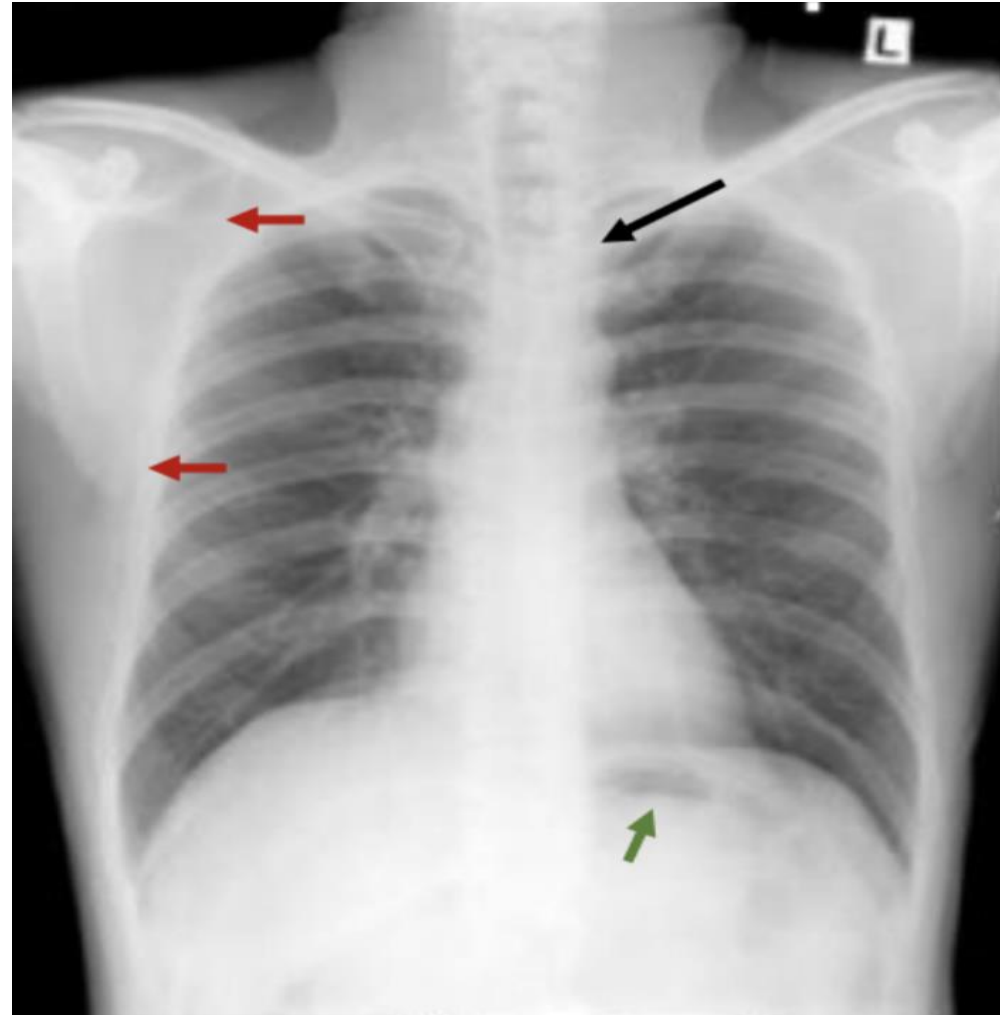


Decubitus



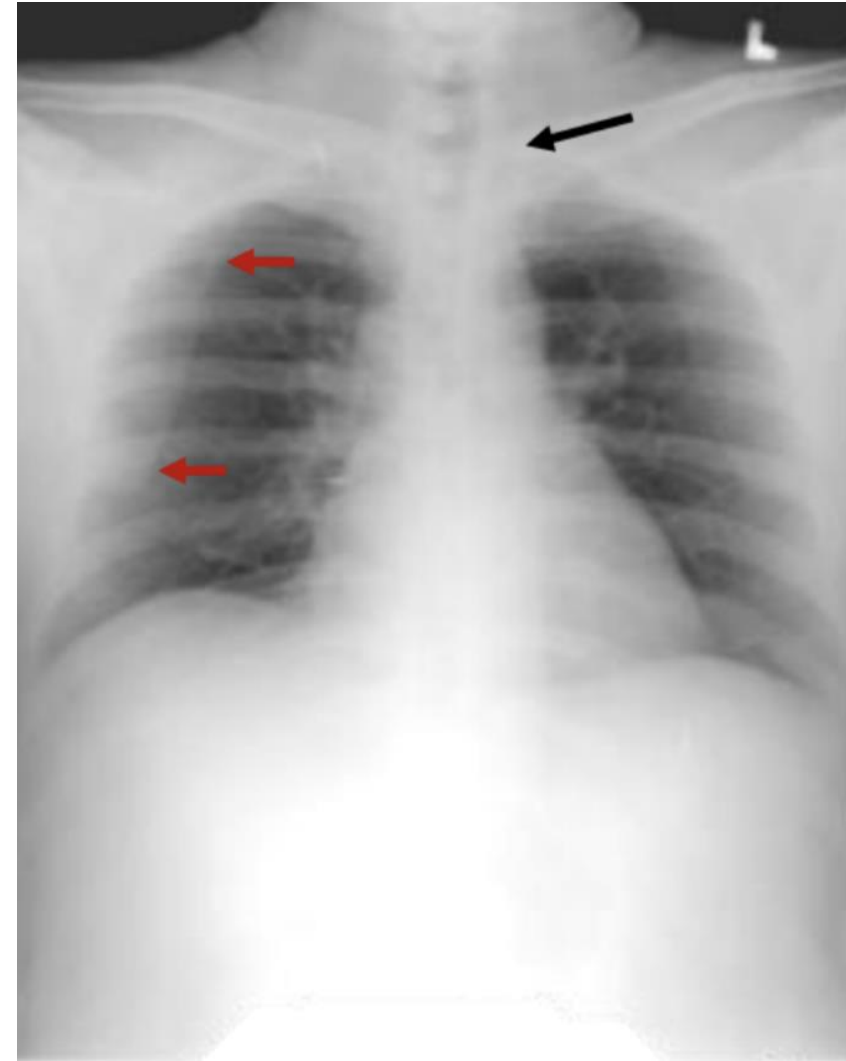
PA (Posteroanterior) Chest X-Ray

- Most frequently requested imaging test
- PA view gives a good assessment of the **Cardiac Size**.
- It avoids magnification of the heart as the film is close to the anterior chest wall.
- **Scapulae lie in the periphery**
- Clavicles project over lung fields
- Fundal gas bubbles are seen



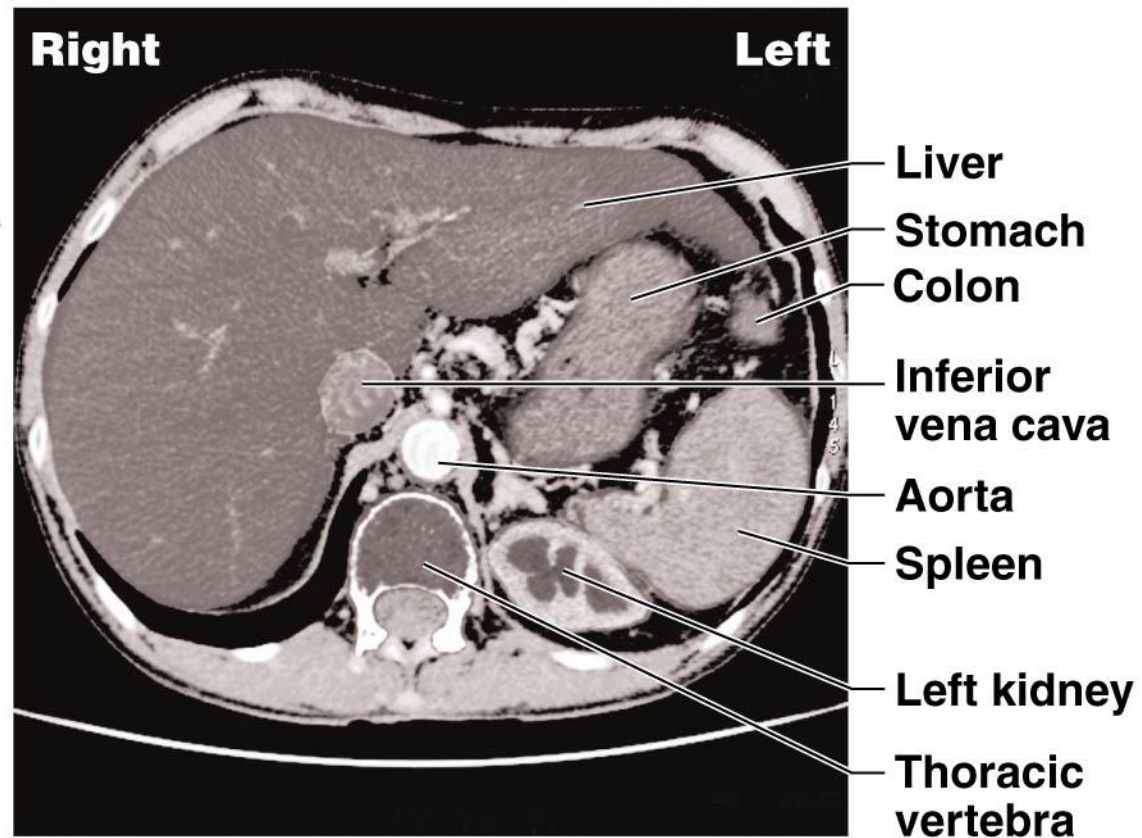
AP (Anteroposterior) Chest X-Ray

- AP chest x-rays are done where it is difficult for the patient to obtain a normal chest x-ray, such as when the patient **cannot get out of bed**.
- It is often used frequently to aid diagnosis of acute and chronic conditions in intensive care units and wards.
- Heart is larger than PA view
- Scapulae are projecting over lung fields
- Clavicles are above the apex of lung fields





© 2014 Pearson Education, Inc.



Claustrophobia?

Computed tomography (CT)

- Uses multiple X-rays to produce cross-sectional layers that show detailed images of soft tissues including bones, organs, tissues, and tumours
- Differing tissue densities show up as various shades of gray.
- typically targets the torso and appears to provide the most benefit in screening for lung cancers, coronary artery disease, and kidney cancers
- *Less useful for nervous tissue structures and for joint images, because bone can obscure the joint details*

G 154721 AGE 44
FRONT



ING 154721 AGE 44
FRONT



MRI is gold standard for soft tissue injuries, spinal cord injury

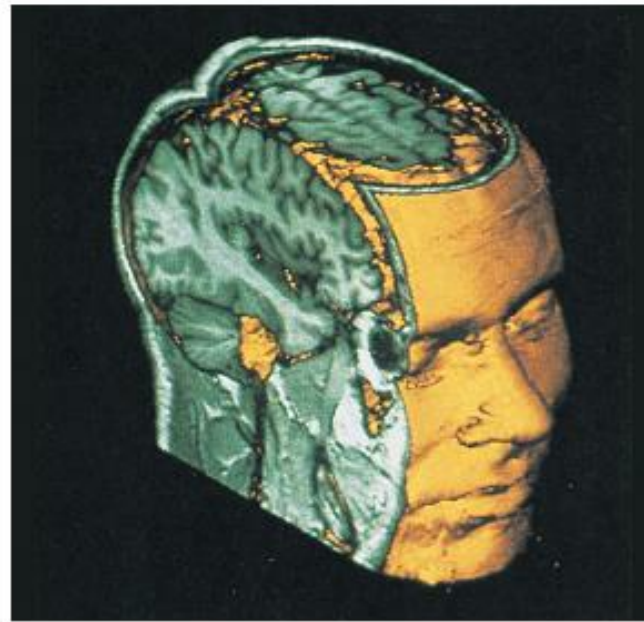
Magnetic Resonance Imaging (MRI)

- Uses magnetic fields and radio waves to show detailed images of soft tissues, ligaments and cartilage. Give images in different planes.
- MRI distinguishes body tissues from one another on the basis of differences **in water content (Hydrogen)**.
- Does not visualize bone or other tissues with low water content
- No radiation exposure
- Cannot be used in patients with metal implants (cardiac pacemakers, heart monitors, defibrillators and other battery-operated devices)
- Most useful for differentiating between normal and abnormal tissues.
- Use to detect tumors and artery-clogging fatty plaques; reveal brain abnormalities; measure blood flow; and detect a variety of musculoskeletal, liver, and kidney disorders.
- *Joints and many tumours, eg. Brain tumours can easily be visualized.*

Functional MRIs (fMRIs), which detect the concentration of blood flow in certain parts of the body,

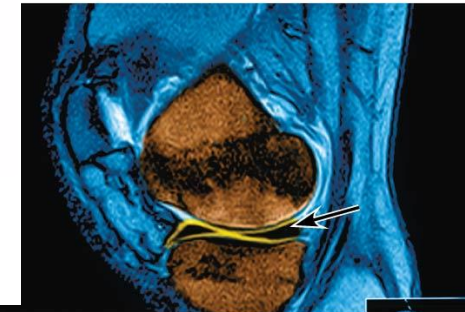


(a) MRI of the head; sagittal section, artificially colored



(b) Volume rendering of an MRI of the head

FIGURE 1.15 Magnetic resonance image (MRI). The flat surfaces in (b) show the original MRI data.

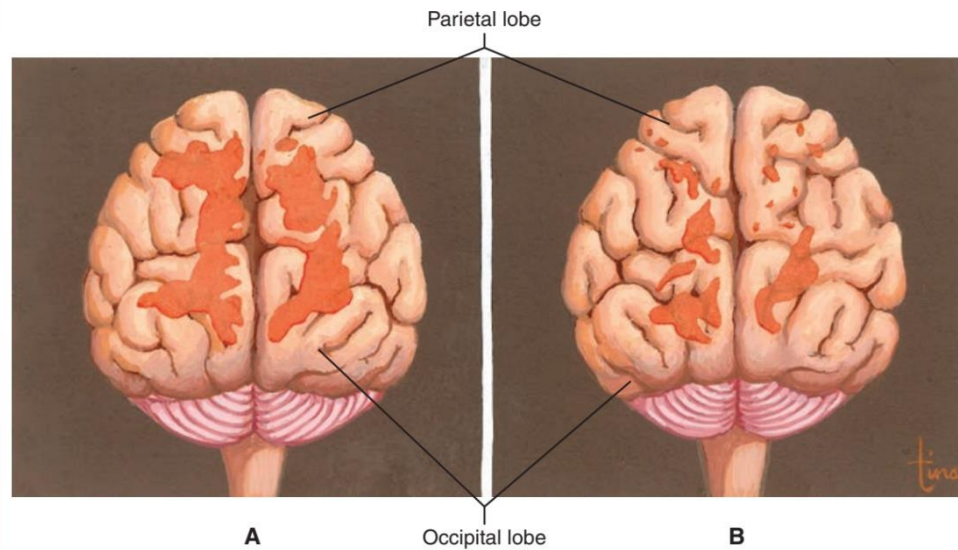


Normal knee, meniscus intact



Injured knee, torn meniscus

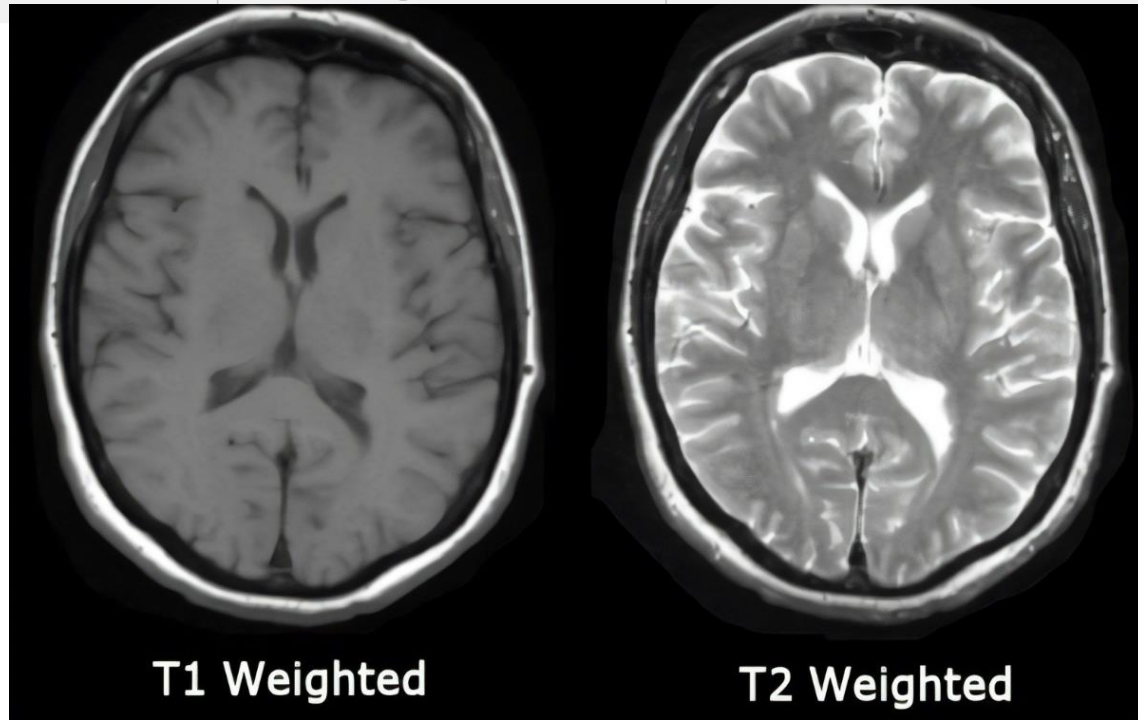
(a) MRI of knee, sagittal section. Arrow indicates meniscus. Note tear in meniscus in bottom image.



Box Figure 1-3 Depiction of a functional MRI, with increased brain activity shown in orange. (A) The posterior cerebrum engaged in driving a car. (B) The posterior cerebrum, driving and talking on a phone.

The magnetic fields produced by the MRI scanner can be manipulated to produce two distinct types of image – T1 weighted and T2 weighted.

Appearance	T1 Weighted Image	T2 Weighted Image
White	Fat Protein Rich Fluid	Water Content E.g. Inflammation, Tumour, Haemorrhage, Infection
Intermediate	Gray Spinal Matter darker than White	White Spinal matter darker than gray spinal matter.
Dark	Bone Air Water Content e.g. Inflammation, Tumour, Haemorrhage	Bone Air Fat

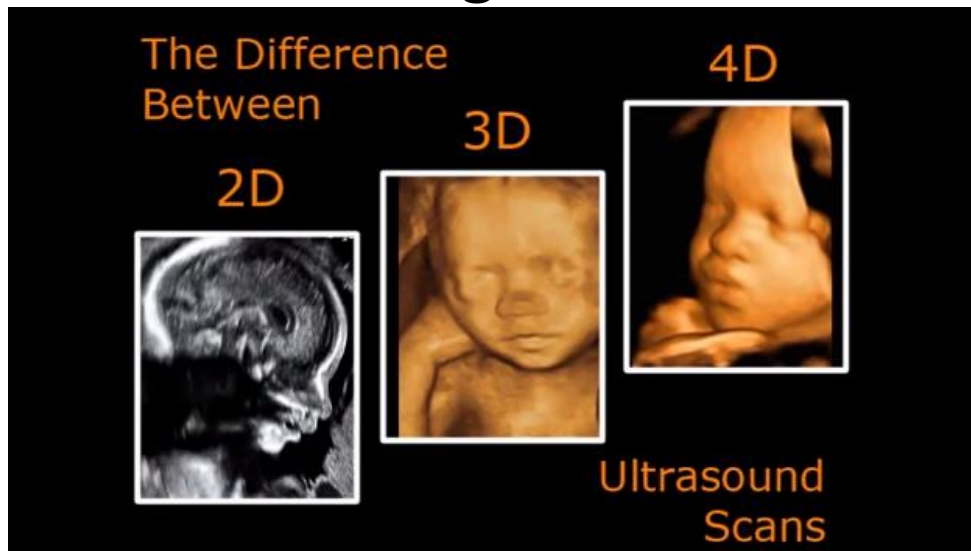


Factor	CT (CT abdo used as example)	MRI	X-ray (CXR used as example)	Ultrasound
Duration	3-7 minutes	30-45 min	2-3 min	5-10 minutes
Cost	Cheaper	Expensive	Cheap	Cheap
Dimensions	3	3	2	2
Soft tissue	Poor detail	Excellent detail	Poor detail	Poor detail
Bone	Excellent detail	Poor detail	Excellent detail	Poor detail
Radiation	10mSv	None	0.15mSv	None

A mSv (milliSievert) is used to measure the average amount of radiation absorbed over the entire body. Exposure to around 100 mSv in a year increases the chance of developing cancer. Additionally, about 5% of people who receive 1,000 mSv of radiation in their lifetime could develop a fatal cancer later in life

Ultrasound

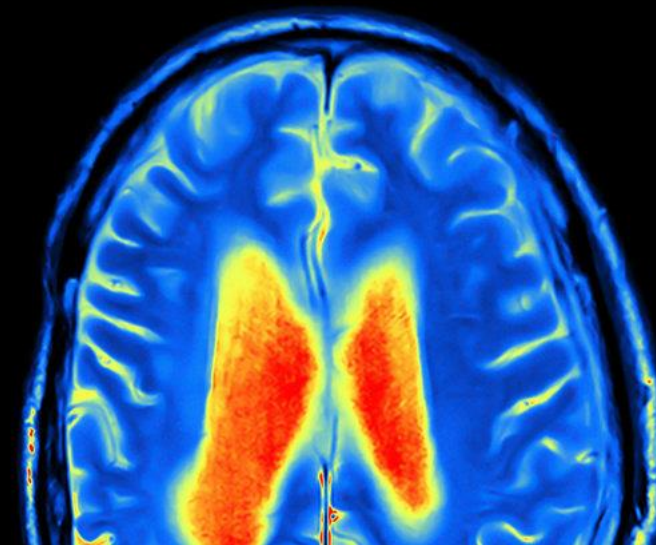
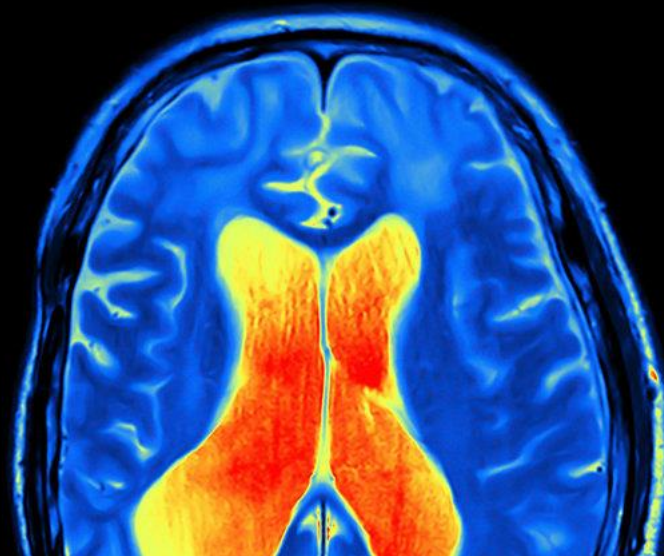
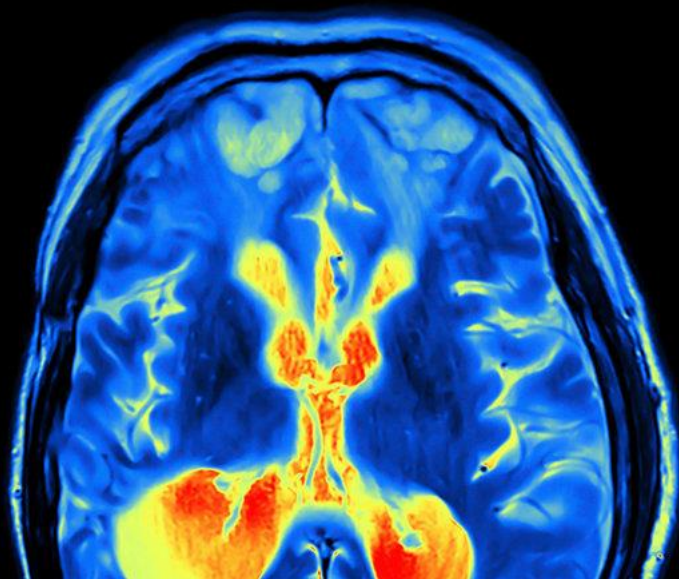
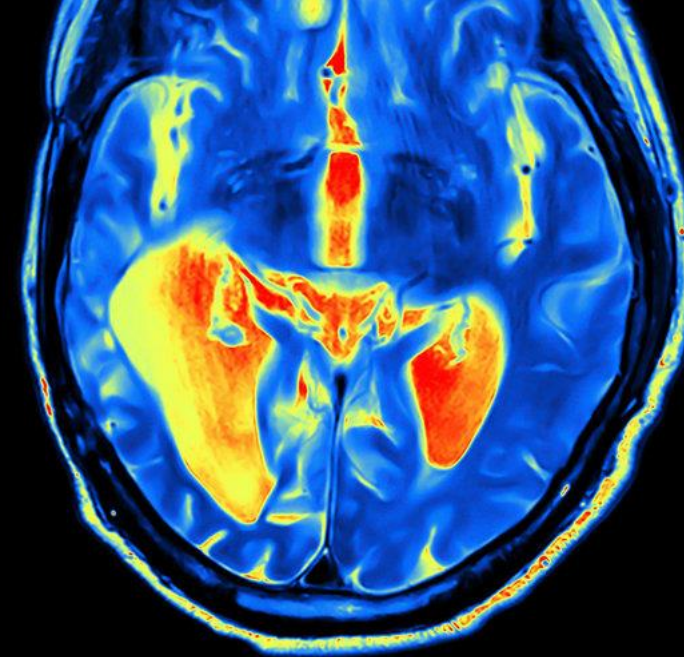
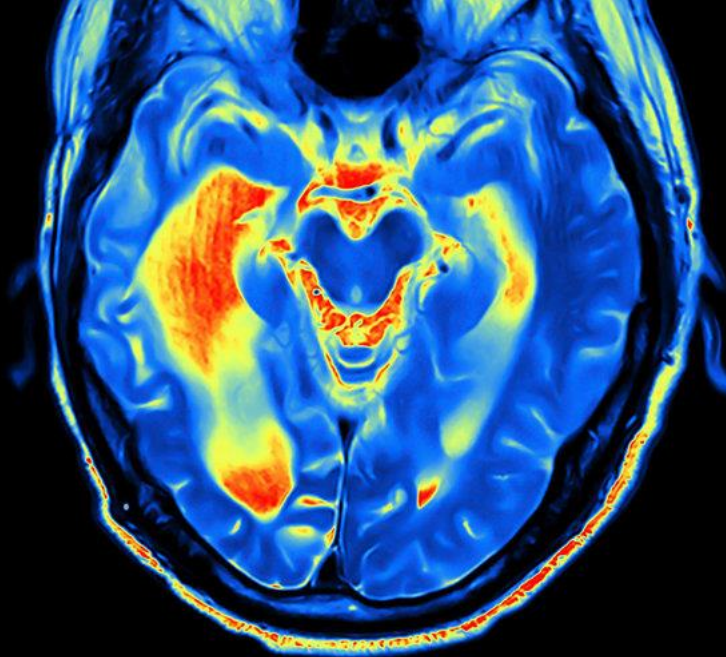
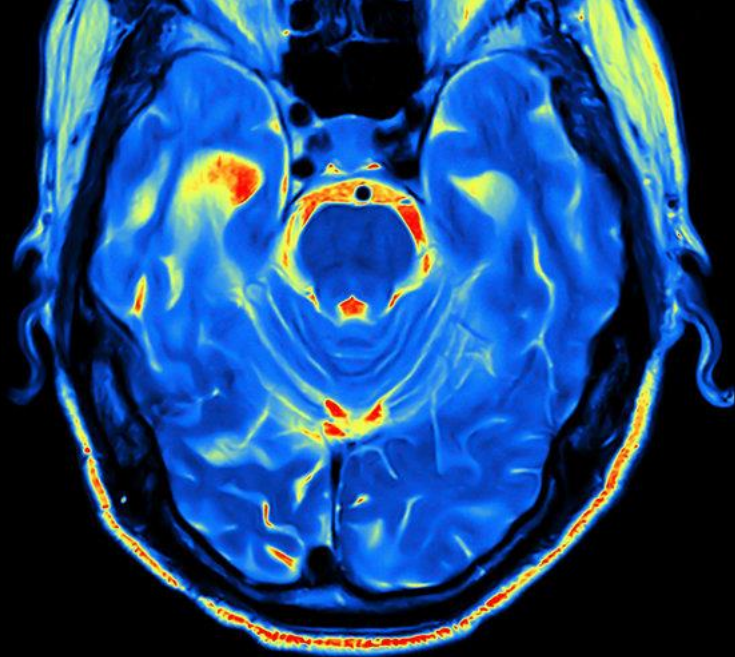
- Uses pulses of high-frequency (ultrasonic) sound waves that reflect (echo) off the body's tissues
- Safe, noninvasive, painless, and uses no dyes.
- Most commonly used to visualize the foetus during pregnancy. Also used to observe the size, location, and actions of organs and blood flow through blood vessels (**Doppler ultrasound**).
- Echoes reflected from tissues of differing densities is picked by a transducer and converted into electrical signals which are recorded to form cross-sectional images.



MEDICAL IMAGING TECHNIQUES-BRIEF OVERVIEW

Nuclear medicine imaging

- Uses unsealed radioactivity to produce functional images by intravenous injection of small doses of radioactive material.

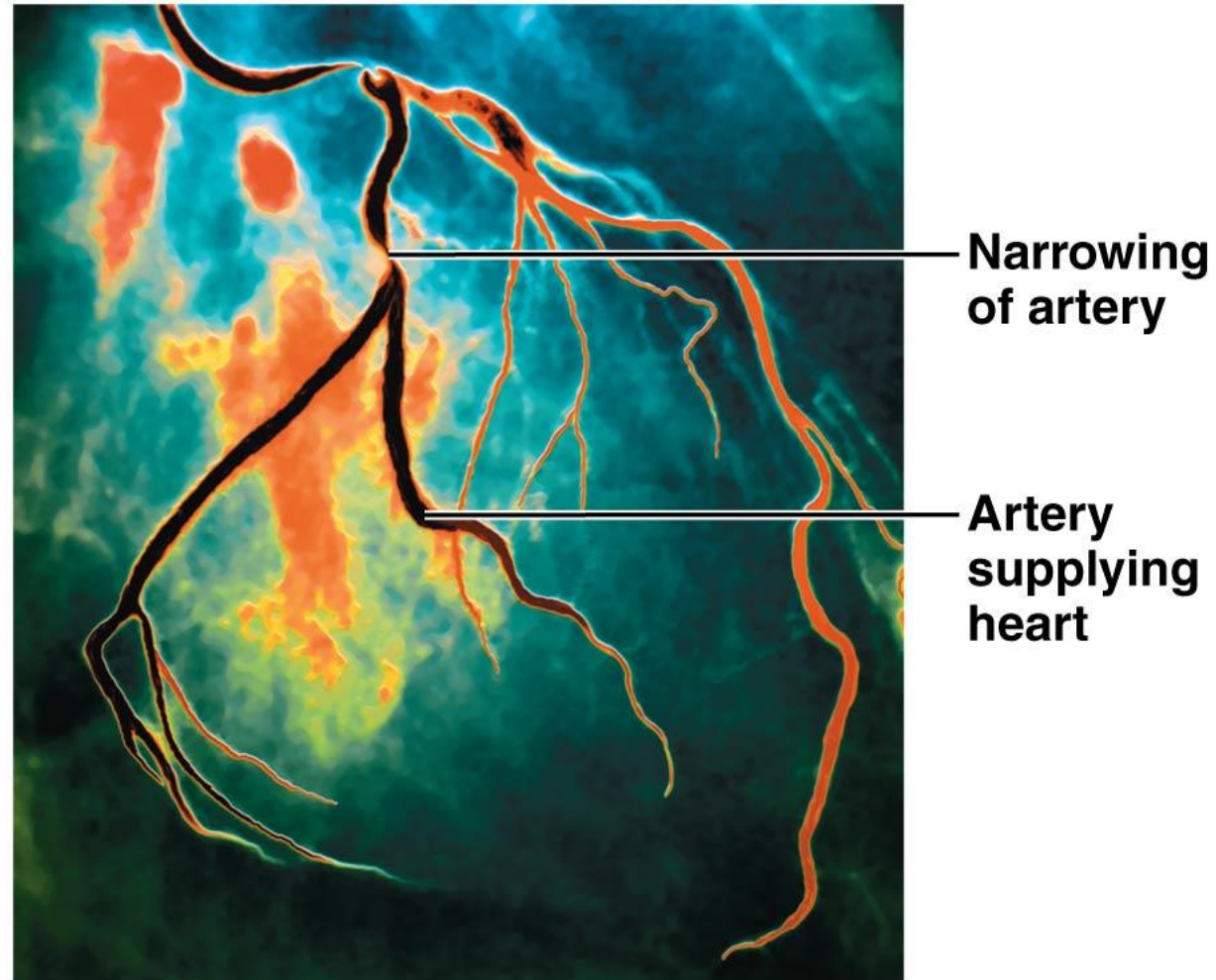


Contrast media

- Group of chemical agents developed to aid in the characterization of pathology by improving the contrast resolution of an imaging modality.
- Examples: X-rays, iodine and barium-sulfate are the most common types of contrast agent, Gadolinium is used in MRI, and microbubbles are used as contrast agents for sonographic examination, specifically echocardiograms
- May be introduced by injection, orally, or rectally, depending on the structure to be imaged
- Adverse reactions to radiographic contrast media include rash, skin redness, and skin swelling, sometimes associated with nausea, vomiting, and dizziness

Angiography (“vessel image”)

- Used for diagnosing aneurysms and atherosclerosis and for identifying a source of internal bleeding.



Digital subtraction angiography (DSA)

- an extension of angiography, is used to monitor blood flow through specific organs, such as the brain, heart, lungs, and kidneys. X-rays are taken before and after radiopaque dye is administered, and a computer “subtracts” details common to both images. The result is a high-contrast image showing the distribution of the dye.



FIGURE 1.12 Digital subtraction angiography (DSA). A DSA image of the arteries that supply the heart.

Positron-emission tomography (PET)

- Involves injecting, inhaling or swallowing a radioactive 'tracer' (a substance that emits positrons (positively charged particles)).
- The collision of positrons with negatively charged electrons in body tissues produces gamma rays (similar to x-rays) that are detected by gamma cameras positioned around the subject

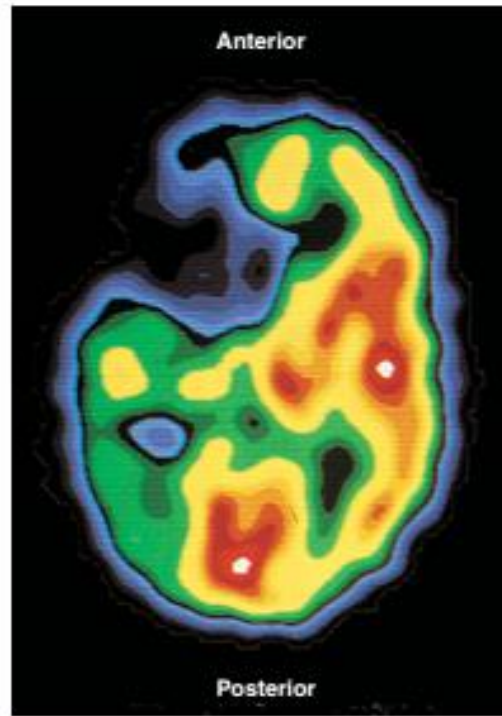


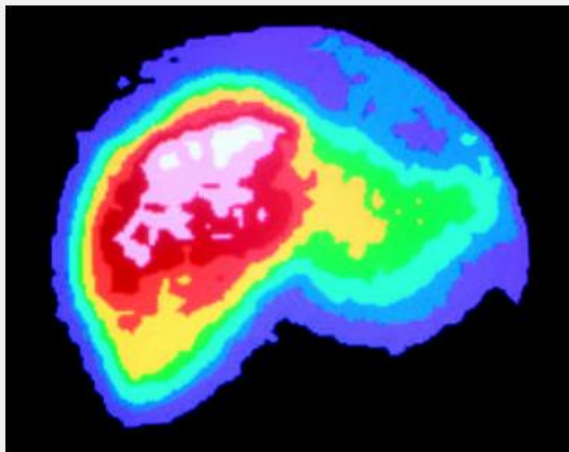
FIGURE 1.13 Positron emission tomography (PET). A PET scan of the brain, in transverse section, shows an area with no neural activity (the frontal region of the brain at upper left). This area was destroyed by a stroke.

Creates images of active blood flow and physiologic activity of the organ or organs being targeted.

The black and blue colors indicate minimal activity; the red, orange, yellow, and white colors indicate areas of increasingly greater activity.

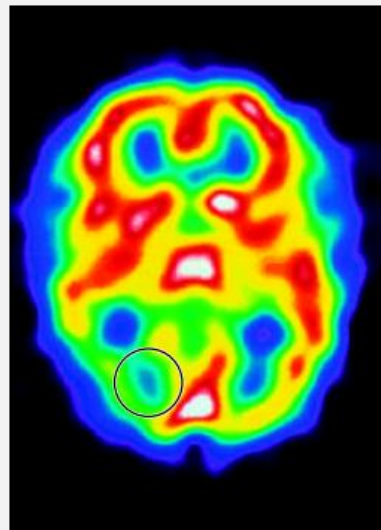
Radionuclide Scanning

- A radionuclide (radioactive substance) is introduced intravenously into the body and carried by the blood to the tissue to be imaged.
- Gamma rays emitted by the radionuclide are detected by a gamma camera outside the subject, and the data are fed into a computer.
- Used to study activity of a tissue or organ, such as searching for malignant tumors in body tissue or scars that may interfere with heart muscle activity.



Publiphoto/Science Source

Radionuclide (nuclear) scan of normal human liver



Charing Cross Hospital/Science Source

Single-photon-emission computed tomography (SPECT) scan of transverse section of the brain (the almost all green area at lower left indicates migraine attack)

Single photon-emission computed tomography (SPECT) scanning is a specialized type of radionuclide scanning that is especially useful for studying the brain, heart, lungs, and liver.

Endoscopy

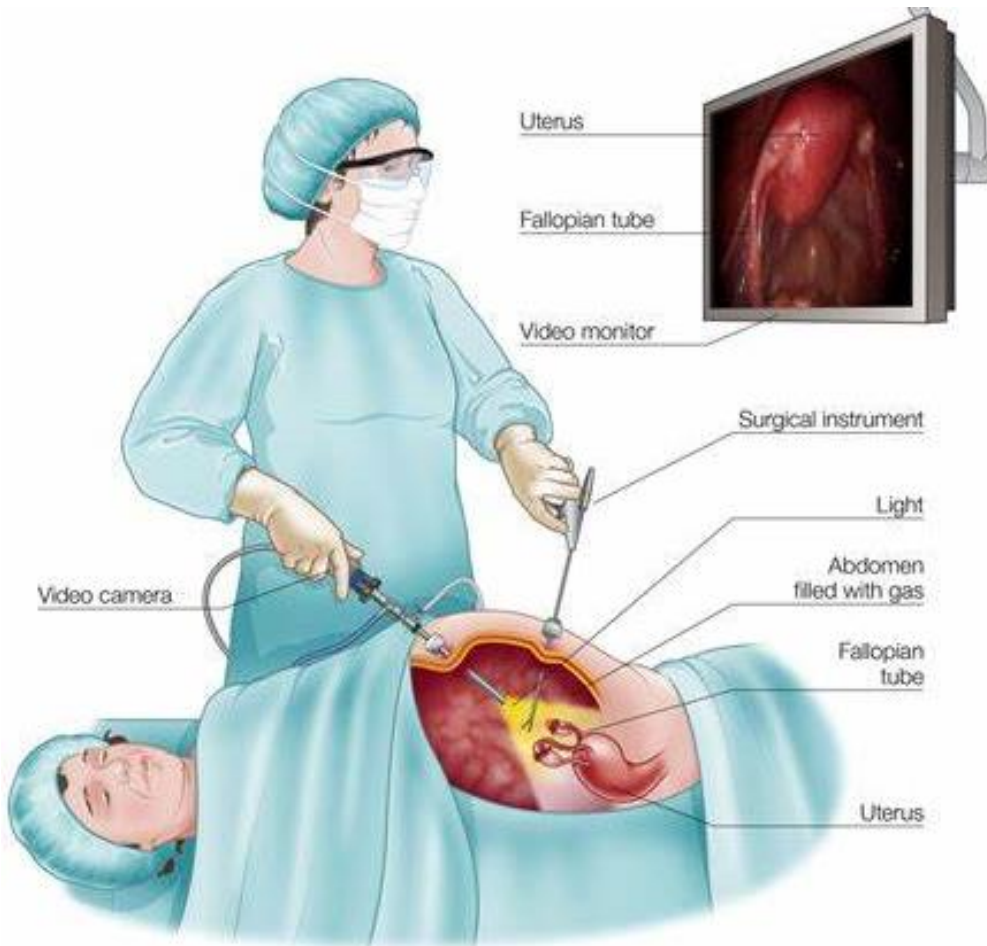
- visual examination of the inside of body organs or cavities using a lighted instrument with lenses called an **endoscope**. Types: colonoscopy, colposcopy, gastroscopy, hysteroscopy, etc
- uses natural orifices such as the anal canal and mouth to access internal body parts.



Interior view of the colon as shown by colonoscopy

Laparoscopy (Keyhole surgery)

- type of surgery that lets a surgeon look inside the body (organs within the abdominopelvic cavity) without making a large incision



Interventional radiology (IR)

- medical specialty that performs minimally invasive procedures using medical imaging guidance.
- **Examples:** Balloon angioplasty, Biopsies, Central venous access, Chemoembolization, Embolization, Radiofrequency (RF) ablation, Gastrostomy tube Insertion, Thrombolysis, TIPS (transjugular intrahepatic portosystemic shunt), Spinal procedures (Vertebroplasty, Myelograms, Facet injections, epidural injections, Discograms, lumbar puncture, Nerve blocks, Rhizotomy).
- Procedures can be both diagnostic and therapeutic.
 - **Diagnostic IR** procedures help make a diagnosis or guide further medical treatment. Examples include image-guided biopsies (sampling tissue from a tumor) and injecting contrast agents into blood vessels or ducts.
 - **Therapeutic IR** procedures provide direct treatment. They involve catheter-based medicine delivery, placement of medical devices (such as stents), and angioplasty
- Benefits of IR include reduced risks, less pain, and faster recovery compared to open surgeries. Real-time imaging also ensures precision during procedures.